

Chapter One

Basic Concepts of Research

1.1. Introduction

Curiosity or questioning is a distinguishing aspect of human beings. As human beings, we may have questions in our mind about business or economic, social, political, environmental, and many other areas of activities. Whenever we encounter problems about these and other areas of concern, we try to find solutions to them. A systematic search for such solutions to problems involves research.

Hence, **Dear learner!**, in this chapter, you will learn introductory issues in research: what research is, a difference between Research Methods and Methodology, characteristics of Research, objectives of Research, motivation in Research, classification of Research, research approaches, and finally Research Process will be highlighted.

1.2. Lesson objectives

Hello learner! After completing this chapter, you will be able to:

- ✓ Properly define and explain what research is
- ✓ Show the difference between Research Methods and Research Methodology
- ✓ Understand the characteristics of a research
- ✓ Know the objectives of and motivation to a research
- ✓ Identify the main classifications or types of research and their characteristics
- ✓ Explain the basic approaches to research, and
- ✓ Understand the main elements in the process of research

1.3. What is research?

Research is composed of two words, “re” and search” which means to search again, or to search for new facts or to modify older ones in any branch of knowledge. In simple words, research can be defined as any scholarly investigation in search for truths, for facts, for certainties.

Research may be defined as the objective and systematic method of finding solution to a problem i.e., systematic collection, recording, analysis, interpretation and reporting of information about various facts of a phenomenon under study. In other words, research refers to the systematic method consisting of enunciating the problem, collecting facts or data, analyzing the facts critically and reaching conclusions based on them. For instance, if we are interested in doing market research, then we should do research concerning with existing or potential markets, marketing strategies and tactics; the interaction between markets, marketing methods and current or potential products or services. In fact, research plays a major role in applying a truly analytical approach to decision making. Besides, it also assists in the evaluation of effect of decisions which have been taken. The social scientists believe that the ultimate aim of research must be the social benefit i.e., they opine that the research workers must solve the problems faced by the society.

Some other Definitions of Research

- Research is sufficiently objective and systematic to make possible classification, generalization and verification of the data observed.
- Research refers to a critical and exhaustive investigation of experimentation having as its aim the revision of accepted conclusions in the light of new discovered facts.
- Research is a careful inquiry or examination in seeking facts or principles, a diligent investigation to ascertain something.
- Research is an honest, exhaustive, intelligent searching for facts and their meanings or implications with reference to given problem. It is the process of arriving at dependable solutions to problems through the planned and systematic collection, analysis and interpretation of data.

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Dear Learner! How do you define research? _____

1.4. Research Methods Vs Methodology

Research methods can be understood as all those methods/techniques that are used for conduction of research. Research methods can be put in to the following three groups:

- i. The methods which are concerned with collection of data

- ii. These statistical techniques which are used for establishing relationships between the data and the unknowns;
- iii. Those methods which are used to evaluate the accuracy of the results obtained.

Research methodology is a way to systematically solve the research problem. It may be understood as a science of studying how research is done scientifically. In it, you find the various steps that are generally adopted by a researcher in studying a research problem along with the logic behind them.

Therefore, researchers need to know how to develop certain indices or tests; how to calculate the mean, the mode, the median or standard deviation or chi-square, how to apply particular research techniques; which of these techniques are relevant or not, and what would they mean and indicate and why they also need to know the assumptions underlying the various techniques, the criteria by which they can decide on the application of certain techniques and not the others. These all constitute research methodology. Moreover, why research study has been undertaken, how the research problem has been defined, in what way and why the hypothesis has been formulated, what data have been adopted, why particular technique of analyzing data has been used and a host of similar other questions are usually answered when we talk of research methodology concerning a research problem or study.

In conclusion, research methodology has many dimensions and research methods constitute a part of the research methodology.

1.5. Characteristics of Research

- Research is directed towards the solution of a problem. It may attempt to answer a question or to determine the relation between two or more variables.
- Research emphasizes the development of generalization, principles of theories that will be helpful in predicting future occurrences.
- Research is based upon observable experience or empirical evidence
- Research demands accurate observations and description.

- Research involves gathering new data from primary or first hand sources or using existing data for a new purpose.
- Research requires expertise. The researcher knows what is already known about the problems and how others have investigated it.
- Research is carefully recorded and reported
- Research sometimes requires courage.

1.6. Objectives of Research

The purpose of research is to discover answers to questions through the application of scientific procedures. Its main aim is to find the truth which is hidden and which has not been discovered as yet. Research objectives may fall in to the following broad groupings.

- i. To gain familiarity with a phenomenon or to achieve new insights in to it (such types of studies are termed as exploratory or formulative research studies);
- ii. To portray accurately the characteristics of a particular individual, situation or a group (such studies are termed as descriptive studies);
- ii. To determine the frequency with which something occurs or with which it is associated with something else (such study is categorized as diagnostic research study);
- iii. To test a hypothesis of a causal relationship between variables (such studies are known as hypothesis-testing research studies).

1.7. Motivation in Research

Why people take themselves to research is the main question. The possible motives may be either one or more of the following:

- 1) Desire to get a research degree along with its consequential benefits;
- 2) Desire to face the challenges in solving the unsolved problems;
- 3) Desire to get intellectual joy of doing creative work;
- 4) Desire to be of service to the society
- 5) Desire to get respectability

The motivation will, however, determine to a considerable extent the nature, quality, depth and duration of research.

1.8. Classification of Research

There are two broad classification of research

- 1) Research in physical or natural science
- 2) Research in human or social sciences

1) Physical science deals with things, which can be put for laboratory tests under guided conditions. These researches deal with physical phenomena upon which ‘man’ has complete control.

2) Researches in social sciences are based on human behavior, which is influenced by so many factors such as physical, social, temperamental, psychological and economic.

Social research is part of research, which studies human behavior. Social research is to find explanation to unexplained social phenomena, to clarify doubts and correct the misconceived facts or social life. Social research is a systematized investigation to gain new knowledge about social phenomena and surveys.

? **Dear Learner!** What is the difference between social science research and natural science research? _____

A research can also be categorized in the following manner.

1. Policy Research

Researches which are conducted for the specific purpose of application, or researches with policy implications, may be treated as policy researches. The results of such studies are used as indices for policy formulations and implementation. Many management researches are policy researches, because they are not merely of theoretical value. They are more of practical utility than of theoretical knowledge.

2. Descriptive Vs analytical

Descriptive research includes surveys and fact-finding enquiries of different kinds. The major purpose of descriptive research is description of the state of affairs as it exists at present. In social science and business research, we often use the term “*Ex-post facto research*” for descriptive research studies. In such kind of research, the researcher has no

control over variables; he can only report what has happened or what is happening. Most *ex-post facto* research projects are used for descriptive studies in which the researcher seeks to measure such items as, for example, frequency of shopping, and preference of people, to discover causes or similar data.

An analytical study or statistical method is a system of procedures and techniques of analysis applied to quantitative data. It may consist of a system of mathematical models or statistical techniques applicable to numerical data.

Moreover, a descriptive study identifies relevant variables but does not aim at testing hypotheses. But analytical study is primarily concerned with testing hypotheses and specifying and interpreting relationships. A descriptive study is less limited by rigorous requirements of measurement and analysis than an analytical study. An analytical study design approximates to the model of an experimental design. A descriptive study employs simple statistical techniques like averages and percentages, but an analytical study employs advanced statistical techniques such as correlations and multivariate analysis.

3. Applied Vs Fundamental

Applied research aims at finding a solution for an immediate problem facing a society or an industrial or business organization.

Fundamental/pure research is mainly concerned with generalizations and with the formulation of a theory. Fundamental research is original and investigative study of a basic nature. The topic or area of investigation would be new and fresh. Investigation would be carried out to originate a new theory, making additional contribution to knowledge. It may also throw additional light on the existing theory or body of knowledge. Its contribution, therefore, can be two folds: developing a new theory and improving upon the existing theory. Examples of fundamental research include research studies concerning some natural phenomenon, pure mathematics, research studies carried on to make generalizations about human behavior, etc.

A research aimed at certain conclusions or solutions facing a concrete social or business problem are an example of applied or action research. Applied research is meant for testing the known theories and models of the populations. The validity of the established theories may have to be tested under certain conditions. For example, preference theory may be tested in the market conditions in Ethiopia. A theory that is established with the help of research in a particular population, at a particular time or in a given environment, need not be valid in a different situation, and hence the need for testing it elsewhere. First, it makes it possible for us to make comparisons between different populations, communities or organizations. Second, a better generalization becomes possible. Third, the validity of the theory increases. Finally, it makes a greater contribution to knowledge within means.

Therefore, applied research aims to discover a solution for some pressing practical problem, where as basic research directs towards finding information that has a broad base of application and thus, adds to the already existing organized body of scientific knowledge.

Researchers can apply several types of applied researches. Some of these include evaluation research, social impact assessment research, and action research

i. **Evaluation research** is one type of applied research. It is made for assessing the effectiveness of social or economic programs implemented (say, family planning program) or for assessing the impact of development projects (say, irrigation project) on the development of the project area. Therefore, its purpose is directed to assess or appraise the quality and quantify of an activity and its performance and to specify its attributes and conditions required for its success. It is also concerned with changes over time.

The types of evaluation research are three:

- *Concurrent evaluation*:- This is a continuous process and partakes the nature of an inspection or social audit of an on-going program. It evaluates the quality of implementation and it serves as a feedback for improving the performance.

- **Period evaluation:** - this is made after each distinct phase of a project is completed. It can be mid-term (in the middle of the period) evaluation or other form.
- **Terminal evaluation:** - This is done after the completion of a program or a project.

ii. Social impact assessment: - This is aimed to estimate the likely consequences of a planned change. It can be used for planning and making choices among alternative policies. Areas assessed in social impact studies include:

- Community services (E.g. School enrollments, speed of police responses)
- Social conditions (E.g. Crime rates, the ability of elderly to feel that they care for themselves)
- Economic impact (E.g. Changes in income levels)
- Demographic consequences (E.g. Changes in the mix of old and young people)
- Environment (E.g. change in air quality)
- Health outcomes (E.g. changes in the occurrence of disease)
- Psychological well-being (E.g. changes in stress, fear, etc)

iii. Action Research: This is a type of concurrent evaluation study of an action program launched for solving a problem. Action programs may include human resource development programs, rural development programs, social welfare programs, managerial improvement programs, etc. This plethora of development has given impetus to action research.

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Dear Learner! Define the following terms:

- Evaluation research_____
- Social impact assessment_____
- Action Research_____

4. Quantitative Vs qualitative

Quantitative research is based on the measurement of quantity or amount. It is applied to phenomenon that can be expressed in terms of quantity or number. On the other hand, qualitative research is concerned with phenomena relating to or involving quality or kind.

For instance, when we attempt to investigate the reasons for human behavior, we quite often talk of “*motivation research*”, an important type of qualitative research.

Qualitative research is loosely defined category of research designs or models, all of which elicit verbal, visual, tactile, and gustatory data in the form of descriptive narratives like field notes, recordings or other transcriptions from audio and videotapes and other written records and pictures and films. In qualitative research, researchers collect data in the form of words, descriptions or pictures. For example, open-ended interviews, naturalistic observations, document analysis, case studies /life stories, descriptive and self-reflective supplements to experiments and correlational studies are means of gathering qualitative data.

5. Conceptual Vs Empirical Research

Conceptual research is concerned with some abstracts or theory. It is generally used by philosophers and thinkers to develop new concepts or to reinterpret the existing ones. Empirical research, on the other hand, relies on experience or observation alone, often without due regard for system and theory. It is data-based research, coming up with conclusions which are capable of being verified by observation or experiment. It can also be called experimental type of research in which there are control groups and experimental groups of variables. Empirical research is appropriate when proof is sought that certain variables affect other variables in some way.

6. Cross-sectional Vs longitudinal research

These categories are seen from the point of *time*. In the case of cross-sectional research, the research is confined to *a single time period*, whereas in the later case, the research is carried on *over several time-periods*. As a weakness, the cross-sectional research can not capture social process or change.

Dear learner! The longitudinal research can further be categorized in to three:

a. Time series research: - In this case, the same type of information is collected on a group of people or other units across multiple time periods. This serves to observe stability or changes in the features of the units.

b) Panel study: - Panel studies are similar to cohort analyses except the same individuals are interviewed in each period, whereas in cohort studies only a random sample of the same age group is interviewed. In panel designs, such as those built into the National Election Studies, individuals serve as their own controls across survey periods, whereas in cohort analyses, random sampling is assumed to serve a control function, assuring similarity of a cohort by gender, race, etc., as it is measured in successive time periods. Where cohort designs measure *net change*, panel designs measure *gross change* (for instance, many more people may switch party identification in a given period than are measured by the net effect of all such switches). Normally panel designs cannot measure net change for an entire population because only certain birth cohorts are studied as they evolve over time, and new cohorts entering the population are not studied. Unlike cohort studies, panel studies have the special problem of *sensitization*: subjects may react differently on a second survey simply because they have had the experience of the first one. To check for a sensitization effect, panel studies may require control groups matched to the panel groups, further increasing the great expense of this design.

Example: if you want to study the gender difference on Mathematics performance and plan to examine the gap-across the grades in AMU Community primary school, you may give exams while they are in grade 1 and continue to administer the same subject for the same students when they enter grade 2, and you may continue when they join grade 3, and etc.

c) *Cohort analysis*: unlike in the panel study, a category of people who share similar life experience (but not same people) in a specified time period is studied. Here, the focus is on the cohort, or category, not on specific individuals. Example, all people born in the same year (birth cohort), all people hired at the same time, all people graduated in the same year may be contacted to collect data.

7. Case studies Vs Surveys

A case study is an in-depth comprehensive study of a person, a social group, an episode, a process, a situation, a program, a community, an institution, or any other social unit. Its purpose may be to understand the life cycle of the unit under study or the interaction between factors that explain the present states or the development over a period of time.

The examples include social anthropological study of a rural community, a causative study of a successful co-operative society; a study of the financial health of a business undertaking; a study of employee participation in management in a particular enterprise, a study of juvenile delinquency; a study of life style of working women; a study of life in slums; a study of urban poor, a study of economic offenses; a study of refugees from an other country.

Survey is a research method involving collection of data directly from a population or a sample thereof at particular time. Data may be collected by observation, interviewing or mailing questionnaires. The analysis of data may be made by using simple or complex statistical techniques depending up on the objectives of the study. In short, the case study and survey methods are compared as follows:

Case study	Survey
Intensive investigation	Broad-based investigation of a phenomenon
Study of a single unit /group	Covers large number of units (units of universe or a sample of them)
The findings of a case study can not be generalized.	The findings of a survey study can be generalized based on sample
Useful for testing hypotheses about structural and procedural characteristics (e.g. Status relation, interpersonal behavior, managerial style) of a specified social unit (e.g. an organization, a small group or a community)	Useful for testing hypotheses about large social aggregates

8. Experimental research

There are various phenomena such as motivation, productivity, development, and operational efficiency which are influenced by various variables. It may become

necessary to assess the effect of one particular variable or one set of variables on a phenomenon. This need has given rise to experimental research.

Experimental research is designed to assess the effect of particular variables on a phenomenon by keeping the other variables constant or controlled. It aims at determining whether and in what manner variables are related to each other. *The factor which is influenced by other factors is called a dependent variable, and the other factors which influence it are known as independent variables.* E.g. agricultural productivity, i.e. crop yield per hectare is a dependent variable and the factors such as soil fertility, irrigation, quality of seed, manuring, and cultural practices which influence the yield are independent variables.

The nature of relationship between independent variables and dependent variables is perceived and stated in the form of causal hypotheses. A closely controlled procedure is adopted to test them.

Now let us see the procedures in experimental research.

Procedures in Experimental Research

Two identical groups are selected. These should be identical in terms of the characteristics of the phenomenon under study. For example, in a farm productivity experiment two plots of farm land with same soil composition, soil fertility, same size, same climate, and same irrigation facility should be selected for the study. One of the groups is used as experimental group, and the other as control group. *Experimental group is exposed to the experimental variable or stimulus. Control group is not exposed to the experimental variable.* The difference between the experimental and control groups' outcome is attributed to the effect of the experimental variable.

♣ *Illustration: An agricultural scientist desires to study the impact of application of a particular chemical fertilizer (say, urea) on the yield of a crop. He selects two plots of land in a farm. All conditions –soil fertility, climate, irrigation, seed, and cultural practices-are the same. Thus all variables are kept constant. He applies chemical fertilizer in the experimental plot alone. This is the only variation between the two plots. Hence, the difference in yield is attributable to the application of the fertilizer.*

Again note that, in some cases, a single group may be used as both control and experimental group. For example, if you want to know the effect of a member-education program on the cooperative knowledge of the members of a cooperative, you can randomly select a cross section of a society. First measure the type and extent of cooperative knowledge of the group. Then expose the same group to a member-education program and again measure their knowledge. The difference in knowledge can be attributed to the program.

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Dear learner! What is the difference between Experimental group and Control group in an experimental research? _____

Under the following heading, the typology of experimental research will be discussed.

Types of experimental Designs

There are many types of experimental design. Let us treat three of them for our purpose.

- a. After-only design
- b. One group Before-after design
- c. Before-after design with control group

a) **After-only design:** This is the simplest type of experimental design. The effect on a dependent variable (Y) is assessed by measuring it in both Experimental (E) and Control (C) groups after E has been exposed to an independent variable (X). In this design, information on base line (pre-test or before observation) is usually “constructed” on the basis of respondents’ recall of the situation before the intervention, or from information available in existing records. The change in the dependent variable is measured by the difference between the ‘before’ (base-line) and “after” observations. Example, from the illustration cited above, the crop yield (Y) is measured in both E and C plots, after experiment is over. The difference is attributed to the application of chemical fertilizer (x). This application is briefly illustrated as follows.

Measurement after experimentation

E	Y_e
C	Y_c

$$\text{Difference} = Y_e - Y_c$$

However, the design has some weaknesses. There may not be proper base-line data to compare the ‘after’ observation. Accordingly, some of the changes in the dependent variable may be attributable to the difference in the way the two sets of data were compiled. The other problem is that it measures total change, including change attributable to extraneous variables (see page 56 and 57); hence, it can not identify the net effect of intervention.

b) One group before-after design: In this category, the same group is used as E and C, measuring before and after the group has been exposed to X. Suppose you want to evaluate the effectiveness of a film in changing attitudes towards drinking and driving. The change in score on an attitude test given on the target group before and after watching the film measures the film’s effect. In short, observe from the following table.

	Before	After
Target group	Y_1	Y_2

$$\text{Difference} = Y_2 - Y_1$$

c) Before after design with control group: In this design, both the E and C groups are measured before and after E is exposed to X. The design is better than the previous ones because it eliminates the impact of extraneous factors occurring during time period.

	Before	After
E	Y_{e1}	Y_{e2}
C	Y_{c1}	Y_{c2}

$$\text{Difference in E} = Y_{e2} - Y_{e1} = D_e$$

Difference in C = $Y_{c2} - Y_{c1} = D_c$

New difference = $D_e - D_c$

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Dear Learner! Briefly explain the differences among the various types of experimental research design_____

Under the following heading, the merits and demerits of experimental design are seen.

Evaluation of Experimental design

The use of experimental design in social sciences is complicated. It is difficult to establish comparable experimental and control groups. Experiment is often difficult to design on human behavior; it is expensive and time consuming. It can be used to study the present only and can not be used for studies of the past or the future. However, in objectivity and in studying causal relationships between variables, no other method can equal experimentation. The effect of extraneous variables is more effectively controlled in this design than in other designs. Above all, it provides means for testing social values.

9. Some other Types of Research

Research can be *field-setting research or laboratory or simulation research*, depending up on the environment in which it is to be carried out. Research can as well be understood as clinical or diagnostic research. Such researches follow case-study methods or in-depth approaches to reach the basic causal relations. They use very small samples and very deep probing data gathering devices.

The research may be *exploratory or formalized*. Exploratory research aims to the development of hypotheses rather than their testing, where as formalized research studies are those with substantial structure and with specific hypothesis to be tested.

Historical research is that which utilizes historical sources like documents, remains, etc to study events or ideas of the past, including the philosophy of persons and groups at any remote point of time.

Research can also be classified as *conclusion-oriented and decision-oriented*. While doing conclusion-oriented research, a researcher is free to pick up a problem, redesign the enquiry as he proceeds and is prepared to conceptualize as he wishes. Decision-oriented research is always for the need of a decision maker and the researcher in this case is not free to embark upon research according to his own inclination. Operations research is an example of decision-oriented research since it is a scientific method of providing executive departments with a quantitative basis for decisions regarding operations under their control.

Explanatory research is also the other type. When you encounter an issue that is already known and have a description of it, you may begin to wonder “why” things are the way they are. The desire to know “why” to explain is the purpose of explanatory research. It builds on exploratory and descriptive research and goes on to identify the reason why something occurs. Therefore, it looks for causes and reasons. For example, a descriptive researcher may discover that 10% of parents abuse their children, whereas the explanatory researcher is more interested in learning why parents abuse their children.

The common types of explanatory research are *case study and tracer study*. As already discussed, in case study, you examine many features of a very few cases in-depth either at one time period or across time periods. The tracer study is also known as follow-up study. It aims at investigating the subsequent development of individuals after a specified treatment or condition. It is used to investigate the direction of movements and predicts what is likely to take place in the future. It is designed to establish patterns of changes in the past in order to predict future patterns or conditions by analyzing data collected about subjects and environment. For instance, you can conduct a tracer study of the former graduates of Arba Minch TVET to know:

1. Whether all its graduates are employed or absorbed by the labor market
2. Whether there is mismatch between the training graduates received and the types of jobs they are handling
3. Whether there are new environmental trends that justify the need of change.

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Dear learner! 1. Mention the various types of research you have understood? _____

1.9. Research Approaches

There are two basic approaches to research: *quantitative approach* and *qualitative approach*. The former involves the generation of data in quantitative form which can be subjected to rigorous quantitative analysis in a formal and rigid fashion. The approach can be further sub-classified in to inferential, experimental, and simulation approaches to research. The inferential approach forms database from which to infer characteristics or relationships of a population. This usually means survey research where a sample of population is studied (questioned or observed) to determine its characteristics, and it is then inferred that the population has the same characteristics. Experimental approach is characterized by much greater control over the research environment and in this case some variables are manipulated to observe their affect on other variables. Simulation approach involves the construction of an artificial environment within which relevant information and data can be generated. This permits an observation of the dynamic behavior of a system (or a sub-system) under controlled conditions. The term “simulation” in the context of business and social sciences applications refers to the operation of a numerical model that represents the structure of a dynamic process. Given the values of initial conditions, parameters and exogenous variables, a simulation is run to represent the behavior of the process over time. It can also be used to build models for understanding future conditions.

Qualitative approach to research is concerned with subjective assessment of attitudes, opinions and behavior. Research in such a situation is a function of the insights and impressions of a researcher. It generates results in non-quantitative form or in the form which are not subjected to rigorous quantitative analysis. Focus group interviews, projective techniques and in-depth interviews are used to collect qualitative data.

1.10. Research Process

Before embarking on the details of research methodology and techniques, it is appropriate to present a brief over view of the research process. Research process consists

of series of actions or steps necessary to effectively carry out research and the desired sequencing of these steps. A research process consists of a number of closely related activities. The activities overlap continuously rather than following a strictly prescribed sequence. However, the following order provides a useful procedural guideline regarding the research process.

1. Formulation of a research problem
 - a. Identifying, selecting, and formulating a research problem
 - b. Extensive literature survey
 - c. Developing hypotheses
 - d. Identifying variables
2. Preparing the research design
3. Determining sample design
4. Collecting the data
5. Analysis of data
6. Hypotheses testing (if any)
7. Generalizations and interpretations, and
8. Preparation of report or presentation of the results, i.e., formal write-up of conclusions reached.

Now if you have understood a brief process of research, you will grasp an in-depth knowledge in the succeeding chapters. Chapter three will begin with the first step of the process.

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Dear learner! List down the major steps in a research process_____

Chapter Summary

From among many ways of obtaining information to find answers to your questions, research is a systematic way. Research can broadly be categorized as research in physical or natural sciences and research in human or social sciences. While social science researches study about social phenomenon, natural science researches deal with things which can be put for laboratory tests under guided conditions. Specifically, some of the typology of research includes policy research, descriptive or analytical research, applied or fundamental research, quantitative or qualitative research, conceptual or empirical research, cross-sectional or longitudinal research, case studies or surveys, experimental research, field-setting research, simulation research, exploratory research, historical research, conclusion-oriented and decision-oriented research, explanatory research, case study and tracer study.

A research applies two approaches: *quantitative approach and qualitative approach*. Quantitative approach to research generates data in quantitative form which is subject to quantitative analysis. This approach can also be sub-classified as inferential, experimental, and simulation approaches to research. Secondly, qualitative approach to research is concerned with the subjective assessment of attitudes, opinions and behavior of subjects. Qualitative data can be generated through Focus group discussions, projective techniques and in-depth interviews.

Since various activities in a research process overlap, the number of steps in a research process varies according to different authors. In this module, you learned an eight-step process the details of which are found in the succeeding chapters. These steps cover the total spectrum of research endeavor, starting from problem formulation and ending with writing-up of a research report.

Exercises

Part I. Choose your answers from the given choices as follows.

1. One of the following statements *does not* characterize a research.
 - a. Research is directed towards the solution of a problem
 - b. Research involves gathering new data from primary or secondary sources
 - c. Research requires expertise
 - d. None of the above is the answer
2. The aim of research can be
 - a. to find the truth which has not been discovered as yet
 - b. to discover answers to questions through the application of scientific procedures
 - c. To describe accurately the characteristics of a particular individual or situation
 - d. All are answers
3. One of the following factors can not motivate an individual to a research
 - a. The motive to face the challenges in solving the unsolved problems;
 - b. The motive to get intellectual joy of doing creative work;
 - c. The motive to be of service to the society
 - d. The motive to get respectability
 - e. None
4. One of the following statements is false regarding descriptive research
 - a. Its purpose is to describe the state of affairs as it exists at present
 - b. Its prime concern is to test hypotheses
 - c. It employs highly advanced statistical techniques
 - d. All are answers
5. An applied research is not concerned with
 - a. finding a solution for a pressing problem facing a society or a firm
 - b. generalizations and formulation of a theory
 - c. testing of the known theories and models

- d. All
 - e. Only a & c
6. The quantitative research strategy places a value on:
- a. Using numbers, measurements and statistical techniques
 - b. Fact relating to or involving quality or kind
 - c. Usually employing open-ended interviews, observations, document analysis, life stories, and such other means
 - d. All of the above
7. In an experimental research, the dependent variable is:
- a) The one that is not manipulated and in which any changes are observed
 - b) The one that is manipulated in order to observe any effects on the other
 - c) A measure of the extent to which personal values affect research
 - d) An ambiguous concept whose meaning depends on how it is defined

Part II. Answer the following questions

1. Mention the different types of applied researches
2. What types of evaluation research did you understand? What are their differences?
3. Differentiate time series, panel and cohort studies.
4. Define the following research types:
 - a. Exploratory research
 - b. *Historical* research
 - c. *conclusion-oriented* and *decision-oriented*
 - d. *Explanatory* research
 - e. *case study and tracer study*

Chapter Two

Formulation of Research Problem

2.1. Introduction

Dear learner! From the previous chapter, you remember that formulation of a research problem is the first step in a scientific inquiry. It is like the identification of a destination before undertaking a journey. As in the absence of destination it is impossible to identify any route, in the absence of clear research problem, it is impossible to have clear and economical plan. A researcher must find the problem and formulate it so that it becomes susceptible to research. A researcher must examine all the symptoms (presented to him or observed by him) concerning a problem before he can diagnose correctly. To define a problem correctly, a researcher must know what a problem is. A research problem is like the foundation of a building. A research problem serves as the foundation a research study: if it is well-formulated, you can expect a good study to follow. Accordingly, in this chapter, issues such as research problem and sources, criteria of a good research problem, problem formulation /definition and the need of definition, necessary conditions and techniques for formulation of research problem, and related issues like operational definition of concepts, and variables and their classifications, research hypothesis and sources, types of hypothesis, rules of constructing hypothesis, forms of stating hypothesis, characteristics of a good hypothesis, and difficulties in the formulation of hypothesis will be discussed.

2.2. Lesson objectives

Dear learner! After completing this chapter, you will be able to:

- ✓ Define a research problem
- ✓ Understand how to select a research problem
- ✓ Identify the sources of a research problem
- ✓ Explain the components of a research problem
- ✓ Mention the criteria of a good research problem
- ✓ Know how and why to formulate or state a research problem

- ✓ Identify the necessary conditions for formulation of a research problem
- ✓ Indicate the techniques involved in the problem definition/formulation process
- ✓ Give operational/ working definitions of concepts, and identify variables
- ✓ Distinguish the differences of concepts, indicators and variables
- ✓ Know what a research hypothesis is, its roles, the rules to construct hypothesis, the types of hypothesis, the forms to state hypothesis, the characteristics of a good hypothesis, and the difficulties to formulate a hypothesis.

2.3. Formulating a Research Problem

1. Research problem, problem identification and selection

Primarily, a research problem refers to some difficulty, which a researcher experiences in the context of either a theoretical or practical situation and wants to obtain a solution for the same. One of the most difficult phases of any research project is the choice of a suitable problem. The beginner is likely to take a very long time in making his choice. In this first step of any research, the research worker should not take a hasty decision. Every problem which comes to his/her mind or even the one suggested by a more experienced person may not fit research problem. The identification of a good research problem should be considered a discovery in itself.

Personal values play important role in the selection of a topic for research. Social scientists with different values tend to choose different topics for investigation. Of course, personal values are not the only determinants in selecting a topic for inquiry. Social conditions do often shape the preference of investigators in a subtle and imperceptible way. The selection of a topic for research is only half a step forward. This general topic does not help a researcher to see what data are relevant to his/her purpose. What methods he/she would employ in securing them and how he/she would organize these? Before he/she can consider these aspects, he/she need to formulate a specific problem. The problem defines the goal of the researcher in clear terms. It is obvious that without a clear out idea of the goal to be reached, research activities would only become a meaningless exercise. A research like any other human activity is goal-directed. If the goal itself is unknown or ill-defined, the whole research operations will lead us nowhere.

Thus, without a problem, research cannot proceed because there is nothing to proceed from and proceed towards. There is nothing but wisdom in the saying “If you start from nowhere you will generally reach there”.

As already said, in the research process, the *first and foremost step* happens to be that of identifying and selecting and properly defining a research problem. The task is a difficult one. Help may be taken from a research guide in this regard. However, every researcher must find out his own solution for research problems cannot be borrowed. A problem must spring from the mind of the researcher as a plant springs from own seed.

The following points may be observed by a researcher in selecting a research problem or a subject for research.

- a) Subject which is overdone should not be normally chosen, for it will be a difficult task to throw a new light in such a case
- b) Controversial subject should not be chosen
- c) Too narrow or too vague problem should be avoided
- d) The subject selected for research should be familiar and feasible so that the related research material or sources of research are within one's reach
- e) The importance of the subject, the qualifications of the researcher, co-operations from participants, the costs and time factors are the other factors should be considered.
- f) The selection of a problem must be preceded by a preliminary study.

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Dear learners! What did you understand about research problem? _____

The other aspects of a Research Problem are dealt under the following headings:

1. Sources of research problem

The research problems may be selected from the following sources:-

- 1) A researcher may select a problem for investigation from a given *theory in which he/she has considerable interest*. In such situations, the researcher must have thorough

knowledge of what theory says and should be sufficiently inquisitive to explore some unexplained aspects or assumptions of that theory.

- 2) Research problem can also be selected on the basis of *daily experience and problems* of a researcher. Every day problems constantly present something new and worthy of investigation and it depends on the sharpness of the researcher's intellect to knit his daily experiences into a research problem. If we are alert, inquisitive and sensitive to our every day life situations, we may hit up on questions worth of investigation. "It is a mark to scientific genius to be sensitive to difficulties where less gifted people pass untroubled by doubt".

The story about the discovery of Newton testifies this reality. Apples have fallen on the heads of people before Newton's time, but it was the sensitive Newton alone who raised the question regarding fall of apple which lead to his discovery of the law of gravitation.

- 3) *Technological and social changes* in fast changing society are constantly bringing forth new problems and new opportunities for research. For example, what is the impact of a changed technology on the existing socio-economic setup or on the existing system?
- 4) Research problems can be both abstract and applied interest. These may also be selected from those *areas which have not been explored so far*, such areas may be theoretical or empirical in nature. For example, an economic analysis of hill farming systems in a region where such an investigation has not been undertaken earlier is a useful piece of research.
- 5) Sometimes the researcher, while *discussing* his interests with the proposed supervisor, may come across a problem that can be researched by the investigator. The problem may relate to any source as discussed above.
- 6) *Reading* assignments in textbooks, special assignments, research reports and term papers may also suggest some additional areas of needed research. Many research articles suggest problems for further investigation that may prove fruitful.
- 7) *Academic experience*: Class room lectures, class discussions, seminar discussions, and out-of-class exchange of ideas with fellow students and lecturers will stimulate problems to be studied.

- 8) *Exposure to field situations*: Field visits internship training and extension work can be sources of research problem.
- 9) *Consultations*: Discussions with experts, researchers, administrators and business executives will help identify research topics.
- 10) *Brain storming*: Intensified discussions within a group of interested persons may help initiate important question
- 11) *Research*: research on one problem may suggest problem for further research
- 12) *Intuition*: Sometimes, new ideas may strike to one's mind like a flash. Reflective mind is a spring of knowledge.

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Dear learners! Mention the possible sources of a research problem_____

2. The Components of a Research Problem

- i. There must be an individual or a group which has some difficulty or problem
- ii. There must be some objectives to be attained. If one wants nothing, one can not have a problem.
- iii. There must be alternative means (or the course of action for obtaining the objectives) one wishes to attain, i.e., at least two means are available to a researcher for if he has no choice of means, he can not have a problem.
- iv. There must remain some doubt in the mind of a researcher with regard to the selection of alternatives. This means that research must answer the question concerning the relative efficiency of possible alternatives.
- v. There must be some environments to which the difficulty pertains.

3. Criteria of a Good Research Problem

Before deciding a problem to be subject to research, a researcher has to be sure that the problem has fulfilled certain qualities. A number of criteria/qualities in the form of conditions can be listed for guidance in the selection of a topic.

I. Novelty or originality: It should be sufficiently original so that it does not involve objectionable duplication. Originality is the basic credit point of any research. Ignorance

of prior studies may lead a student to spend time on a problem already investigated by some other worker. Moreover, the study should employ the most recent data.

While originality is an important consideration, the fact that a problem has been investigated in the past does not mean that it is no longer fit for study. There is constant need for verification of the findings of previous investigations, using newer and better devices and procedures. There is also a need for the testing of former findings under changed cultural condition.

II. Interest: The problem should be interesting for the investigator him/herself. If s/he is not interested in it, s/he will not be able to face and overcome the obstacles which come at every step in research. S/he should have a strong inherent motivation for it. If it seems to her/him dull and boring, there is little hope that s/he would do justice to it. Interest may also sometimes develop with long familiarity, but it is not likely that the student can do his best work on a topic that has no personal meaning for him. A major reason for encouraging a researcher to select her/his own research problem is seen in terms of the growth of the investigator and the quality of her/his work. Her/his interest should be purely intellectual and should not be there only for a reward, material benefit, advancement in position, increased authority and so forth.

III. Importance: It should be significant enough and involve an important principle or practice. If it is not worthwhile, it neither adds to knowledge nor leads to any improvements in the current practices, it would be in vain. It should add to the development of social set-up as a discipline and to the previous research findings in any way. No research project should be undertaken unless its consequences given promise of improving significantly in that area. The difference between the significant and non-significant projects is not in the amount of work required in the case of the former but the amount of thought involved.

IV. Feasibility or Amenability: The next question to be asked is whether research into the problem would be feasible. The suitability of the problem for a particular research worker is the matter of its feasibility. It may be a very good problem but it should be good for the investigator. He would be able to carry it to a successful conclusion. He should possess the required competence, knowledge and understanding. He should be

skilful enough to develop, administer and interpret the necessary data-gathering devise and procedures. He should possess a reasonable grounding in the necessary statistical techniques. He should be competent enough to carry out all the steps of a project. It should match his special qualifications, training and experience.

V. Availability of Data: The research worker should ensure the availability of valid and reliable data gathering devices and procedures. In case the study demands confidential, sensitive and classified information, will it be possible for him to obtain it?

VI. Availability of co-operation: The study may require co-operation from various institutions, authorities and individuals. It may need administration of tests to pupils, interviewing employees, observation of students at play, meeting the citizens, writing for data, going through records, conduction of necessary experiments and going in to prolonged study of exceptional cases. The investigator must make sure that necessary permission and co-operation will be readily available.

VII .Availability of other Facilities: Will I have the necessary financial resources to carry out the study? The investigators should be able to meet the expense involved in data gathering equipment, printing, test machines, travel, computerization, clerical help, postage, preparation of the reports, and etc. If the project is expensive in nature, he will explore the possibility of obtaining financial help or grant from some agency.

Will I have enough time to complete the project? Selecting a problem, library study, preparing the data-gathering devices, collecting and analyzing the data, writing the research report, etc are all time-consuming processes. Most of the research programs impose time limitation. Certain worthwhile studies of a longitudinal type are, therefore, out of question.

VIII. Immediate Application: Will my research help in solving an urgent problem? Supposing I choose to work on education of unfair means in the examinations, how far my research will solve the problem at hand?

IX. Aim of Research: The aim of research also influences the selection of the problem.

X. Level of Research: It is another criterion to help in the selection of a problem. The nature and scope of a study will be determined in the light of levels like, Master's

Degree, M.Phil.Degree and Ph.D. It may simply be an action research or a research to produce a research paper or an experimental project.

XI. Experience and Creativity: Good research problems stem from a clear understanding of the theoretical, empirical and practical aspects of the subject derived from personal experience and from a thorough review of the literature. Conversely, lack of familiarity with the subject is almost sure to result in a poor choice.

Another major contribution to the wise choice of a problem is creativity and the personality factors that make for originality, flexibility, initiative, ingenuity and foresight. Thorough familiarity with a given field is conducive to original thinking.

XII. Courage and Confidence: In this case, such questions as “Will the researcher have the courage and confidence to pursue the study in spite of the difficulties and social hazards that may be involved? Will he/she be able to work aggressively when data are difficult to gather and when others are reluctant to co-operate?” are asked.

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Dear learner! What are the requirements of a good research problem did you understand?_____

2. Formulating and stating/defining the Problem

Introduction

The problem selected for research may initially be vague topic. The question to be studied or the problem to be solved may not be known. The reason why the answer is wanted may not be known as well. Hence, the selected topic should be defined and formulated. If it is to serve as a guide in planning the study and interpreting its result, it is essential that the problem is stated in precise terms. This is a difficult process. It requires intensive reading of related literatures in order to understand the nature of the selected problem. The researcher should read selected literatures, digest, think, and reflect up on what is read and digested. He/she should also discuss with experienced persons.

With regard to the formulation of research problems, the following points should also be noted.

1. What is formulation /defining?

Formulation means translating and transforming the selected research problem /topic in to a scientifically researchable question. It is concerned with specifying exactly *what* the research problem is or *why* it is studied. It involves the task of laying down boundaries within which a researcher shall study the problem with a predetermined objective in view.

Moreover, problem definition implies the separation of the problem from the complex of difficulties and needs. It means to put a fence around it, to separate it by careful distinctions from like question found in related situations of need.

To define a problem means to specify it in detail and with precision. Each question and subordinate question to be answered is to be specified. Sometimes it is necessary to formulate the point of view on which the investigation is to be based. If certain assumptions are made, they are explicitly noted.

It is important to define and elucidate the problem as a whole and further define all the technical and unusual terms employed in the statement. By this, the research worker removes the chance of misinterpretation of any of these crucial terms. The definition helps to establish the frame of reference with which the researcher approaches the problem.

Three principal components in the progressive formulation of a problem for research are identified as follows:

1. The originating question (what one wants to know)
2. The rationale (why aspect)
3. The specifying question (possible answers to the originating question)

1. The originating question: This indicates what the problem is. It may be of different kinds. It may call for discovering new and more decisive facts relating to the subject-matter of study; it may put to question the adequacy of certain concepts, may be related to empirical validity; or it may be related to the structure of an organization.

2. *Rationale for the question:* This is the statement of reasons why a particular question is posed. It states how the answer to the question will contribute to theory and /or practice. It helps to make discrimination between scientifically important and trivial question. In short, it “states the case for the question in the court of scientific opinion”.

3. *Specifying questions:* the originating question is decomposed in to several specific questions in order to identify the observations or data that will provide answers to them. These questions should be simple, pointed, clear, and empirically verifiable. They are known as ‘investigative’ questions. It is only these questions (when synthesized) can afford the solution to the problem selected for research. This solution has implication for theory or systematic knowledge and /or for practice.

2. Types of Problem Statement

The type of problem statement to be employed depends on the preference of the research worker and the nature of the problem. Preferably it should be set as a question or in such form that the question to be answered is clearly indicated. There are two different ways of stating a problem:-

- 1- Posing question/questions
- 2- Making declaration statement/statements

One may choose any of these ways remembering that the question form has an advantage in sharpening and focusing the issue, but the declarative form is perhaps more common, and both of the ways may be combined easily in an initial statement.

3. Necessity of Defining the Problem

Quite often, we all hear that a problem clearly stated is a problem half solved. This statement signifies the need for defining a research problem. The problem to be investigated must be defined unambiguously for that will help to discriminate relevant data from irrelevant ones. Proper definition will enable a researcher to be on the right track where as ill-defined problems will create hurdles. Questions like “What data are to be collected? What characteristics of data are relevant and need to be studied? What relations are to be explored? What techniques are to be used for the purpose?” and similar other questions crop up in the mind of the researcher who can well plan his/her strategy and find answers to all such questions only when the research problem has been well

defined. Thus, defining a research problem is a prerequisite to any study and is a step of the highest importance. In fact, formulation of a problem is often more essential than its solution.

4. Necessary conditions for formulation of a research problem

We can list some of the conditions that experience has proved to be conducive to formulation of significant research problem.

(i) Systematic immersion in the subject matter through first-hand observation

The researcher must immerse himself thoroughly in the subject matter area with in which he wishes to pose a specific problem. For example, if the researcher was interested in the general area of juvenile delinquency, it would serve him well if he visited remand homes, juvenile centers, juvenile courts, the families of the delinquents and the locality where the incidence is high. He should try to know at first hand the various aspects of the lives of delinquents by interviewing them, their family members and by observing relevant aspects about their lives, and etc. There is often no alternative to such field observation , no experience is more rewarding in terms of getting a deep feel of the specific questions that may be posed for the study to answer. This process is known by various names, e.g., Pilot survey, preliminary survey, or exploratory study.

(ii) Study of relevant literature on the subject

The researcher must be well equipped to experience some difficulty or challenge to be able to pose a problem. This, in turn, would depend upon the researcher being well conversant with the relevant theories in the filed reports, records and etc. This will help the researcher to know if there are certain gaps in the theories (his/her problem then will be to bridge these) or whether the prevailing theories applicable to the problem are inconsistent with each other or whether the findings of different studies do not follow a pattern consistent with theoretical expectations and so on. All these will afford occasions for institutions of research problems. This is also an aspect of exploration. For more understanding, chapter three will detail the same issue.

(iii) Discussions with persons having rich practical experience in the field of study

This is often known as an experience survey, which again is an exercise at exploration.

Administrators, social workers, community leaders, etc. are persons who have rich practical experience in different fields of social life. These persons, therefore, are in a position to enlighten the researcher on different aspects of the field of his proposed study. Such persons represent in a nutshell years of experience, hence their advice, components, information and judgment are usually invaluable to the researcher. They help him sharpen his focus of attention on specific aspects within the field.

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Dear learner! Briefly discuss the necessary conditions for formulation of a research problem? _____

5. Techniques involved in defining the problem or formulation process

Defining a research problem properly and clearly is a crucial part of a research study and must in no case be accomplished hurriedly. The techniques for this purpose involves the undertaking of the following steps generally one after the other:

i. Developing title and Statement of the problem in a general way: The title should be carefully worded. It should indicate the core of the study, reflect the real intention of the researcher, and show on what the focus is.

E.g.” Financing small-scale industries by commercial banks”. Here the focus is on commercial banks and not on small scale industries.

The title may also indicate the geographical area of the study. Moreover, the topic chosen should neither be too vague nor too narrow in scope. It should not also be too narrow in which it becomes insignificant and trivial.

A problem should be stated in a broad general way, keeping in view either some practical concern or some scientific or intellectual interest. To this effect, the researcher must immerse in the subject matter concerning which he wishes to pose a problem. In case of social research, it is advisable to do some field observation for preliminary survey or what is often called pilot survey. Then the researcher has to state the problem either him self or through the guidance of an expert or guide. Often, the guide puts forth the problem in general terms, and it is the duty of the researcher to narrow it down and phrase the problem in operational terms. The problem can be stated in accordance with the directive

of the organizational authority. The ambiguities in the general statement of the problem can be resolved by cool thinking and rethinking over the problem. At the same time, the feasibility of a particular solution has to be considered and the same should be kept in view while stating the problem.

ii. Building a conceptual model: On the basis of our theoretical knowledge of the phenomenon under study, the nature of the phenomenon, its properties /elements and the inter-relationships should be identified and structured in to a framework. This conceptual model gives an exact idea of the research problem and shows its various properties and variables to be studied. It is the basis for formulation of the objectives of the study and the hypothesis to be tested. To workout a conceptual model, a critical study of available literature on the subject-matter has to be made. Without adequate background knowledge, a researcher can not grasp and comprehend the nature of the research problem. The following example will illustrate the process of developing a conceptual model.

Example: Conceptual model for a Study of profitability of Hawas Bank

Profit is the difference between income and expenditure. Hawas Bank's main source of income is interest earned on loans and advances. It also derives some non-interest income in the form of commission, service charges, etc. Its major expenditure is interest paid on deposits; it incurs manpower and other expenses for providing services to the customers.

The difference between interest earned and interest paid may be called "spread", and the difference between non-interest expenses and non-interest income is called "Burden". The difference between the Spread and Burden determines the net profit (or loss). By identifying the factors which influence Spread and Burden an approach for profit management can be developed.

iii. Defining the objective of the study: The objectives state which elements in the conceptual model are to be investigated, but it is the conceptual model that defines, describes, and states the assumptions underlying these elements. Objectives may be specified in the form of either statements or questions.

iv. Setting Investigative Questions: Once the objectives of the research/ or the general research questions have been defined, the formulation moves to the next level, i.e.

investigative questions. These sub-questions are set up for each of the major research objectives/ questions. These specific questions guide the development of concepts, operational definitions, and measurement devices.

E.g. Research objective: “To identify the major factors affecting the bank’s profitability”

Investigative questions:

- a. What are the factors affecting interest earned on advances?
- b. What are the factors affecting interest paid on deposits and borrowings?
- c. What are the factors affecting manpower expenses? and so on.

v. Formulation of Hypotheses: These are tentative propositions relating to investigative questions. For detail discussion, *see* section 2.4 below.

vi. Operational definition of concepts: The next step in the formulation process is to define operationally the concepts involved in the title, objectives, investigative questions and hypotheses. For more explanation on the same issue, *see* section 2.5 below.

vii. Delimiting the Scope of the Study: Since a complete study of any phenomenon is non-manageable, a research should take a close view of the scope and dimensions of the selected problem and keep the demarcation of the study within manageable limits of time, finance, the ability of mind to grasp the implications, and other resources. The purpose of delimitation is to avoid confusion and superficiality and prompt clarification and successful completion within resource limits.

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Dear learner! What techniques are employed in defining research problem or formulation process? _____

2.4. Research Hypothesis

Introduction

The formulation of hypotheses or propositions as the possible answers to the research questions is an important step in the process of formulation of the research problem as explained in the previous section. Keen observation, creative thinking, hunch, wit, imagination, vision, insight and sound judgment are of great importance in setting up reasonable hypotheses. When the mind has before it a number of observed facts about

some phenomenon, there is a need to form some generalization relative to the phenomenon concerned.

Having introduced this much, let us now see other aspects of a research hypothesis as follows.

1.The Meaning of Hypothesis

A hypothesis is alternatively defined in the following manner. A hypothesis is a statement temporarily accepted as true in the light of what is, at the time, known about a phenomenon, and it is employed as a basis for action in the search of new truth.

A hypothesis is a tentative assumption drawn from knowledge and theory which is used as a guide in the investigation of other facts and theories that are yet unknown. It is a guess, supposition or tentative inference as to the existence of some fact, condition or relationship relative to some phenomenon which serves to explain such facts as already are known to exist in a given area of research and to guide the search for new truth.

A hypothesis is a tentative supposition or provisional guess which seeks to explain the situation under observation.

A hypothesis states what we are looking for. A hypothesis looks forward. It is a proposition, which can be put to a test to determine its validity. It may prove to be correct or incorrect.

A hypothesis is a tentative generalization the validity of which remains to be seen. In its most elementary stage, the hypothesis may be any hunch, guess, imaginative idea which becomes the basis for further investigation.

A hypothesis is an assumption or proposition whose tenability is to be tested on the basis of the compatibility of its implications with empirical evidence and with previous knowledge.

A hypothesis is, therefore, a shrewd and intelligent guess, a supposition, inference, hunch, provisional statement or tentative generalization as to the existence of some fact, condition or relationship relative to some phenomenon which serves to explain already known facts in a given area or research and to guide the research for new truth on the basis of empirical evidence. The hypothesis is put to test for its tenability and for determining its validity.

The testing of a hypothesis is the important characteristics of the scientific method. It is a prerequisite of any successful research, for it enables us to get rid of vague approaches and meaningless interpretations. It establishes the relationship of concept with theory, and specifies the test to be applied especially in the context of a meaningful value judgment. The hypothesis, therefore, plays a very pivotal role in the scientific research method.

The formulation of hypothesis, thus, is very crucial and the success or the failure of a research study depends upon how best it has been formulated by the researcher. We may conclude by saying that it is hard to conceive modern science in all its rigorous and disciplined fertility without the guiding power of hypothesis.

2. Hypothesis: Necessary or not?

Is the formulation of useful hypotheses always necessary and possible? Despite its importance, can hypotheses be set up in all cases? In a mere fact-finding investigation, no problems may be raised and the need for formulating hypotheses may not arise. Similarly, it may not be possible to set up a hypothesis in exploratory studies because the very purpose of this study may be to formulate a hypothesis. However, in all analytical and experimental studies, hypothesis should be set up in order to give a proper direction to them.

3. Sources of Hypothesis

i. Theory: - This is one of the main sources of hypotheses. It gives direction to research by stating what is known. Logical deduction from theory leads to new hypotheses. For example, profit /wealth maximization is considered as the goal of private enterprises. You

can derive various hypotheses from this assumption. “The rate of return on capital employed is an index of business success”; “The higher the earning per share, more favorable is the financial leverage”; “The optimum capital structure is that combination of debt and equity which leads to the maximum value of the firm”.

ii.**Observation:** From the observation of price behavior in a market, for example, the relationship between the price and demand for an article is hypothesized.

iii.**Analogies:** As Julian Huxley points out, casual observation in nature or in the framework of another science may be a fertile source of hypotheses.

iv.**Intuition and personal experiences:** personal life and experiences of persons determine their perception and conception which direct the persons to certain hypotheses. The story of Newton and the falling apple, the flash of wisdom to Buddha under a Banyan tree illustrate this individual accidental process.

v.**Findings of studies:** this may be made in order to replicate and test the findings of other studies.

vi.**State of knowledge:** where formal theories exist, hypotheses can be deduced. If the hypotheses are rejected, theories would be modified. Where formal theories are scarce, hypotheses are generated from conceptual frameworks. In either case, the hypotheses are related to the conceptual- theoretical level.

vii.**Culture:** For example, in India in socio-economic and leadership studies, hypotheses based on caste and economic status are common, because Indian society is caste-ridden, hierarchical and segmental, and the Indian Economic system is riddled with inequalities and privileges.

viii. Continuity of research in a field it self.

4. Rules of Constructing Hypothesis

According to [Smith](#), the following rules apply:

- i. Search for variable measurements with the most quantitative characteristics available. Hage gives the following techniques of searching for and creating variables out of non-variable concepts. First, the researcher can search for implied dimensions under non-variable concepts. Campbell provides a good example for this method in studying the non-variable concept “social group”. He identifies four underlying dimensions: degrees of proximity, similarity, perceived common fate, and perceived spatial pattern. Second, one can create new variables by comparing conceptual synonyms or analogies. For example, concepts like “Participation in decision making”, “organization control”, “power” and “influence” refer to the degree of organizational centralization. Third, one can search the literature for rarely occurring associations between phenomena. Fourth, one can generate new variables through ordering many concepts from more to less abstract.
- ii. Make the variable scale properties explicit by stating all of the variable’s mutually exclusive and totally inclusive categories by degree. For example, a variable like “Income” may be categorized (per month) as (i) up to Birr 500, (ii) Birr 501-1000, (iii) Birr 1001-2000, (iv) Birr 2001 and above.
- iii. Describe the means used to sort observation in to your variable categories in sufficient detail so that your methods may be evaluated and replicated by others. “Personality disintegration” is a good example of poorly operationalized variable. It is an unreliable measure and can not be replicated.
- iv. Always consider alternative operations which might be more appropriate for a given variable.
- v. Analyze variables through their relations.
- vi. Link two or more formal propositions through a share independent or dependent variable where possible.

For example, from the following concrete observations-

- I. Married persons are less likely than unmarried persons to commit suicide.

II. Married persons with children are less likely than married persons without children to commit suicide- the following abstract formal hypotheses may be inductively produced:

- a. Suicide rates vary directly with the degree of individualism
- b. Suicide rates vary indirectly with the degree of group cohesion

5. Types of Hypothesis

Hypotheses are classified in to several ways. Based on their function, hypotheses are two types: a) Descriptive hypothesis, and b) Relational hypothesis. They can also be classified in to: c) Working hypothesis, d) Null hypothesis, and e) Statistical hypothesis. Thirdly, they can be categorized in to three based on the level of abstraction: f) Common-sense hypothesis, g) complex hypothesis and h) analytical hypothesis.

a. Descriptive hypothesis: These are propositions that *describe the characteristics* (such as size, form or distribution) of a variable. The variable can be an object, person, organization, situation or event. Some examples are given below.

“The rate of unemployment among diploma graduates is higher than that of degree graduates.”

“Public enterprises are more amenable for centralized planning.”

“The Educational system is not oriented to human resource needs of a country.”

b. Relational hypothesis: Such hypotheses describe the *relationship between two* variables. The relationship can be positive or negative or, in short, causal relationship.

Examples: “Families with higher incomes spend more for recreation.”

“Participative management promotes motivation among the executives.”

“The lower the rate of job turnover in a work group, the higher the worker Productivity.”

”Upper-class people have fewer children than the lower-class people.”

“The productivity of labor decreases as the working duration increases.”

c. Causal hypothesis: These hypotheses state that the existence or the change in one variable causes or leads to an effect on another variable. The first variable is called an *independent* variable while the second is a *dependent* variable. In this respect, the direction of the relationship flow should be noted, i.e., which is cause and which is effect.

d. Working hypotheses: These are the hypotheses formed while planning the study of a problem and they may not be specific. Such hypotheses are subject to modification as the investigation proceeds.

e. Null hypotheses: These are the hypothetical statements denying what are explicitly indicated in working hypotheses. They state that no difference exists between the parameter and the statistic being compared to it. For example, even though there is a relationship between a family's income and expenditure on recreation, a null hypothesis may state: "There is no relationship between families' income level and expenditure on recreation." Null hypotheses are formulated for testing statistical significance since this form is a *convenient approach* to statistical analysis. As the test would nullify the null hypotheses, they are so called.

There is some justification for using null hypotheses. They conform to the qualities of *detachment* and *objectivity* to be possessed by a researcher. If he attempts to test hypothesis which he assumes to be true, it would appear as if he is not behaving objectively. This problem does not arise when he uses null hypotheses.

Moreover, null hypotheses are more exact. It is easier to reject the contrary of a hypothesis than to confirm it with complete certainty. Hence, the null hypothesis is found to be useful.

f. Statistical hypotheses: Such statements are about a *statistical population*. They are derived from sample. They are quantitative in nature in that they are numerically measurable. E.g. "Group A is older than Group B."

Statistical hypothesis may be hypothesis of difference or hypothesis of association. The latter specify the relations between variables. This relation is measured by the co-efficient of correlation, e.g. if the co-efficient of correlation between bonus and productivity is +1.0, then there is a perfect positive correlation between the two variables.

g. Common sense hypotheses: These represent the common sense ideas. They state the existence of empirical uniformities perceived through day to day observations. These observation can be in business establishments, the social background of workers, and the behavior patterns of specific groups like students, e.g. "Shop-assistants in small shops lack motivation."

“Soldiers from upper-class are less adjusted in the army than lower-class men.”

“Fresh students conform to the conventions set up by seniors.”

h. Complex hypothesis: These statements aim at testing the existence of *logically derived relationships* between empirical uniformities. For example, in the early stage of Human ecology described empirical uniformities in the distribution of land values, industrial concentrations, types of businesses, and other phenomena. Further study and logical analysis of these and other related findings led to the formulation of complex hypotheses such as “The concentric growth circles characterize a city”, “Members of minority groups suffer from oppression psychosis”, etc. Such hypotheses are purposeful distortions of empirical exactness. Because of their removal from empirical reality, these constructs are termed “*ideal types*”. The function of such hypotheses is to correct tools and problems for further research in other wise very complex areas of investigation.

i. Analytical hypotheses: These are concerned with the relationship of *analytic variables*. The hypotheses occur at the higher level of abstraction. These specify relationship between changes in one property and changes in another. For instance, the study of human fertility might show empirical regularities by wealth, education, region, and religion.

E.g. “There are two high-fertility population segments in India, viz., low-income urban Muslims and low-income rural caste Hindus.” In this case, taking the effects of the region, education and religion on fertility constant, now it is possible to measure the relationship between the variables of wealth and fertility.

These kinds of hypotheses in general are the most sophisticated type for formulation and development of “brilliant” abstract theories.

6. Forms of stating hypothesis

The statement of a research hypothesis can take declarative (positive) form, negative form, the null form or the question form.

Examples on

➤ Positive form

H₁: The academic achievements of extroverts is significantly higher than that of introverts

H₂: Students who learn in small class size will perform significantly better in mathematics test than who learn in large class size

H₃: Teaching children through programmed instructional material will decrease their test anxiety.

➤ Negative form

H₁: The academic achievements of extroverts is not significantly higher than that of introverts

H₂: Students who learn in small class size will not perform significantly better in mathematics test than who learn in large class size

H₃: Teaching children through programmed instructional material will not decrease their test anxiety

➤ Null form

H₁: There is no significant difference between the academic achievements of extroverts and introverts

H₂: There is no significant difference between students who learn in small class size and those who learn in large class size in performing a mathematics test.

H₃: Teaching children through programmed instructional material will have no effect on their test anxiety

➤ Question form

H₁: Will the academic achievements of extroverts be higher than that of introverts?

H₂: Will students who learn in small class size perform better in mathematics test than who learn in large class size?

H₃: Will teaching children through programmed instructional material decrease their test anxiety?

7. Importance of Hypothesis

Hypothesis has very important place in research although it occupies a very small place in the body of a thesis. It is almost impossible for a research worker not to have one or more hypothesis before proceeding with his work. If he is not capable of formulating a hypothesis about his problem, he may not be ready to undertake the investigation. The aimless collection of data is not likely to lead him anywhere. The importance of hypothesis can be more specifically stated as under:-

- It provides direction to research. It defines what is relevant and what is irrelevant. Thus it prevents the review of irrelevant literature and the collection of useless or excess data. It not only prevents wastage in the collection of data, but also ensures the collection of the data necessary to answer the question posed in the statement of the problem.
- It sensitizes the investigator to certain aspects of situations, which are relevant from the standpoint of the problem in hand. It spells the difference between precision and haphazardness, between fruitful and fruitless research
- It is guide to the thinking process and the process of discovery. It is the investigator's eye-a short of guiding light in the world of darkness
- It focuses research. Without it, research would be like a random and aimless wandering
- It prevents blind research. It also prevents indiscriminate gathering of data which may later turn out to be irrelevant
- It sensitizes the individual to facts and conditions that might otherwise be overlooked
- It places clear and specific goals before us. The clear and specific goals provide the investigator with a basis for selecting samples and research procedures to meet them
- It serves the function of linking together related facts and information and organizing them in to one comprehensible whole
- It enables the investigator to understand, with greater clarity, his/her problem and its ramification. It farther enables a researcher to clarify the procedures and methods to be used in solving his problem and to rule out methods which are incapable of providing the necessary data
- It serves as a framework for drawing conclusions. It makes possible the investigation of data in the light of tentative proposition or provisional guess. It provides the outline for setting conclusions in a meaningful way.

8. Characteristics of a good Hypothesis

The following criteria are suggested to frame a good hypothesis:

- i. The hypothesis must be an adequate answer to the specific problem that demanded an answer
- ii. The hypothesis should have logical simplicity

- iii. The hypothesis should be expressed in a quantified form or be susceptible to convenient quantification
- iv. The hypothesis must be verifiable
- v. The hypothesis must be stated in such a way that allows it to be refuted

Moreover, a “good” hypothesis is the one which satisfies any of the following criteria.

i. Conceptual clarity: A hypothesis should be conceptually clear. It should consist of clearly defined and understandable concepts. This means that the concepts found in the hypothesis should be formally and operationally defined. *Formal definition or explication* of the concepts will clarify what a particular concept stands for, while the *operational definition* will leave no ambiguity about what would constitute the empirical evidence or indicator of the concept in the field. An ambiguous hypothesis characterized by undefined or ill-defined concepts cannot be tested since, understandably, there is no standards basis for knowing what observable facts would contribute to its test. It is advisable that the concepts embodied in the hypothesis be defined in a manner commonly accepted and combinable. This would ensure continuity in researches and go a long way in bringing about a cumulative growth of scientific knowledge.

ii. Specificity: A hypothesis should be specific and explain the relations between variables and the conditions under which these relations will hold. For example, “when there is dissatisfaction and no care is taken, deprivation will engender violence.”

Often the researchers are tempted to express their hypothesis in terms so general and with so grandiose a scope that they are simply not amenable to test. The temptation is *suicidal*. The researchers, therefore, would do all to avoid concepts in their hypothesis for which suitable tangible indexes have not been developed. A hypothesis would include a clear statement of indexes, which are to be used. For example, the concept of social class needs to be explicated in terms of indexes such as income, occupation, education, etc. Such specific formulations have the obvious advantage of assuring that research is practicable and significant. It also helps increase the validity of the results because the more specific the statement or predication, the smaller the probability that will actually be borne out as a result of mere accident or chance.

iii. Testability: A hypothesis should be testable and should not be a *moral judgment*.

It should be possible to collect *empirical evidences* to test the hypothesis. Statements like “Capitalists exploit their workers”, “Bad parents produce bad children” are common place generalizations and can not be tested, as they merely express sentiments and their concepts are vague.

iv. Availability of techniques: Hypothesis should be related to available techniques. Otherwise the research problems will not be researchable. The researcher who does not know that techniques are available to test his hypothesis is in a poor way to formulate usable questions. In other words, the hypothesis should be formulated only after due thought has been given to the methods and techniques that can be used to measure the concepts and variables incorporated in the hypothesis.

v. Theoretical relevance: A hypothesis should be related to a body of theory or some theoretical orientation. This requirement concerns the theoretical rationale of a hypothesis, i.e., what will be the theoretical gains of testing the hypothesis? If the hypothesis is related to some theory, research will help to qualify, support, correct or refuse the theory. A science can be cumulative only by building on an existing body of facts and theory. It can not develop if each study is an isolated investigation. When a research is systematically based on a body of existing theory, a genuine contribution to knowledge is more likely to result. Therefore, a hypothesis should possess theoretical relevance.

vi. Consistency: A hypothesis should be logically consistent. Two or more propositions logically derived from the same theory must not be mutually contradictory.

vii. Objectivity: Scientific hypothesis should be free from value-judgment. In principle, the researcher’s system of values has no place in scientific method. However, as social phenomenon are affected by the milieu in which they take place, the researcher must be aware of his values and state them explicitly.

viii. Simplicity: A hypothesis should be a simple one requiring fewer conditions or assumptions. But “Simple” does not mean obvious. Simplicity demands insight. The more insight the researcher, has in to the problem, the simpler will be for his hypothesis about it.

ix. A good hypothesis does not conflict with any law of nature, which is known to be true. We know that there are a number of conflicting theories of learning and teaching.

These conflicts arise partly out of the failure of research workers to produce hypothesis that are in agreement with fact, and are not in conflict with any law of nature. When we say that there can be memory tablet for strengthening memory, we are trying to disprove well-established laws of nature.

9. Difficulties in the formulation of Hypothesis

There are a number of difficulties from which a beginner may suffer at the stage of formulation of a good hypothesis:-

- Lack of knowledge and clarity of the theoretical framework of the area in which the investigator chooses to work
- Lack of ability to make use of the theoretical framework logically
- Lack of familiarity (acquaintance) with available research technique resulting in failure to be able to phrase the hypothesis properly
- Vagueness of the statement: For example, “A course in ethics will make a student a more ethical adult”

?

Dear learner! Based on your previous lesson, answer the following questions.

1. What did you understand by research hypothesis? _____
2. Mention the different types of Hypotheses _____
3. Identify the various forms of stating hypothesis _____

2.5. Operational/ working definition of concepts, and identifying variables

The operational definitions specify the measurement parameters of variables. Scales/indices are constructed for measuring abstract concepts. The operational definitions of concepts and measurement devices enable us to decide exactly the data needs of the research. Suppose the following objectives are stated for a certain study. The main objectives are:

1. To measure the *effectiveness* of a retraining program designed to help *young people*.
2. To find out the number of *children* living below the *poverty line* in Ethiopia.
3. To ascertain the impact of immigration on *family roles* among *immigrants*.

These objectives are not specific in terms of the main concepts to be studied and study populations. You can not count the number of children living below the poverty line until

you decide what constitutes the poverty line and how to determine it; you can not find out the impact of immigration on family roles unless you identify which roles constitute family roles; you can not measure effectiveness until you define what effectiveness is. On the other hand, it is equally important to decide exactly what you mean by “children”, “immigrants” or “young”. Up to what age will you consider a person to be child, i.e., 5, 10, or 15? Up to what age would you consider a person to be young, i. e, 15, 20, or 25? Who would you consider to be an immigrant, i.e., those who immigrated 40, 20, or 5 years ago? Are you considering immigrants from every country or only a few?

In a research study, you need to define those concepts and study population to avoid ambiguity and confusion. Operational definitions may differ from dictionary definitions and from day to day meanings.

?

Dear student! What is meant by operational definition of concepts? _____

Let us have a further discussion on *concepts and variables* as follows.

The differences between concepts, indicators and variables

Concepts are mental images or perceptions and, therefore, their meanings vary markedly from individual to individual, (i.e. they are **subjective**). Therefore, you need to consider its operationalization, i.e., how it will be measured. To operationalize it, you first need to go through the process of identifying indicators. Indicators are set of criteria reflective of the concept which then can be converted in to variables. A variable refers to an event or process that can be changed or can change on its own accord. A variable can have more than one value (e.g. sex, age, weight, income and religion, ESLCE scores). A variable should have operational definitions clearly stated. Variables are subjected to measurability, of course, with varying degrees of accuracy. Hence, measurability is the main difference of a concept and a variable. For more understanding, look to the following examples.

Table 2. 1. Converting concepts in to variables

Concepts $\rightleftarrows$ Indicators $\rightleftarrows$ Variables

Concepts $\Rightarrow$	Indicators $\Rightarrow$	Variables	Decision level (operation definition)
Rich	a) Income b) Assets	a) Income per year b) Total value of Home (s), Boat (s), car (s), investment(s)	a) If > Br 100,000 b) If > Br 250,000
Effectiveness (of a health program)	Number of patients Changes in morbidity I. Changes in the extent of morbidity II. Changes in the pattern of morbidity Changes in the mortality I. Changes in the Crude Death Rate(CDR)	a) Number of patients served in a month / year b) I. Changes in morbidity rate (Number of illness episodes per 1000 population) II. Changes in the morbidity typology c) I. Changes in CDR	Whether the difference in before and after level is statistically significant or point-prevalence increase or decrease in each variable as decided by the researcher or other experts

	II. Changes in the Age Specific Death Rate(ASDR) d. Changes in nutritional status 1. Changes in weight 2. Changes in illness episodes 3. Changes in morbidity	II. Changes in ASDR d) 1) Changes in weight 2) Illness episodes in a year 3) Changes in the morbidity type	
High academic achievement	a) Average marks obtained in examination b) Average marks obtained in practical work c) Aggregate marks	a) Percentages of marks b) Percentages of marks c) Percentages of marks	If >75% If >80% If > 78%

?

Dear student! Show the differences among concepts, indicators and variables_____

Classifications of variables

The variables are classified in several ways. A few of them are mentioned below.

Classification I

The variables can be categorized into qualitative and quantitative variables.

- *Qualitative variables* are usually discrete in their nature. For example, sex, methods of teaching, etc.
- *Quantitative variables* are usually continuous in their nature and are broadly categorized into four types: The nominal, ordinal, interval and ratio variables.

i) Nominal Variables (i.e., variables measured by nominal scales): no order, distance or origin. Nominal variables make it possible to establish relations to equality or diversity for any pair of objects but they do not enable us to ascertain that one object belonging to a particular group has a given feature in a higher degree than another object belonging to the same group. The numerals are arbitrarily given for identification of mutually exclusive categories. E.g. Male and female variables.

ii) Ordinal or Ranking Variables: Order, but no distance or origin. These variables enable us to state whether two objects, compared from a given point of view, possess a given property to the same degree or whether one has it to a higher degree. The numerical value of the scale reflects differing amounts of the characteristics being measured.

E.g. The ranks given to women athletes as 1st, 2nd, 3rd ...

iii) Interval Variables: Both order and distance, but no origin. These variables make it possible to establish *by how much* one of the two objects possesses a given feature in greater measure than the other. However, interval variables does not permit to determine how many times a given object has a given feature to a more intensive degree, e.g. *Calendar time, temperature scale, grade points*, where an *arbitrary zero* has been set by convention, i.e., it has no true zero value.

iv) Ratio variables: Order, distance and origin. Those attributing a certain absolute value of intensity to a certain variable, thus permitting comparison not only of the distances between different values but also the ratio between them, e.g. speed of a vehicle, when expressed in terms of KMPH, the speed has an *absolute zero* value if the vehicle is immobile. There are true zero value, equal units, and equality of ratios for the ratio variables. All mathematical operations can meaningfully be performed on ratio variables because there are equal intervals between the numbers on the scale as well as true zero point.

Classification II

In experimental research, the following terminologies are commonly used: - Independent variables, dependent variables and interfering variables.

i. Independent variables are those variables, which are manipulated by the experimenter in his attempt to ascertain their relationships to an observed phenomenon. Independent variables are also known as stimulus variables and antecedent variables. They are also sometimes referred to as the causal or predictor variables. Or they are the causes supposed to be responsible for bringing about changes in a phenomenon or situation.

ii. Dependent Variables or response variables or consequent variables are those events, which are hypothesized as dependent on the changes in the independent variables. The variables are some times referred to as the resultant or criteria variables.

For instance, a researcher could examine the impact of heavy drinking alcohol on liver. Here, **heavy drinking alcohol** would be an independent variable and the effect like **liver disease** could be considered as dependent variable. Similarly, **ready-made films and lectures** are considered as independent variables, whereas the behavioral changes occurring as a result of the environmental manipulations are examples of dependent variables.

In other instances, some variables can be used as either independent or dependent in any given analysis. For example, one researcher could examine the influence of gender on higher education entrance examination scores. Here, *gender* is the independent variable

while *higher education entrance examination score/performance* will be a dependent variable. However, in other analysis, the influence of higher education entrance examination scores and gender on University choice would be explored. In this case, both *gender and entrance examination scores* are the independent variables while the *University choice* will be the dependent variable.

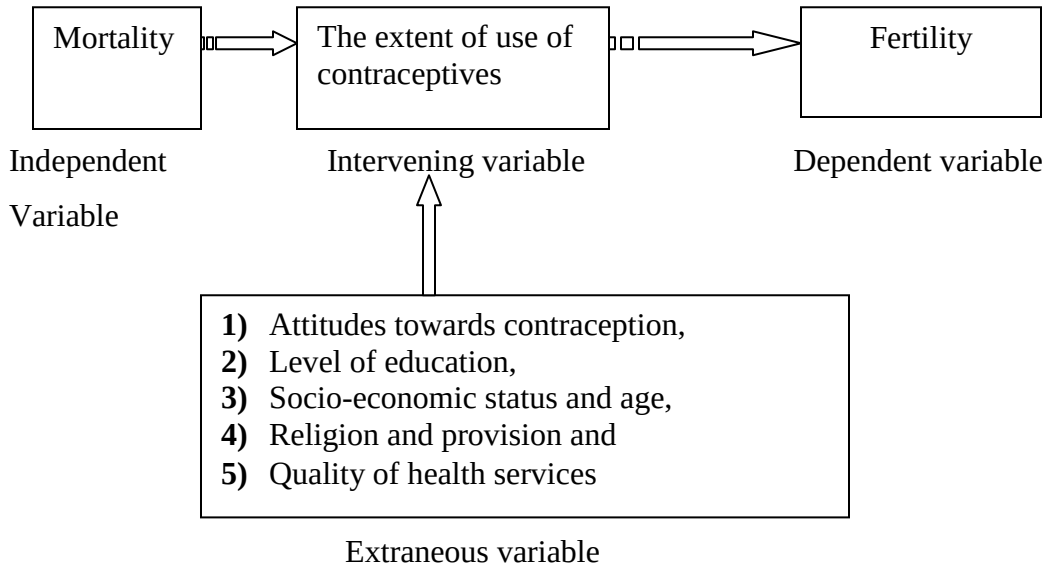
iii. Connecting or mediating variables are those variables which distort the relationship between independent and the dependent variables. They affect the changes attributed to the independent variable. The connecting variables can be extraneous or intervening/confounding variables. Extraneous variables may increase or decrease the magnitude of the relationship between the independent and dependent variables. Intervening/confounding variables link both variables in which, in certain situations, the cause variable will have the assumed effect only in the presence of an intervening variable.

Suppose you want to study the relationship between smoking and cancer. You assume that smoking is the cause of cancer. Studies have shown that there are many factors affecting these relationships: number of cigars or the amount of tobacco smoked every day, the duration of smoking, the age of the smoker, dietary habits, and the amount of exercise undertaken by the individual. While these variables are extraneous variables, smoking and cancer are the independent and dependent variables, respectively. They may increase or decrease the magnitude of the relationships but you are not interested in measuring their impact.

Let us take another example. Suppose you want to study the relationship between fertility and mortality. Your aim is to explore what happens to fertility when mortality declines. The history of demographic transition has shown that a reduction in the fertility level follows a decline in the mortality level, though the time taken to attain the same level of reduction in fertility varies markedly from country to country. As such, there is no direct relationship between fertility and mortality. With the reduction in mortality, fertility will only decline if people attempt to limit their family size. People use one method or another to control their fertility, resulting in lower fertility levels. It is thus the intervention of

contraceptive methods that complete the relationship: the greater the use of contraceptives, the greater the decline in the fertility level. The extent of the use of contraceptives is also affected by a number of other factors such as attitudes towards contraception, level of education, socio-economic status and age, religion and provision and quality of health services. These are classified as extraneous variables.

Fig. 2.1. Example on Independent, dependent, intervening, and extraneous variables



From the above figure, it is understood that with out the intervening variable, the relationship between the independent and dependent variables will not be complete.

? **Dear student!** Mention the various kinds of Variables _____

Chapter Summary

The formulation of a research problem is the most important step in the research process. Any defect in it will adversely affect the validity and reliability of a study. Even though there are no specifically set guidelines, in this chapter, you learned a seven-step model in the formulation process of a research problem. These steps follow a logical sequence that takes a research beginner through the formulation process of a research problem.

In the formulation process, it is important to articulate the objectives of your study clearly. Objectives should be specific, free from ambiguity and each one should relate to only one aspect of the study.

Statement of hypothesis can also be one important element (though not essential) in the formulation of a research problem. A perfectly valid study can be conducted without constructing a single hypothesis. Hypotheses are important for bringing clarity, specificity and focus to a research study. A hypothesis is a speculative statement that is subjected to verification through a research study. In formulating a hypothesis, it is important to ensure that it is simple, specific, conceptually clear, able to be verified, rooted in a body of knowledge and able to be operationalized. A hypothesis can be categorized as descriptive hypothesis, relational hypothesis, working hypothesis, null hypothesis, statistical hypothesis, common-sense hypothesis, complex hypothesis, analytical hypothesis, research hypothesis and alternative hypothesis.

Operationalization of concepts and identification of variables are also key factors in the formulation process of a research problem. Concepts are mental images whose meanings vary from person to person. This necessitates their operationalization. To do this, you first need to identify indicators. Indicators are set of criteria reflective of the concept which then can be converted into variables. A variable is an event or process that can be changed or subject to change. Variables can be classified as qualitative and quantitative. While *qualitative variables* are discrete, quantitative variables are usually continuous. Quantitative variables are sub-classified as nominal, ordinal, interval and ratio variables. In experimental research, again, variables are classified as independent variables, dependent variables, extraneous and intervening variables.

Exercises**Part I. Multiple choice questions** (The answers are provided at the end of the module.)

1. A difficulty or question which a researcher experiences in a theoretical or practical situation and wants to obtain a solution for it is termed as _____.
 - a. Research problem
 - b. Hypothesis
 - c. Variable
 - d. Concept
2. Sources of researchable problems does include:
 - a. Researchers' own experiences
 - b. Practical issues that require solutions
 - c. Theory and past research
 - d. All of the above
3. Formulation of a research problem doe not involve which one of the following?
 - a. The separation of a research problem from like question found in related situations of need
 - b. Specifying what the research problem is or why it is studied
 - c. Helping to establish the frame of reference with which the researcher approaches the problem.
 - d. None of the above
4. One of the following conditions may not be important in the formulation of a research problem
 - a. Systematic immersion in the subject matter
 - b. Study of relevant literature on the subject
 - c. Discussions with persons having practical experience in the field of study
 - d. None of the above
5. Research hypotheses are _____.
 - a. Formulated prior to a review of the literature
 - b. Statements of predicted relationships between variables
 - c. Stated such that they can be confirmed or refuted
 - d. b and c
6. One of the following alternatives can not be the source of Hypotheses
 - a. Findings of research documents
 - b. personal life and experiences
 - c. Personal observation
 - d. conceptual frameworks
 - e. None

7. The kind of hypothesis which states that no difference exists between the parameter and the statistic being compared to it is termed as_____.
- a. Relational hypothesis b. Descriptive hypothesis c. Working hypotheses
d. Null hypotheses
8. If an alternative/positive hypothesis is stated as “A firm with higher fringe benefits retains its qualified employees better than firms with lesser fringe benefits.”, which one of the following is a null hypothesis to this?
- a. There is no relation between fringe benefits and employee retention
b. There is a relationship between fringe benefits and employee retention
c. A firm with higher fringe benefits retains its qualified employees lesser than firms with lesser fringe benefits
d. A firm with higher fringe benefits retains its qualified employees better than firms with lesser fringe benefits
9. Which of the following statements does not show the importance of hypothesis?
- a. It spells the difference between fruitful and fruitless research
b. It is the investigator’s eye-a short of guiding light in the world of darkness
c. It makes possible the investigation of data in the light of tentative proposition or provisional guess
d. It prevents haphazard gathering of data
e. All of the above statements show the importance of hypothesis
10. One of the following criteria does meet the requirement of a hypothesis
- a. A hypothesis should go in contrary with any law of nature
b. A hypothesis should be simple
c. A hypothesis is not related to a body of theory
d. A hypothesis must be stated in such a way that does not allow it to be refuted
11. One of the following statements does define a variable
- a. Variables are subjected to measurability
b. Variables are mental images or perceptions and therefore their meanings vary markedly from individual to individual
c. Variables are set of criteria reflective of the concept which then can be converted in to indicators

d. All of the above

12. An ordinal scale is used to rank order people, objects, or characteristics.

a. True

b. False

13. Which scale is the simplest form of measurement?

a. Nominal

b. Ordinal

c. Interval

d. Ratio

14. If a researcher calculates per-day temperature averages, what scale would be used?

a. Interval scale

c. Nominal scale

b. Ratio scale

d. Ordinal scale

Chapter Three

Review of Literature

3.1. Introduction

Knowledge is growing rapidly. Scholars, researchers and writers go on adding knowledge through their studies and writings. Publications have tremendously increased throughout countries. To make a worthwhile contribution, a researcher has to survey the available literature relating to his/her field of study. He/she must keep him/herself update in his/her field and related areas.

Hello student! From the previous chapter, you learned about the formulation of a research problem. Now you are to deal with the review of related literature. It is one input in the formulation process of a research problem. Of course, related literature is reviewed before, during, and after the problem is defined and goes on until the completion of a study. In this chapter, you will learn about what the literature and literature review are, components of literature, the purposes of review, Literature Search Procedure, Sources of Literature, how to record what is read and how to use index cards and how to present bibliography, and lastly how to take note and related issues are discussed.

3.2. Objective of the Lesson

Hello learner! After completing this chapter, you will be able to

- Define what literature and literature review are
- List the components of literature
- Appreciate the purposes of literature review
- Understand the procedures followed for Literature Search,
- Know how to record what is read
- Understand how to present bibliography
- Develop understanding on how to paraphrase and/ or incorporate direct Quotes

3.3. Literature, when defined

Literature in this context consists of

- a. Books:-
 - 1) Encyclopedias
 - 2) Year Books e.g. published as supplements to Encyclopedias.
 - 3) Textbooks.
 - 4) Reference Books.
- a. Journals: published monthly, quarterly, or Half-yearly or Annually.
- b. Reports:- Reports of Committees/Commissions appointed by Governments and public institutions.
 - Seminar Reports and Conference proceedings.
- c. Research Dissertations and thesis.
- d. Newspapers.
- e. Micro Forms: Audio and Video tapes; Micro Card; Micro Film.

3.4. What to Review and for What Purpose

The review of literature is not mere reading for reading sake; it is also not a causal reading like reading of a story or novel. It is focused and directed towards specific purposes. It is also selective. A researcher has to select the kinds of literature to be reviewed and determine the purposes for which he/she has to study them. The literature review starts with the selection of a problem for research, continues through the various stages of the research process and ends with report-writing.

a. The purposes of Review

The reasons for review of related literature are:

- 5) To gain a background knowledge of the research topic,
- 6) To identify the concepts relating to it, potential relationships between them and to formulate researchable hypotheses,
- 7) To identify appropriate methodology, research design, methods of measuring concepts and techniques of analysis,
- 4) To identify data sources used by other researchers, and
- 5) To learn how others structured the reports.

For further illustration, the kinds of literature to be reviewed at the various stages of the research processes and the specific purposes of the review are indicated below:

Table 3.1. Purposes of literature reviewed at different stages of research process

Steps in the research Process	Kinds of literatures to be reviewed	Purposes of review
1. Identification and selection of Problem	- Source materials, encyclopedia, Text books, Reference Books - Latest publications on the discipline - Survey of Research in the related field -Bibliography of Doctoral Dissertations - Theses in the selected discipline	- To gain preliminary orientation and background knowledge - To gain up-to-date knowledge in the field - To know the work already done on the subject - to identify research gaps in the field -To avoid duplication
2. Formulation of the Selected problem	-Previous studies in the field . Journals, published and unpublished theses	- To become familiar with with appropriate methodology and research techniques relevant to the study
3. Operationalization of concepts	- Same as in 2	-For clarifying concepts and knowing measurement techniques
4. Preparation of research Proposals	*-Illustrated Books on Methodology of Research	- To develop alternative designs

- * -Published and unpublished theses - For formatting hypothesis & deciding sample design, etc
- 5. Construction of Tools -Same as 4* - For gaining a thorough knowledge of the process of tools and measurement techniques
 - Copies of tools and Scales furnished in the Appendix of theses
- 6. Drafting the theoretical Chapters and theory Parts of other chapters -Journals, text books, reference books and reports on the subject matter -To take notes and compile bibliography

3.5. Literature Search Procedure

The literature search procedure involves a series of steps. But the exact sequence varies based upon the subject and the knowledge of the researcher. In our case, there are four steps involved in conducting a literature review.

1. Search for existing literature in your area of interest

In support for this, the following general approaches are also useful procedures.

- i. Request professors/teachers/instructors, librarians or others familiar with the field to suggest relevant references.
- ii. Find out whether any bibliography already prepared on the subject is available in a library. See the bibliographic index, if any, maintained in the library.
- iii. Consult bibliographies in the theses on the topic and related topics.
- iv. See the card catalogues of the library:
 - a. Author catalogue;
 - b) Subject catalogue cards to locate the titles and other details of the book on the subject.
- v. Examine periodicals, monographs, reports and conference proceedings and other materials including microfilms available in the library.

- vi. Consult references cited in the books and articles already located. Each book or article will be a means for locating additional references.
 - vii. Consult the abstract journal on the subject or 'Abstracts' section in the journals relating to the subject
 - viii. See the 'Book-Review' pages in the daily newspapers and in the journals.
2. Review the literature selected
 3. Develop a theoretical framework, and
 4. Develop a conceptual framework, and finally writing-up the reviewed literature

?

Dear learner! Write the steps required to conduct a literature review_____

Good! If you have answered this question, let us now have a detail discussion on some of the steps under the following headings.

1. Sources of Literature

As already mentioned, the sources of documentary information to aid the process of locating literatures relating to selected topic are:

- i. Subject Catalogues of Libraries: These catalogues furnish lists of books available on various subjects. Subject index in libraries also serve the same purpose.
- ii. Documentation Service: There are a number of documentation centers attached to national institutions.
- iii. Bibliographies of books and published papers
- iv. Lists of Books and Publishers' Bulletins: These can be brought out by leading publishers, news papers, book review column of journals, and world list of books.
- v. Journals
- vi. Government Reports
- vii. Research abstracts
- viii. Information on research already done

2. Review the literature selected

Now you have identified several books and articles as useful, the next step is to start reading them critically to pull together themes and issues that belong together. If you do

not have a theoretical framework in mind, to start with, use separate sheets of paper for each article or book. Once you develop a rough framework, fit the findings from the material so far reviewed into that framework, using a separate sheet of paper for each heading of that framework. As you read further, go on fitting the information where it logically belongs under the headings so far developed. You may need to add more headings as you go. In doing so, read critically with particular reference to the following aspects:

- Note whether the knowledge relevant to your theoretical framework has been confirmed beyond doubt;
- Note the theories put forward, the criticisms of these and their basis, the methodologies adopted (study design, sample size and its characteristics, measurement procedures, etc.) and the criticisms of them;
- Examine to what extent the findings can be generalized to other situations;
- Notice where there are significant differences of opinion among researchers and give your opinion about the validity of these differences;
- Ascertain the areas in which little or nothing is known-the gaps that exist in the body of knowledge.

After reading your related literatures, recording follows. See the following discussion!

a. Recording Your Reading

Finding information in the first place can be hard enough. Finding it again some time after wards can be even harder unless your recording and filing are systematic. Therefore, *you should resolve right from start to note down full details of every thing you read.*

These details should include:

- The author or authors;
- The title of the paper, report or book;
- The date of publication;
- If it is a book or report, the publisher and place of publication;
- If it is a chapter in an edited book, the title and editor of the book, and the page numbers of the chapter;
- If it is a paper in a journal, title of the journal, volume, issue number, and pages.

There are a number of ways in which you might collect and store this information. These include *notebooks, loose sheets of paper and note cards (index cards)*. Now let us explain about the card index.

The Card index

A card index is a collection of lined cards of various sizes (e.g. 5 "x3", 6"X4", or 8"X5", etc), which is meant to keep review notes and reference information. To make things convenient, choose one size and use it always. You are, however, strongly advised to avoid the habit of taking notes on a piece of paper of different sizes as such a habit makes your work inefficient and time consuming.

➤ **What to put on your index cards**

The cards should include all the information you are likely to need for inter-library loan applications and for drawing up the references or bibliography for your project report. They may include brief notes about the content of book or article and reminders that a certain chapter or page had useful information about some topic. They will be your stock-in-trade and you should start building up your stock as soon as you begin your studies.

More specifically, the following points should be noted:

Note: -The index cards can be used in two separate cards: *source cards and note cards*.

- The recoding of bibliographical information in the source cards should be made in Proper bibliographical format (this will be discussed in section 4.6 below).
- On the left-hand top corner of the source card, the letter 'B' or 'J' or 'R' (B=Book, J=Journal, R=Report) may be marked according to the reference.
- On the left-hand top corner of note cards, the source reference code may be marked, and on the right-hand top corner, the number of the researcher's proposed chapter to which the information is related may be entered.
- On the space between the top edge and the first horizontal line, the title of the broad theme (e.g. Supply analysis) may be written, and in the next space the title of the specific idea or fact (e.g. Definition of Supply analysis) may be written.
- In the body of the card, the name of the author and the title of the book /article may be noted. Then the fact extracted from the concerned reference is recorded

For illustration, look at the following example of index cards and what they contain.

Box 1: Sample Source card

Author (s) _____ No _____
 Title of the book/article _____
 Date of publication _____
 Place of publication _____
 Volume number _____
 Page number _____
 Publisher _____
 Edition number _____
 Issue number _____

Box 2: Example on how to use index card

B

Dr. O.R. Krishnaswami (1993), *Methodology of Research in Social Science*, Mumbai: Himalaya Publishing House.

It should be noted that it is best to make only one point on each card and to use only one side of the card. You will then have maximum flexibility in sorting out the cards at the writing stage.

?

Dear learner! As a researcher, about what elements do you record on your index card?

b. The Presentation of the Bibliography

Different publishers adopt different styles of citing information from the source (e.g. books, journals, reports, etc.). The citations are listed as either bibliography or references which are then arranged alphabetically by author's names and then, where necessary, by

date and letter. Single-author books and articles appear before co-authored books and articles by the same author with earliest dated first. Where there is more than one reference for an author in any one year, the order is alphabetical by title, and lower case letters added to denote more than one publication in that year.

The citations should follow a standard format. At this particular juncture, we shall limit ourselves to using the Harvard method, as it is believed to be highly practiced by many authors. It has also an advantage of excluding footnotes, which are awkward to deal with. Now, let you understand from the discussion here under.

1) Citation of books

The citation of books follows the use of either **underlining** or **italics**. The presentation should be: Author's family name, comma, initials of first name (s), year of publication in brackets, title of the book in upper and lower case underlined or in italics, comma, edition (if applicable, comma follows), place of publication, colon, publisher's name, full stop.

Books and articles often have more than one author. The names of all the authors should be given in the bibliography in the order presented by the authors themselves, linked by the word 'and' for the last named. But when the work has three or more authors, the abbreviation "*et al.*" meaning "and others", may be used after the name of the first author.

Ethiopian Authors should be listed in the alphabetical order of first name. For example, Amaya Laku should be listed under A and not under L. Write Ethiopian names in full in the bibliography as they are given in the publication you are citing. Avoid using honor titles such as Ato, Obbo, Dr., W/ro, Mr, etc. in citation or references.

Examples

- Atkinson, Atkinson RC and Hilgard Er. (1983) Introduction to psychology, 5th ed., New York: Harcourt Brace Jovanovich
- Blyth, WAL and Derricot, R (1977) The Social Significance of Middle Schools, London. Batsford.

- Hawi Lawe (1979) Business Development in Arba Minch town: Prospects & Challenges, Ann Arbor: Michigan University press.
- Koontz, Harold, *et al* (1998) Introduction to Management, 1st ed., New Delhi: Mc Graw-Hill International Book Company.
- Trudgill, P (1974a) The Social Differentiation of English in Norwich, Cambridge: Cambridge University press.
- Trudgill, P (1974b) Socio-linguistics: An Introduction, Harmondsworth: Penguin.
- Trudgill, P (1978) Socio-linguistics patterns in British English, London: Edwards Arnold.

2) Citation of articles or journals

A slightly different format is used when citing contributions to journals: author's family name, comma, initials of first name (s), comma, year of publication in brackets, title of contribution in upper and lower case and in quotation marks, comma, the journal's name, comma, volume number in bold (heavy) type, issue number in brackets, comma, the page numbers of the article prefaced by pp, full stop.

Examples

- Best, D (1985) "Primary and Secondary Qualities: Waiting for an educational Go dot", Oxford Reviews of Education, **1**(1), pp 73-84.
- Dachasa Hundara (1992) "Brain Drain Among academicians in Higher Education institutions in Ethiopia ", The Ethiopian Journal of education, **8**(2), pp 27-46.

3) Citation of Contributions in books

Where books are made of contributions from a variety of authors, the following order and format should be used. Cite author's surname, comma, initials of first name (s), comma, year of publication in brackets, then the title of the contribution in upper and lowercase and in quotation marks, comma, followed by "in", the editor's surname followed by comma, the initials then (ed), the title of the book in upper and lower case underlined or in italics, comma, publisher, comma, place of publication, comma, page numbers ,and finally full-stop.

Example

Labov, W. (1968) "The Reflection of Social Processes in Linguistic Structures", in Fishman, JA (ed) Readings in the Sociology of Language, Mouton, The Hague, pp 47-63.

4) Citation of Government Publications

This should be cited as follows: name of the responsible department, year of publications in brackets, title of the publication in upper and lower case underlined or in italics, comma, publisher, comma, place of publication. If there are official series codes, these should be cited after the title.

Example

Department of Education and science (1985) Education for All: Report of the Committee of Inquiry into the Education of Children from Ethnic minority groups, Cmnd 9453 HMSO, London.

5) Citation of Thesis

Occasionally, you may wish to quote from a thesis and should follow this format: author's surname, comma, initial (s), year of award in brackets, title in upper and lower case underlined or in italics, comma, degree awarded followed by thesis, comma and the name of the awarding institution.

Example

Misganu Getahun(2009), Assessment of Small and Micro Business Environment in Arba Minch town, MA Thesis, AAU.

6) Mimeographed materials

Mimeographs refer to duplicated materials, such as papers presented to conferences but not formally published. The format required to site such sources is: name of the author (s) and initials, the date, the title of the paper, comma, the circumstances under which the paper was presented, and the fact that it was mimeographed (duplicated). The title of the paper is in quotation marks and has initial capitals.

Example

Adams, RG (1968) “Aspects of documentation in the social sciences”, paper presented to the 1968 Annual conference of the British Education Management and Administration society, mimeo.

?

Dear learner! According to the Harvard style, show the correct format to cite books and journals_____

4. Developing a theoretical framework

Examining the literature can be a never-ending task but as you have limited time, it is important to set parameters by reviewing the literature in relation to some main themes pertinent to your research topic. As you start reading the literature, you will soon discover that the problem you wish to investigate has its roots in a number of theories that have been developed from different perspectives. The information obtained from different books and journals now needs to be sorted under the main themes and theories, highlighting agreements and disagreements among the authors and identifying the unanswered questions or gaps. You will also realize that the literature deals with a number of aspects that have a direct or indirect bearing on your research topic. Use these aspects as a basis for developing your theoretical framework. Your review of the literature should sort out the information within this framework. Unless you review the literature in relation to this framework, you will not be able to develop a focus in your literature search: i.e., your theoretical framework provides you with a guide as you read. This brings us to this paradox: until you go through the literature you cannot develop a theoretical framework and until you have developed a theoretical framework, you cannot effectively review the literature. The solution is to read some of the literature, then attempt to develop a framework, even a loose one, within which you can organize the rest of the literature you read. As you read more about the area, you are likely to change the framework. However, without it, you will get bogged down in a great deal of unnecessary reading and note-taking that may not be relevant to your study.

If you want to study the relationship between mortality and fertility, you should review literature about:

- Fertility-trends, theories, some of the indices and critiques of them, factors affecting fertility, methods of controlling fertility, factors affecting acceptance of contraceptives, etc.;
- Mortality-factors affecting mortality, mortality indices and their sensitivity in measuring change in mortality levels of a population, trends in mortality, etc.; and, most importantly
- The relationship between fertility and mortality-theories that have been put forward to explain the relationship, implications of the relationship.

Out of this literature review, you need to develop the theoretical framework for your study. Primarily this should revolve around theories about the relationship between mortality and fertility. You will discover that a number of theories have been proposed to explain this relationship. For example, it has been explained from economic, religious, medical, and psychological perspectives.

Your literature review should be written under the following headings, with most of the review involving examining the relationships between fertility and mortality:

- Fertility theories;
- The theory of demographic transition;
- Trends in fertility (global, then narrow it to national and local levels);
- Methods of contraception (their acceptance and effectiveness);
- Factors affecting mortality;
- Trends in mortality (and their implications);
- Measurement of mortality indices (their sensitivity), and
- Relationships between fertility and mortality (different theories such as ‘insurance’, ‘fear of non-survival’, ‘replacement’, ‘price’, ‘utility’, ‘risk’, ‘hoarding’).

Literature pertinent to your study may deal with two types of information:

1. Universal; and
2. More specific, i.e., local trends, or a specific program.

In writing about such information, you should start with the general information, gradually narrowing it down to the specific as, for example, shown above.

5. Developing a Conceptual Framework

The conceptual framework stems from the theoretical framework and concentrates, usually, on one section of that theoretical framework. The latter consists of the theories or issues in which your study is embedded whereas the former (i.e. conceptual framework) describes the aspects you selected from the theoretical framework to become the basis of your study. The conceptual framework is the basis of your research problem. For instance, in the example cited in section 4.6, the theoretical framework includes all the theories that have been put forward to explain the relationship between fertility and mortality. However, out of these, you may be planning to test only one, say, and ‘the fear of non-survival’. *Hence, the conceptual framework grows out of the theoretical framework and relates to the specific research problem, concerning the fear of non-survival.*

?

Dear learner! Shortly write the difference between theoretical framework and conceptual framework _____

6. Note-Taking (Writing up the literature reviewed)

Now, all that remains to be done is to write about the literature you have reviewed. Some researchers write about it under one heading: ‘Review of the literature’ or ‘Literature review’. But also the literature review should be written around themes that have emerged from reading the literature. The headings displaying themes should be precise, descriptive of the contents, and should follow a logical progression. Findings from the literature should be organized under these themes, providing references for substantiations or contradictions. Your arguments should be conceptually clear, stressing the reasons for and against, and referring to the main findings, gaps, and issues.

The other point is that the process of note-taking can be done either in the form of *paraphrasing or directly quoting the author’s ideas*. See the section that follows!

a. Paraphrasing

Paraphrasing may be defined as restating or rewording a passage from a text, giving the same meaning in another form. The main objective of paraphrasing is to present an author's ideas in your own words. Often paraphrasing fails due to:

- i) misunderstanding of the passage by the reader, or
- ii) Partial understanding of the passage and trying to guess the meaning

Therefore, accurate paraphrasing can be achieved through close reading and complete understanding of what is read. The following five guidelines help in this respect.

- i. Place the information found in the source in a new order
- ii. Break the complex ideas into smaller units of meaning
- iii. Use concrete, direct vocabulary in place of technical jargon found in the original source.
- iv. Vary the sentence patterns.
- v. Use synonyms for the words in the source.

➤ How to paraphrase appropriately

Broaden your understanding from the following examples

Original Passage

During the last two years of my medical course and the period which I spent in the hospitals as house physician, I found time, by means of serious encroachment on my night's rest, to bring to completion a work on the history of scientific research into the thought word of St. Paul, to revise and enlarge the Question of the Historical Jesus for the second edition, and together with Wider to prepare an edition of Bach's preludes and fugues for the organ, giving with each piece directions for its rendering. (Albert Schweitzer, Out of My life and Thought. New York: Mentor, 1963, p.94)

A Poor Paraphrase

Schweitzer said that during the last two years of his medical course and the period he spent in the hospitals as house physician he found time, by encroaching on his night's rest, to bring to completion several works.

(Note: This paraphrase uses too many words and phrases from the original without putting them in quotation marks and thus is considered plagiarism. (**Plagiarism** is unauthorized use of an author's thoughts or ideas and presenting them as one's own). Furthermore, many of the ideas of the author have been left out, making the paraphrase incomplete. Finally, the one who is paraphrasing has neglected to acknowledge the source through a parenthetical citation.)

A Good Paraphrase

Albert Schweitzer observed that by staying up late at night, first as a medical student and then as a "house physician," he was able to finish several major works, including a historical book on the intellectual work of St. Paul, a revised and expanded second edition of Question of the Historical Jesus, and a new edition of Bach's organ preludes and fugues complete with interpretive notes, written collaboratively with Wider.

(Note: This paraphrase is very complete and appropriate; it does not use the author's own words, except in one instance, which is acknowledged by quotation marks.)

b. Incorporating direct Quotes

At times you may want to use direct quotes in addition to paraphrases and summaries. To incorporate direct quotes smoothly, the following general principles hold.

- ❖ When your quotations are four lines in length or less, surround them with quotation marks and incorporate them into your text. When your quotations are longer than four lines, set them off from the rest of the text by indenting five spaces from the left and right margins and triple spacing above and below them. You do not need to use quotation marks with such block quotes. Follow the block quote with the punctuation found in the source. Then skip two spaces before parenthetical citation. Do not include a period after the parentheses.
- ❖ Introduce quotes using a verb tense that is consistent with the tense of the quote. Change a capital letter to a lower-case letter (or vice versa) within the quote if necessary. Use brackets for explanations or interpretations not in the original quote. E.g. ("Evidence reveals that boys are higher on conduct disorder (behavior directed toward the environment) than girls.") Use ellipses (three spaced dots) to indicate that

material has been omitted from the quote. It is not necessary to use ellipses for material omitted before the quote begins. E.g. (“Fifteen to twenty percent of anorexia victims die of direct starvation or related illnesses... [Which] their weak body immunity cannot combat.”)

- ❖ Punctuate a direct quote as the original was punctuated. However, change the end punctuation to fit the context. (For example, a quotation that ends with a period may require a comma instead of the period when it is integrated in to your own sentence.)
- ❖ A period, or a comma if the sentence continues after the quote, goes inside the quotation marks E.g. (Although Almaz tries to disguise “her innate evil nature, it reveals itself at the slightest loss of control, as when she has a little alcohol”) when the quote is followed with a parenthetical citation, omit the punctuation before the quotation mark and follow the parentheses with a period or comma. E.g. Alemu has” recognized the evil in himself, (and) is ready to act for good.”
- ❖ If an ellipsis occurs at the end of the quoted material, add a period before the dots. E.g. (Almaz is “more than a Woman, who not only succumbs to the Serpent, but becomes the serpent itself... as she triumphs over her victims...”)
- ❖ Place question marks and exclamation points outside the quotation marks if the entire sentence is a question or an exclamation. E.g. (Has Sara read the article “Alienation in east of Eden”?)
- ❖ Place question marks and exclamation points inside the quotation marks if and only if the quote itself is a question or an exclamation. (Mary attendee the lecture entitled” Is Cathy Really Eve?”)
- ❖ Use colon to introduce a quote if the introductory material prior to the quote is long or if the quote itself is more than a sentence or too long.
E.g. Taylor puts it this way: (Long quote indented from margin)
- ❖ Use comma to introduce a short quote. (Stein Beck explains, “If Cathy were simply a monster, that would not bring her in the story.”)

?

Dear learner! What is the difference between paraphrasing and direct quoting? _____

c. Referring to others in the text

In Harvard system, at every point in the text at which reference is made to other writers, the name of the writer and the year of publication should be included. It is also advisable to include page number.

- If the surname of the author is part of the sentence, then the year of the publication will appear in brackets.

Example

Bloom (1963, p 16) describes this...

- If the name of the author is not part of the sentence, then both the surname and the year of publication with page number be in brackets;

Example

In a recent study (Smith, 1990, p36), it is described as...

- If there are three or less authors, then their family names should be given; if there are more than three authors, the first author's family name should be given, followed by *et al.*

Example

Tolera, Barabaran and Jones (1991, p33) suggest that...

In a recent study (Barabaran and Jones, 1993, p10), it is suggested....

The most recent work (Barabaran *et al*, 1995, p16) shows that...

If the same author has published two or more works in the same year, then each work should be referred to individually by the year followed by lower case letters (a, b, c, etc). (These different references should be included in the bibliography).

Example

Barabaran (1996a, pp35-7) shows how...

Chapter Summary

Reviewing the literature is a continuous process. It begins before a research problem is finalized and goes on until the last report is written up. However, there is a paradox in literature review: you cannot undertake an effective literature review unless you have formulated a research problem yet your literature search plays an extremely important role in helping you to formulate your research problem. At the various stages of the research processes, literature review brings multi-purposes. For instance, it brings clarity and focus to your research problem, improves your methodology, and broadens your knowledge base.

Reviewing the literature involves a number of steps: searching for existing literature in your area of study, reviewing the selected literature, using it to develop a theoretical framework from which your study emerges, and to develop a conceptual framework which will become the basis of your investigation. The sources for identifying literature include books, journals and many others.

Exercises

Part I. Chose the best answer from among the alternatives given and check your answers against the answers given for this part at the back.

1. Why do you need to review the existing literature?
 - a) To make sure you have a long list of references
 - b) Because without it, you could never reach the required word-count
 - c) To find out what is already known about your area of interest
 - d) To help in your general studying
2. Which of the following can not be a component of Literature?
 - a. Encyclopedias
 - b. Journals
 - c. Newspapers
 - d. Micro Film
 - e. None
3. Which of the following order does not show the steps in the review of a literature?
 - a. Searching for literature-Reviewing literature - Developing theoretical framework- Developing conceptual framework- writing-up the reviewed literature
 - b. Reviewing literature - Searching for literature- Developing conceptual framework- Developing theoretical framework- writing-up the reviewed literature
 - c. Searching for literature- Developing theoretical framework- Developing conceptual framework- writing-up the reviewed literature-Reviewing literature
 - d. Developing theoretical framework- Reviewing literature - Searching for literature- Developing conceptual framework- writing-up the reviewed literature
4. What ways do you use to collect and store relevant information for your research?
 - a. Notebook
 - b. loose sheets of paper
 - c. Source cards
 - d. Note cards
 - e. All
5. Of the following, one is a true statement
 - a. the theoretical framework grows out of the conceptual framework and relates to the specific research problem
 - b. Theoretical framework provides with a guide to read relevant literature
 - c. Without theoretical framework, it will be easier to get what to read and write-up notes that are relevant to for a particular study
 - d. Theoretical framework describes the aspects you select from the conceptual

- framework to become the basis of your study
- e. All
6. From the following methods, one does not help paraphrase a text
- a. Putting the information found in the source in a new order
 - b. Breaking the complex ideas in the source in to smaller units of meaning
 - c. Replacing technical jargon with direct vocabulary
 - d. Using synonyms for the words in the source
 - e. None
7. According to Harvard system, identify the right way of citing sources in a text
- a. Kilole, Bati and Keraji (2008, p33) suggest that...
 - b. The most recent work, (Barabaran *et al*, 1999, p16), shows that...
 - c. Kuman (p 16, 1963) describes this...
 - c. (Jara, 1996, p34) describes that...
8. According to the Harvard referencing convention, which is the correct reference?
- a)** Bryman, A. (2008, 3e) *Social Research Methods*, Oxford; Oxford University Press
 - b)** Bryman (2008, second edition), Oxford University Press
 - c)** Bryman, Alan, *Social Research Methods* (2008: OUP)
 - d)** Bryman, A. Social Research Methods (2008)
9. A review of the literature prior to formulating research questions allows the researcher to do which of the following?
- a. To become familiar with prior research on the phenomenon of interest
 - b. To identify potential methodological problems in the research area
 - c. To develop a list of pertinent problems relative to the phenomenon of interest
 - d. All of the above
10. Sometimes, a comprehensive review of the literature prior to data collection is not recommended by theorists.
- a. True
 - b. False

Chapter Four

Research Design or Plan

4.1. Introduction

Dear learner! Having decided what you want to study about and its proper formulation, the next question is how are you going to conduct your study? What procedures will you adopt to obtain answers to research questions? How will you carry out the tasks needed to complete the different components of the research process? What should you do and what should you not do in the process of undertaking the study? These are some of the questions that need to be answered (in your research plan) before you pursue your study.

4.2. Lesson objectives

Up on the successful completion of the chapter, the learner will be able to:

- ✓ Define a research design
- ✓ Confidently explain the functions of and rationale for a research design
- ✓ Understand the classifications of a research design
- ✓ Know the qualities/characteristics of a good research design
- ✓ Identify and understand the essence in each component of a research design

4.3. The Definitions of a Research Design

Research design has been defined by different social scientists in different terms. All these definitions emphasize systematic methodology in collecting accurate information for interpretation with economy in procedure. To mention a few definitions, [Miller](#) has defined “research design” as “the planned sequence of the entire process involved in conducting a research study”. According to [Selltiz](#) et al, “Research design is a catalogue of the various phases and facts relating to the formulation of a research effort. It is an arrangement of the essential conditions for collection and analysis of data in a form that aims to combine relevance to research purpose with economy in the procedure”.

A few other definitions of research design are “a research design designates the logical manner in which individuals or other units are compared and analyzed; it is the basis of making interpretations from the data.”

“The challenge of research design is to translate the general scientific model into a practical research operation. Research design will refer to the entire process of planning and carrying out a research study.”

“Research design is the plan, structure and strategy of investigation conceived so as to obtain answers to research questions and control variance”. The plan is the complete program of the research. It includes an outline of what the investigator will do from writing the hypothesis and their operational implications to the final analysis of data (Kerlinger, 1986:279).

More explicitly, the design decisions happen to be in respect of:

1. What is the study about?
2. Why is the study being made?
3. What are the objectives of the study?
4. What are the propositions to be tested (if any?)
5. What is its scope?
6. What are the major concepts to be defined operationally?
7. On the basis of what criteria/measurements are the operational definitions made?
8. Where will the study be carried out?
9. What periods of time will the study include?
10. What type of data is required?
11. Where can the required data be found?
12. What is the study population?
13. What will be the sample design?
14. What techniques of data collection will be used?
15. How will the data be processed and analyzed?
16. What is the significance of the study?
17. To what target audience the reporting of the finding is meant?
18. In what style will the report be prepared?
19. What is the time period required for each stage of research work?
20. What is the cost involved?

Any kind of research incorporates all the essential components of research design but more importance is given to certain aspects of a research design in a particular type of research compared to the other.

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Dear learner! How do you define a research design? _____

4.4. The Functions of a Research Design

Research design has two main functions. These are

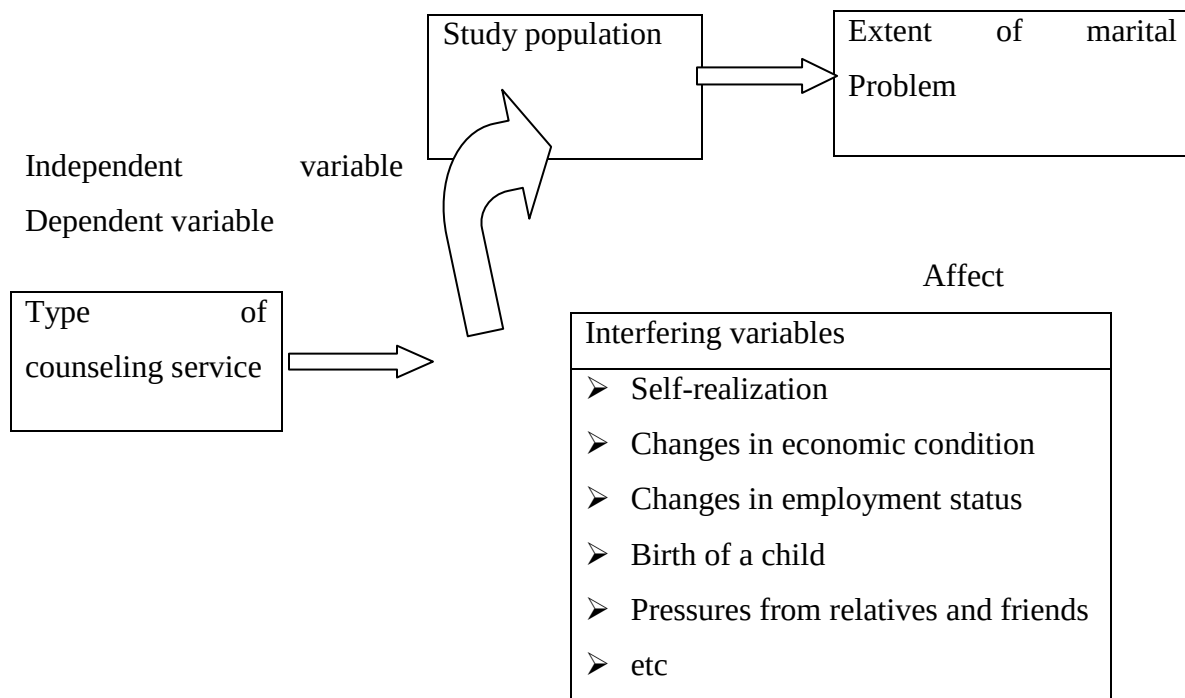
1. Identifying and/or developing procedures and logistical arrangements required to undertake a study,
2. Giving emphasis on the importance of quality in these procedures to ensure their validity, objectivity, and accuracy.

The first function is meant to conceptualize the operational plan. It details for you, your advisor, and other readers all procedures you plan to use and the tasks you are going to perform to obtain answers to your research questions. One of the most important requirements of the research design is to specify every thing clearly so that a reader will understand what procedures to follow and how to follow them. A research design, therefore, should:

- Name the study design *per se* (Cross sectional, Comparative, etc)
- Provide detail information about the following aspects of the study:
 - ✓ The study population and how it is identified,
 - ✓ A sample (if any) and how it is selected,
 - ✓ Data collection methods and why,
 - ✓ How to contact respondents(if any), and etc

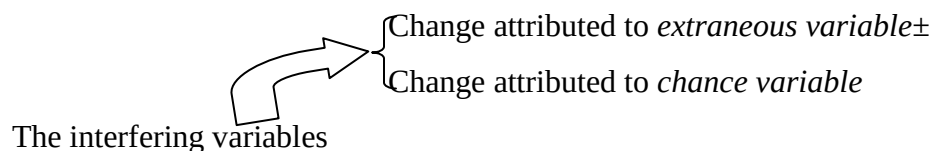
Let us explain the second function by example. Suppose you want to study the effectiveness of marriage counseling service provided by an agency (i.e. the extent to which the service has been able to resolve the marital problems of its clients). In studying this relationship, a number of factors can affect the dependent variable (the extent of marital problem):

- The independent variable (the type of counseling service);
- Interfering variable (the items listed in the box of the following diagram);and
- Chance or random variables (variables that can be associated with the respondents and/or the research instrument). These relationships are summarized as follows.



The interfering variables can affect the dependent variable in both positive and negative ways. Here, it may help in resolving marital problem (positive effect), adding to the effect of the independent variable on the dependent variable. On the other hand, they may stand in the way of resolving the marital, thereby decreasing the effect of the independent variable on the dependent variable (negative effect).

In short, the change in the marital problem= Change attributed to counseling service ±



Thus, in order to obtain the real relationship between the independent and dependent variables, the influence of the intervening variables is to be controlled. In experimental research, the main purpose having a control group is to measure the change that is a result of extraneous variables. The effect of the chance variables is often assumed to be none or negligible as it is assumed that if some respondents affect the dependent variable positively, others will affect it negatively. Hence, the effect cancels each other out.

There are two methods used to ensure that extraneous variables have similar effect on control and experimental groups and two methods for eliminating extraneous variables.

1. Ensure that extraneous variables have a similar impact on control and experimental groups. It is assumed that if two groups are comparable, the extent to which the extraneous variables will affect the dependent variables will be similar in both groups. Whether the groups are comparable are ensured by the following two methods:

- ❖ **Randomization**-ensures that the two groups are comparable with respect to the variable(s). It is assumed that if the groups are comparable, the extent to which extraneous variables are going to affect the dependent variable is the same in each group.
- ❖ **Matching**- this is an other way of ensuring that the groups are comparable so that the effect of extraneous variables will be the same in both groups.

2. Eliminate extraneous variable(s). Sometimes, it is possible to eliminate the extraneous variable, or build it in to the study design. Especially, this is done when the extraneous variables have strong correlation with the dependent variables, or when you want to isolate the impact of the variables separately. Two methods are used to achieve this method:

- ❖ **Build the affecting variable on to the design of the study.** Suppose you want to study the impact of maternal health services on the infant mortality of a population. Assume that the nutritional status of children also have a marked effect on infant mortality. To study the impact of maternal health services *per se*, you adopt a two-by-two factorial design as explained in the table that follows.

		Maternal Health	
		Yes	No
Nutritional supplements	Yes	Maternal Health + Nutritional supplement	Nutritional supplement
	No	Maternal Health	Control

❖ **Eliminate the variable-** Suppose you want to study the impact of a health education program on the attitudes toward and beliefs about the causation and treatment of a certain illness among Arab and Ethiopian people living in a particular community. As attitudes and beliefs vary markedly from culture to culture, studying Arab and Ethiopian people as one group will not provide an accurate picture. In such studies, it is important to eliminate the cultural variation in the study population by selecting and studying the populations separately or by constructing culture-specific cohorts at the time of analysis.

?

Dear learner! 1. Site the main functions of a research design _____.

2. What are the methods used to eliminate totally and/or to ensure that extraneous variables have similar effect on control and experimental groups? _____

4.5. Needs for research design

Just as we need a blue print (or the map of a house) well thought-out and prepared by an expert or architect, so we need a research design or plan in advance of data collection and analysis. Generally speaking, a thoroughly thought research design is needed for the following reasons:

- It helps the researcher to organize his ideas in a form whereby it will be possible for him to look for flaws and inadequacies;
- It facilitates the smooth running of various research operations;

- It makes a research as efficient as possible yielding maximal information with minimal expenditure of effort, time and money;
- It serves as framework for the process of reliable and valid data collection and analysis;
- It saves the researcher from offering hasty generalizations or misleading conclusions;
- It serves as a basis for others to provide their genuine comments and comprehensive review of the proposed study.

Therefore, preparation of the research design should be made with greater care as any error in it may upset the entire project.

4.6. Classifications of Research Designs

As you learnt in the first chapter, there are different classifications of research designs:

1. Experimental, Historical, and inferential designs (American Marketing Association),
2. Exploratory, Descriptive, and causal designs (Selltiz, Jahoda, Deutsch),
3. Experimental, and ex-post fact (Kerlinger),
4. Exploratory, Case and clinical studies (Goode and Scates),
5. Sample surveys, field studies, experiments in field settings , and laboratory experiments (Festinger and Katz),
6. Exploratory, Descriptive and Experimental studies (Body and Westfall),
7. Exploratory, Descriptive and Causal (Green and Tool),
8. Experimental, “Quasi-Experimental Designs” (Nachmias and Nachmias),
9. True experimental, Quasi Experimental and Non-experimental Designs (Smith),
10. Experimental, Pre-experimental, quasi-experimental designs and Survey Research (Kidder and Judd).

These different categorizations exist because “research design” is a complex concept.

Note: Recall the previous discussions made on various research types.

In fact, there are different perspectives from which any given study can be viewed:

- ❖ The degree of formulation of the problem(*exploratory or formalized*)
- ❖ The topical scope-the depth and breadth-of the study (*a case or statistical study*)

- ❖ The research Environment: field setting(survey) or laboratory experiment
- ❖ The time dimension(*one time or longitudinal*)
- ❖ The mode of data collection(*Observational or Survey*)
- ❖ The manipulation of the variables under study (*Experimental or Ex-post facto*)
- ❖ The nature of the relationship among Variables (*Descriptive and causal*)

4.7. Characteristics of a Good Research Design

The selection of a method or methods of logic and the planning of the design in time do not guarantee sound results. These are only first steps. The research design at the most, is a blueprint, and therefore at best it is only tentative.

Every design has its own strengths and weaknesses and at the same time there is no such thing as a single correct design. A good research design should satisfy the following four conditions: objectivity, reliability, validity and generalizability of findings.

a) Objectivity: The objectivity of the findings pertains to the methods of collection of data and scoring of the responses. The objectivity of the procedure (either collection of data or scoring of the responses or both) may be judged by the degree of agreement between the final scores assigned to different individuals by more than one independent observers. The more subjective the observation, recording and evaluation of the responses; the less the different observers agree.

The investigators who use close-ended questionnaires are said to be collecting data with the help of objective tools because all the scorers can apply a scoring key and agree perfectly on the result. In contrast, the open-ended questionnaires allow room for great disagreement among the scorers. However, in certain instances, the questionnaires consisting of both the types of items are used purposefully. In such instances, free response items and observations can be made fairly objective by giving careful instructions and guidelines to the observer or scorer.

Thus, any research design should permit the use of measuring instruments which are fairly objective in which every observer or judge seeing a performance arrives at

precisely the same report. This ensures the objectivity of the collected data, which will be used for the analysis, inferences and generalizations.

b) Reliability: Reliability refers to “Consistency” throughout a series of measurements. That is to say, if a respondent gives out a response to a particular item, he is expected to give the same response to that item whenever he is asked subsequently. On the contrary, if the respondent keeps on changing his responses to the same item when he is asked repeatedly, then the investigator will be facing a difficulty in considering which one of these responses is the genuine response of the respondent. So, the investigator should frame his items in such a way that the respondent can give only one genuine response. There are different methods in determining the reliability of the responses given out by a respondent. Some of these methods are: - Using “check items”, administering the same test repeatedly, using a series of “parallel” forms, etc.

c) Validity: Any measuring instrument is said to be valid when it measures what it purports to measure. For example, an intelligence test, constructed for measuring intelligence should measure only intelligence and nothing else. As in the case of reliability, there are a good number of procedures for establishing the validity of test. Some such procedures are: validating the present data against a “concurrent” criterion or a future criterion or a theory, etc.

d) Generalization/Generalizability: Once it is ensured that the measuring instruments used in a research investigation yield objective, reliable and valid data, the next important problem a well planned research design has to answer is the “generalizability” of the findings of the present study. That is *how best the data collected from a sample can be utilized for drawing certain generalization applicable to a larger group (population) from which the sample is drawn*. In other words, with how much authority and confidence an investigator can say that the same findings will be obtained even though the data is collected from the total population from which the sample is selected. A research design, thus helps an investigator in his attempt to generalize the findings, provided he has taken due care in defining the population, selecting the sample and using the appropriate statistical analysis while planning his research design. Thus a good research design should ensure that:-

- a) The measuring instruments can yield objective, reliable and valid data
- b) The 'population' is defined in unequivocal terms
- c) The requisite size of the sample is collected by using the most appropriate technique of sample selection
- d) The appropriate statistical analysis has been employed, and
- e) The findings of the present study can be 'generalized' without being contaminated by the errors of measurement or sampling errors or any other interfering factors.

?

Dear learner! Write the conditions that a good research design should satisfy _____

4.8. Components of a Research Design

Any investigator, before getting on to the job of planning a research design or proposal for a proposed study, should be clear about the following aspects:

- | | |
|--------------------|--------------------------------------|
| ❖ What to observe? | ❖ How to record the observations? |
| ❖ Whom to observe? | ❖ How to analyze the observations? |
| ❖ How to observe? | ❖ What inferences can be drawn?, and |
| ❖ Why to observe? | etc |

Once the investigator has sufficient understanding about the above-mentioned aspects, s/he can resort to the job of planning a **research design or proposal**. In doing so, he proceeds step by step in the following order.

1) Title of the investigation: The title of any research project:-

- Should be as brief as possible
- Should be as precise as possible
- Should project the scope of the problem in generalized terms

2) Introduction: The introduction of a *research plan or proposal* should place the research problem in its historical perspective; state the need for studying it, and the researcher's precise interests in the study of the problem. Remember that some of the contents suggested in this section may not be relevant to certain studies, so use your

discretion in selecting only what is pertinent to your study. The literature review is of central importance in this section.

Start with a very broad perspective of the main subject area, before gradually narrowing the focus to the central problem under investigation. In doing so, cover the following aspects of your study area:

- An overview of the main area under study;
- A historical perspective (development, growth, etc) pertinent to the study area;
- Philosophical or ideological issues relating to the topic;
- Trends in terms of prevalence, if appropriate;
- Major theories, if any;
- The main issues, problems and advances in the subject area under study;
- Important theoretical and practical issues relating to the central problem under study; and
- The main findings relating to the core issue(s)

Example I: *Suppose that you plan to study the relationship between advertising and product demand (the case of Arba Minch town). The preamble or introduction would include the following:*

- The role of advertising in the market place
 - Major changes in the philosophy of advertising over time
 - Factors affecting attitudes toward advertising business
 - The development of advertising in Ethiopia
 - Trends in the effectiveness of advertising in Ethiopia with particular reference to the area/region in which the study is being carried out
 - Changing advertising values, etc
-

3) Statement of the Problem

Having provided a broad introduction to the area under study, now focus on the issues relating to its central theme, identifying some of the gaps in the existing body of

knowledge. A rationale for your study with particular reference to how your study will fill the identified gaps should be stated as well. The main unanswered questions need to be identified. Some of the main research questions that you would like to answer through your study should also be raised. Knowledge gained from other studies and the literature about the issues you are proposing to investigate should be an integral part of this section. In other words, once you are able to collect different studies conducted in your field of investigation from various sources and examine each one of them critically for the points mentioned above, you will be in a position to state your problem in unambiguous and more precise terms. Moreover, in this section, you as researcher should:

- Identify the issues that are the basis of your study;
- Specify the various aspects of/ perspectives on these issues;
- Identify what knowledge is available concerning your questions, specifying the difference of opinion in the literature regarding these questions if differences exist.

Usually, the problem of investigation is stated in two ways:-

- a) In the form of statement or b) In question form

In some cases, the problem is stated with the help of more than one statement or a question.

For example I above, the issues you should include in this section are the following:

- What theories have been developed to explain the relationship between advertising and product demand?
- What is the relationship between advertising and product demand: What theoretical model will be the basis of your study?
- What do previous theories and researches have to say regarding the components of the theoretical model and advertising, for example, the relationship between product advertising and:
 - Motivation of customers;
 - Purchase frequency of customers;
 - Attitudes of customers for a product
 - Rate of production and sale with in a firm; and etc

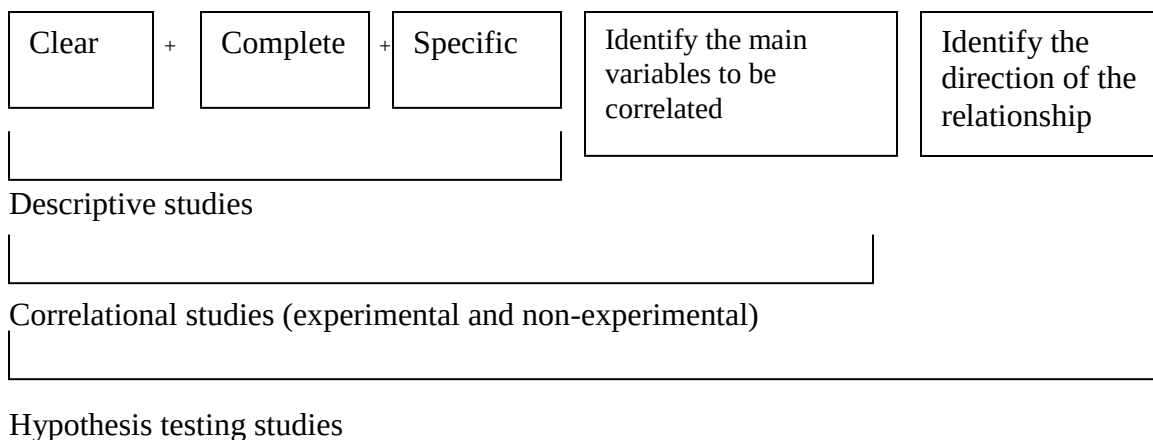
4) Objectives of the Study

Objectives inform a reader of what you want to achieve through the study. Therefore, it is extremely important to word them clearly and specifically. You spell out the objectives/purposes of the present study in the form of statements and while doing so, you take care that the objectives mentioned are well within the scope of the investigation envisaged by you.

The objectives of the study are listed under two headings:

- ❖ *Main objective; and*
- ❖ *Sub-objectives/Specific objectives*

The *main objective* is an overall statement of the thrust of your study. It is also a statement of the main associations and relationships that you seek to discover or establish. The *sub-objectives* are the specific aspects of the topic that you want to investigate within the main framework of your study. Sub-objectives should be numerically listed. They should be worded clearly and unambiguously. Make sure that it contains only one aspect of the study. The objectives should start with such action words as “to determine”, “to find out”, “to ascertain”, “to measure”, “to explore”, etc. The wording of your objectives determines the type of research design (e.g., descriptive, correlational or experimental, or others) you need to adopt to achieve them. The following figure displays the characteristics of the wording of objectives in relation to the type of research study:



If your study is descriptive, the main objective of the study should clearly describe the main focus of your study, even mentioning the organization and its location unless these are to be kept confidential (e.g., *to find out the opinion of the community about the services provided by Small and Micro Enterprise, SMEs, operating in Arba Minch town*).

If your study is correlational in nature, in addition to the above proprieties, the wording of the main objective should include the main variables to be correlated (e.g, *to ascertain the impact of SMEs on poverty alleviation, or to compare the effectiveness of different advertising media on demand creation*).

If the thrust of your study is to test hypothesis, the wording of the main objective-in addition to the above properties- should indicate the direction of the relationship being tested (e.g., *to ascertain if an increase in youth unemployment will increase the incidence of street crime, or to demonstrate that the provision of maternal and child health services to native people living in rural Ethiopia will reduce infant mortality*).

For example I above, the objectives can be stated in the following manner:

Main objective:

To examine the relationship between advertising and product demand in A/Minch town

Specific objectives:

- To find out the relationship, if any, between advertising and motivation of customers to purchase a product;
 - To ascertain the association between advertising and purchase frequency of customers;
 - To explore the relationship between advertising and attitudes of customers toward a product
 - To examine the links between advertising and rate of production
 - To examine the links between advertising and rate of sale with in a firm;
-

5) Review of the previous studies

On the selected theme, some studies might have been previously made by others so that it will bring out information on the theme. The primary aspects of those studies might be briefly described and the gaps may be pointed out. Does the selected study fill the gap? Or is it a replication of an earlier study made elsewhere? In the later case, the justification for studying the same problem may be stated with reference to environmental differences, time period differences, etc.

6) Significance of the study

While preparing the research proposal, you as a researcher have to incorporate the justification for the need of the research. You should justify the importance and urgency of the study, as to how the results of your study will be useful to the beneficiaries. This step would also prevent wastage of research effort on unimportant or insignificant problems. Problems should be broad-based enough to provide an investigation of real significance. As a research worker, you would assess to what extent the solution of the problem would contribute for the furtherance of human knowledge. The list of the objectives of the study magnifies further its utility and importance.

The following are some of the main components the justification stresses:

- The need for new knowledge, techniques or conditions
- The need to help address those areas that remain untouched or inadequately treated.
- The need to fill the gap in the knowledge pertaining to the given area.

7) Conceptual Model

This section is the heart of the research plan. This is where the researcher formulates and develops the structure of the relationships among the variables he/she is investigating. The logical connections of the variables are delineated; the assumptions, and propositions used to develop the explanatory framework are included. The entire research project rests up on the theoretical framework.

8) Formulation of Hypothesis

Hypothesis is a testable statement of a potential relationship between two or more variables (See the previous discussion on the Hypotheses). If you are going to test hypotheses, list them in this section.

From example I, the following hypotheses can be formulated:

H₁= Customers' demands for a product and product advertising are positively correlated.

H₂= The higher the advertising, the higher the product purchase frequency of customers;

H₃= Product advertising is positively correlated with attitudes of customers toward a product

H_n= etc

9) Conceptualization of the different terms used in the Investigation

The researcher should be clear about the meaning of every term he uses. It not only helps him but also the other investigators, respondents and readers of the research report in understanding the different terms without any ambiguity. Normally, these concepts are explained either in the form of nominal definitions or in the form of operational definitions.

Nominal definitions are used in cases where a given word is synonymous with certain other expressions, the meanings of which are already established.

In the second type of definition, the major concepts used in the title of the study, its objectives, the investigative questions and the hypotheses should be identified. A variable or a concept is said to be **operationally defined** when the steps in its observation or measurement are clearly stated. Thus, an operational definition of a concept helps the investigator in obtaining reliable results, without which he/she can not establish any fact scientifically which in its turn makes the problem of investigator much more complex in his/her attempt to accept or refute the hypothesis under consideration with more authenticity.

Sometimes, the investigators even define a variable operationally with the help of items in the questionnaire or interview schedule.

The investigator usually engages him/her self in the following exercises before arriving at the operational definitions of different terms. The investigator will:-

- a) discover the various dimensions involved within the concept

- b) correlate these different dimensions so that all possible types can be discerned
- c) reduce the number of types to be studied
- d) translate those types selected for study into operational categories and empirical indices

It is always advisable to define each concept both:

- i) In abstract terms given the general meaning it intends to convey, and
- ii) In terms of operational categories by which it is represented in a particular study.

(For more understanding, refer back to the previous discussion on the same issue.)

10) Scope of the Investigation

Here the investigation states the restrictions and limitations which he/she imposes on his/her study. It is statement of the limits or scope of the investigation. This statement will provide information concerning who, what, where and how many. It will determine the boundaries of the project in hand. Again it will be important to mention as to how many subjects will constitute the sample of the study and how they will be distributed over the institutions, geographical areas of time intervals. The age level, grade level, ability level, achievement level, etc. will also be specified.

The scope of any investigation is dependent on several factors such as

- (a) The time and money availability to the investigator,
- (b) A availability of a sample,
- (c) Accessibility of the investigator to the respondents, and
- (d) The amount of time the respondents can spare for the investigator, etc.

11) The study design

Describe the study design (as discussed before) you plan to use to answer your research question. For example, state whether you use “cross sectional, experimental, or any other design). Identify the strengths and weaknesses of your study design. Moreover, at least the following important elements need to be included in your study design:

- Who make up the study population? Can each element of the study population be identified? If yes, how?
- Will a sample or the total population be studied,

- How will you get in touch with the selected sample?
- How will the sample's consent to participate in the study be sought?
- How will the data be collected (e.g., by interview, questionnaire or by observation)?
- In the case of a mailed questionnaire, to what address should the questionnaires be returned?
- Are you planning to send a reminder?
- How will confidentiality be preserved?
- How and where can respondents contact you if they have queries?

Example: Now a different example is considered for explanation.

It is proposed that the study will be carried out in two government high schools in Arba Minch town. The principals of the schools most accessible to the researcher will be contacted to explain the purpose of the study and the help needed from the school, and to seek their permission for the students to participate in the study. As the constraints of time and resources do not permit the researcher to select more than two schools, negotiations with other schools will cease when two schools agree to participate in the study.

It is proposed to select year 9 students as the academic achievement of students in years 8 and 10 could be affected by factors unique to them. Year 8 students may be experiencing anxiety as a result of having just made the transition to a new system. The motivation of students in year 10 could be affected by their being at the stage in their education where they must decide if they will continue on at school.

In order to control the variance attributable to the gender of a student, it is proposed to select only male students.

Once the principal of school agrees to allow the study to be carried out, the researcher will brief the teacher in charge about the study and its relevance, and will arrange a date and time for administering the questionnaire.

When the students are assembled ready to participate in the study, the researcher will explain its purpose and relevance, and then distribute the questionnaire. The researcher will remain in the class to answer any questions the students might have.

12) Selection of the Sample

It is needless to say that no investigator can study the entire population and hence selects a few individuals belonging to a population for the purpose of his investigation. These selected individuals form a sample and while selecting these individuals, the investigator should consider the following facts:-

i) Definition of the population: - The investigator should define population in operational terms, describing it with all the necessary identifiable characteristics. This helps the investigator in selecting the sample depicting the same characteristics with more ease and further helps the investigator in generalizing his findings of the present investigation with more authenticity. Thus the only difference between the population and the sample is their size and except for that, the sample is exactly similar to the population in all the characteristics.

ii) Sample size:- Depending on the mode of research, the sample size varies. For example, in experimental research, the sample size is very small compared to that of descriptive research. It is always advisable to collect data from more number of individuals than required by sample size to take care of any casualty particularly in the case of problems where the data is to be collected from the same sample repeatedly at different times. Sample size is usually determined by estimating the reliability of statistics calculated from the data collected from different sizes of the samples.

iii) Representativeness of the Sample: - When a sample is selected, it is assumed that it represents the population for which it stands. Failure of this is failure of entire research study. This is very fundamental and crucial in any research design to give sense to any research study undertaken. Thus the investigator should use his ingenuity in carefully selecting the appropriate technique of sample selection. The different techniques usually employed by the investigators in selecting the sample are random sampling technique, stratified sampling, quota sampling, incidental sampling, purposive sampling, multistage sampling, etc.

Example

The selection of schools will be done primarily through Quota sampling. Schools will be selected on the basis of their geographical proximity to the researcher. The researcher will prepare a list of schools, in rank order, of accessibility. Once two schools agree to participate in the study, negotiations with other schools will cease.

All year 9 male students will constitute the study population. It is expected that the sample will not exceed 100 students.

13) Data Collection (Methods and Instruments)** Construction/adaptation of Instruments**

The investigator should now find instruments for collecting the data required by the hypothesis. The investigator himself may have to construct these instruments or he may have to adopt the readily available instruments to suit the local conditions. In the latter case, the investigator may make certain necessary changes in the format and etc, with the help of the feedback received by conducting *a pilot study* on a very small sample.

Besides using the instruments like questionnaires, interview schedules, projective techniques, attitude scales, etc, the investigator may adopt other methods like observation, examination of records and so forth for data collection. Each of these various methods has its own advantages and limitations, which will be discussed in detail in the forth-coming chapters.

** Administration of the Questionnaires/Tests**

Depending on the nature of the questionnaire/test and the purpose of the investigation, the investigator decides whether to administer the questionnaire/test individually or in small groups or in large groups.

14) Methods of Data Analysis

Complete planning of analysis in advance is not always possible or even desirable; new idea occurs to the investigator as he/she collects the data. But except in exploratory studies, it is always possible and desirable to workout in advance the basic outlines of the

analysis. Depending on the nature of the data and the information required by the hypothesis, the investigator should subject the data to appropriate statistical analysis.

In this section, you need to describe the strategy you intend to use for data analysis. Specify whether the data will be analyzed manually or by computer. For computer analysis, identify the program and the statistical procedures you plan to perform on the data. Also identify the main variables for cross-tabulation.

Example

Frequency distribution in terms of:

- Customers' motivation;
- Customers' purchase frequency,
- Customers' attitude toward a product, etc...

Cross tabulations:

Advertising

- Customers' motivation;
- Customers' purchase frequency,
- Customers' attitude toward a product, etc...

15) The structure of the report

As clearly as possible, state how you intend to organize the final report (see the last chapter). In organizing your material for the report, specific objectives of your study are of immense help. Plan to develop your chapters around the main themes of your study. The title of each chapter should communicate the main thrust of its contents.

The first chapter, possibly entitled "Introduction", should be an overall introduction to your study, covering most of your project proposal and pointing out deviations, if any, from the original plan. The second chapter should provide some information about the study population itself, i.e., some of its socio-economic-demographic characteristics. Titles for the rest of the chapters will vary from study to study but each chapter should be written around the main theme.

Example

The report will have the following chapters:

Chapter 1: Introduction

Chapter 2: Research design and methodology

Chapter 3: Literature review: Customers' motivation, Customers' attitude toward a product, customer purchase frequency, and etc and product advertising

Chapter 4: Customers' motivation product advertising, Customers' attitude toward a product and product advertising, customer purchase frequency and product advertising,

Chapter 7: Summary, conclusions and recommendations

16) Problems and limitations

This section should list problems you thought you might encounter concerning, for example, the availability of data, securing permission from the agency/ organization to carry out the study, obtaining the sample, or any other aspect of the study.

Any limitations that could affect the validity of your conclusions and recommendations should be communicated to the reader.

Here the problems refer to difficulties relating to logistical details, where as limitations designate structural problems relating to methodological aspects of the study. For example, the study design you choose may not be the best but you may adopt it for a number of reasons. This is classified as limitation of the study. It is also true for sampling or measurement procedures.

17) Work/Time schedule

A time schedule should be prepared so that the researcher may plan his time and energy effectively. Dividing the project in to parts and assigning time for the completion of each part helps to systematize the project. The total time duration of the study are also specified in this section. You can develop a chart as follows for your work plan.

	To be completed by (Months in 2009)											
Tasks	Jan	Feb	Mar	April	May	June	July	Aug	Sep	Oct	Nov	Dec
Proposal writing												
Instrument construction												
Reviewing and writing related literatures												
Data collection												
Coding												
Data analysis												
Report(first draft)												
Report(final draft)												
Typing												
Presentation and communication												

18) Budget

This should include the estimate of the expected costs of the project under various major categories like salary (if any), printing and stationary, postage, travel expenses, computation, secretarial and typing, etc.

19) Bibliography

This is a section where a list of books, journals, and other documents is offered. The researcher should list all reference materials that he/she has used in selecting the problem and which he/she may use during the conduct of study.

?

Dear learner! List the main components of a research proposal_____

Chapter Summary

Dear learner! Let us now summarize the main points covered in this chapter. As you remember, the discussion was all about the research design: a plan, structure, and strategy of investigation so conceived as to obtain answers to a research problem. It acts as a blue print and is prepared before data collection and analysis. A good research design satisfies four important conditions: objectivity, reliability, validity and generalizability of findings. Lastly, the basic components of the research design or proposal were covered and the elements include title of the investigation, introduction/background of the study, statement of the problem, objectives of the study, review of the previous studies, significance of the study, conceptual model, formulation of Hypothesis, Conceptual definitions of the different terms used in the investigation, scope of the Investigation , study design, sampling and population , methods and instruments, the structure of research report, problems and limitations of a study , work/time schedule, budget, and finally bibliographies.

Exercises

Part I. Multiple choice questions

1. What is a research design?
 - a) A way of conducting research that is not grounded in theory
 - b) The choice between using qualitative or quantitative methods
 - c) The style in which you present your research findings, e.g. a graph
 - d) A framework for every stage of the collection and analysis of data
2. A research plan _____.
 - a. Should be detailed
 - b. Should be given to others for review and comments
 - c. Sets out the rationale for a research study
 - d. All of the above
3. One of the following does not show the purpose of a research design
 - a. It shows what the significance of the study is
 - b. It identifies the study design to be employed
 - c. It provides detail information about the study population and how it is identified
 - d. It provides detail information about the data collection methods to be used
 - e. None of the above
4. One step that is not included in planning a research study is:

a. Identifying a researchable problem	d. Conducting a meta-analysis of the research
b. A review of current research	e. Developing a research plan
c. Statement of the research question	
5. The feasibility of a research study should be considered in light of:

a. Cost and time required to conduct the study	c. Potential ethical concerns
b. Skills required of the researcher	d. All
6. A formal statement of the research question or “purpose of research study” generally
 - a. Is made prior to the literature review
 - b. Is made after the literature review

- c. Will help guide the research process e. b and c
- d. All of the above
9. The research participants are described in detail in which section of the research plan?
- a. Introduction b. Method c. Data analysis d. Discussion
10. The Method section of the research plan typically specifies
- a. The research participants
- b. The results of prior studies that address the phenomena of interest
- c. The apparatus, instruments, and materials for the research study
- d. The planned research procedures
- e. a, c and d
12. If you want to study *the effectiveness of salary increment on employee retention*, the variable “salary increment” here is
- a. Dependent variable c. Intervening variable
- b. Independent variable d. Extraneous variable
13. In a research design, when a measuring instrument is said to measure what it purports to measure, it is termed as _____
- a. Objectivity c. Consistency
- b. Generalizability d. Validity
14. One of the following statements defines the Scope of a study, *except one*
- a. It states the restrictions which a researcher imposes on his/her study
- b. It will determine the boundaries of the project in hand
- c. It provides information concerning who, what, where and how many.
- d. to mentions how many subjects will constitute the sample of the study
- e. None
15. Of the following, one can not be instrument of data collection
- a. Questionnaire
- b. Interview schedules
- c. Observation,
- d. Examination of records
- e. None

Chapter Five

Census and Sample Investigation

5.1. Introduction

Dear learner! In the last four chapters, we covered discussions on the first two major steps in a research process. Now you are going to learn the third step: determination of sample design. In this particular chapter, the main issues to be addressed include census and sampling methods and their merits and demerits, the objectives and basis of sampling, the assumptions underlying sampling, types of sampling techniques, steps in sampling design and how to determine a sample size, qualities of a good sampling design, and finally the concepts related to sampling and non-sampling errors will be explained in depth.

5.2. Lesson objectives

Dear learner! Up on the successful completion of the chapter, you will be able to:

- ✓ Differentiate census method and sampling method of a survey
- ✓ Identify the merits and demerits of using census method and sampling methods of undertaking a survey
- ✓ Confidently describe the objectives and assumptions of sampling
- ✓ Identify the various types of sampling techniques
- ✓ Know the necessary steps in sample selection
- ✓ Get a base on how to determine a sample size
- ✓ Explain the qualities of a good sampling design
- ✓ Differentiate between sampling and non-sampling errors

5.3. Census Method and sampling method

Introduction

Statistics deals with large numbers. It does not study a single figure. All the items under consideration in any field of inquiry constitute a universe or population. A complete enumeration of all the items in the “population” is known as a census method of collection

of data. In practice, sometimes, it is not possible to examine every item in the population. But also a complete enumeration or estimation of all the items in the “population” may not be necessary. Sometimes it is possible to obtain sufficiently accurate results by studying only a part of the total “population”. In the case of population census, every household is to be enumerated. But in certain cases, a few items are selected from the population in such a way that they are representatives of the universe. Such a section of the population is called a sample and the process of selection is called sampling. But the size of the sample should be sufficient so as to be representative of the universe. Thus following are the two ways by which the required information in any statistical inquiry can be obtained:

A) *Census method or complete enumeration survey*

B) *Sampling technique*

5.3.1. Census method

The census method deals with the investigation of the entire population, i.e. it is a complete enumeration method. Under this method, the data are collected for each and every unit of the universe. This method provides more accurate and exact information as no unit is left out. Now let you identify the merits and demerits of census method of data collection.

❖ Merits of Census Method

The following are some of the advantages of making use of census method

- ⇒ When all the members are covered by the enquiry, no element of chance is left and highest accuracy is obtained. These advantages as we will observe presently, though seemingly true may not be very material.
- ⇒ The entire mass of data being taken into consideration; all the characteristics of the universe is maintained in original
- ⇒ This method is free from sampling errors. The possibility of personal bias affecting selection of units and the method of sampling are not there.
- ⇒ When the universe is a small one, one cannot resort to sampling method. In such a situation, completeness and accuracy can be ensured only by the census method.

❖ Demerits of Census Method

The following are some of the demerits of census method:-

- ⇒ The census method requires a large *army* of enumerators who will be having different standards of skill, willingness and efficiency and therefore organizational difficulties will be faced.
- ⇒ As it involves a great deal of time, money and energy; small organizations, individuals, etc cannot afford to resort to this method. It is practically beyond the reach of ordinary researchers. Perhaps, government is the only institution, which can get complete enumeration carried out.
- ⇒ The census method is useless in case results are urgently required since by the time these are made available, the entire phase of the problem under study may go a complete change.
- ⇒ As the volume of data involved is large, possibilities of statistical errors creeping in are also very great.
- ⇒ An element of bias, however slight, will get larger and larger as the number of observations increases.
- ⇒ There is no way of checking the element of bias or its extent except through a resurvey or use of sample checks.

In practice, sometimes it is not possible to examine every item in the population. Moreover, a complete enumeration or estimation of all the items in the population may not be necessary. Sometimes it is possible to obtain sufficiently accurate results by studying a part of the total population.

5.3.2. Sampling Method

Sampling is simply the process of learning about population on the basis of a sample drawn from it. Under this method, a small group of the universe is taken as the representative of the whole mass and the results are drawn. It is a method to make social investigation practicable and easy. “A statistical sample is a miniature picture or cross section of the entire group or aggregate from which the sample is taken. A sample is the reflection of the universe and bears all the characteristics of the universe.

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Dear learner! What is the difference between census and sampling methods? _____

Now the other important aspects of sampling methods of data collection are discussed in the sections that follow.

1. Sampling terminologies

- **Population or study population/Universe**-Aggregate of the units pertaining to a study. It is a target group to be studied.
- **Sample**-part of a population from which you obtain the information about the population
- **Sample size**-The number of the sample units from whom you get the required information
- **Sampling design or strategy**-The way we select a sample
- **Sampling unit or element**- each element of the population that becomes the basis to select a sample
- **Sampling frame or source list**-The list of sampling units (members of a population) from which a sample is taken.
- **Sample statistics**-Your findings on the basis of the information obtained from the sample. E.g. Average income of families living in Arba Minch town (calculated from the information obtained from the sampled families). Sampling statistics become the basis of estimating the prevalence of a characteristic in the study population
- **Population parameters** -The population characteristics you want to investigate.

2. Objectives of Sampling

The primary objective of the sample survey is to obtain accurate and reliable information about the universe with minimum cost, time and energy and to set out the limits of accuracy of such estimates.

The census type of enquiry requires a great deal of time, money, and energy. Perhaps, government is the only institution, which can get complete enumeration carried out. In a census enquiry, information is made available for each individual item of the universe. This is neither possible nor required in a sample enquiry. Thus, some information is to be sacrificed in sample method of enquiry. But when the universe is a small one, it is no use

resorting to a sample method. For example, if the expenditure habits of 100 students living in a college hostel are to be studied, then perhaps census method may be resorted to. But if 500 students living in a university hostel are to be studied, then perhaps the sample method may be more convenient and inexpensive; sample method is less costly because it involves a fraction of the total population. But a sample should not contain almost all the items, because then the very purpose of sample is defeated. Moreover, the sample should not contain too few items to be representative of the whole population. A balance is to be struck between the two to arrive at the size of the sample.

3. Advantages/Merits of Sampling

The following are some of the advantages of making use of sampling technique

- With the help of sampling, the volume of data being small, which can be collected and analyzed quickly, one can arrive at certain decisions for problems which are required to be solved urgently.
- Sometimes census method is impossible to be employed. For example, in the case of manufacture of electric bulbs, only a sample can be tested to find out the life span of the whole lot for obvious reasons
- In certain types of enquiries, highly trained personnel or specialized equipment is required for collection of data: complete census in such case may not be practicable. In such a situation, either sampling may be resorted to or enquiry is abandoned altogether.
- In the case of sampling, we are in a position to get much more accurate information than is possible by complete enumeration for the following reasons:-
 - Detailed information can be obtained from a small group of respondents
 - Qualified persons for investigation can be appointed and intensive training given
 - Relatively limited data can be handled much more easily
 - Follow-up is easy in case of poor response
 - Chances of statistical errors are minimized as the data collected and proceed is relatively small. No doubt some sampling errors might creep in but they can be estimated with great precision.
 - It is much more economical method of inquiry than the census method. At times where the purpose of inquiry does not warrant complete enquiry; it would be

wasteful to go in for a complete enquiry. Thus time, energy and money can be saved without sacrificing accuracy of results. That is sampling method makes exhaustive and intensive study possible with much less time, money and material. Sometimes the result drawn from the sampling methods are more reliable and accurate than the results drawn from the census method.

- Where the population is hypothetical, e.g. where a coin is tossed a number of times, the sampling is the only method of estimating the character of the population.
- In respect of a too large a universe like the contents of a mine, sampling is the only way to form an estimate of the characteristics of the population.

4. Demerits of Sampling

Some of the demerits are:-

- The results obtained may be false, inaccurate and misleading as the sample might not have been drawn properly.
- Chance of sampling errors is great. The investigator may have personal bias especially with regard to choice of technique and drawing sampling units
- The size of the sample may not be sufficient so as to be representative of the universe
- When the universe is a small one, it is no use resorting to sampling
- In the absence of qualified and experienced persons, the information obtained from sample surveys cannot be relied upon.

5. Assumptions of Sampling

Selection of a sample as representative of the universe bases upon following assumptions.

- i. **Homogeneity:** A social phenomenon is very complex in nature and every unit appears to be different from another. But at the same time they also possess similarities in many respects. It is, therefore, assumed that there is the possibility of such representative types in the whole population that makes sampling possible.
- ii. **Possibility of Representative Selection:** Sampling has its origin in the mathematical theory of probability and law of statistical regularity. The law of statistical regularity lays down that a group of objects chosen at random from a large group tend to possess the

characteristics of the large group (universe). It is, therefore, assumed that there is a possibility of selection of representative sample.

- iii. **Absolute Accuracy not essential:** The absolute accuracy is not essential in the case of large scale observation and the results of sample studies, although not hundred per cent accurate, are nevertheless sufficiently accurate to draw valid generalizations with a given standards of accuracy. Thus, it is not the absolute accuracy but relative or significant accuracy that are needed in case of large-scale observations. In fact, in any statistical enquiry, perfect accuracy in final results is practically impossible to achieve because of the errors in measurement, collection of data, its analysis and interpretation of the result.

6. Types of sampling Techniques

Different types of sampling techniques are used for drawing a sample plan. The various methods of sampling may broadly be classified into two categories:-

- A. Probability sampling
- B. Non-probability sampling

A. Probability sampling

Probability sampling provides a scientific technique of drawing samples from population according to some laws of chance in which each unit has some definite pre-assigned probability of being chosen in the sample. Random sampling, systematic sampling, stratified sampling, cluster sampling, multi-stage sampling and area sampling are considered as probability sampling.

1) Random Sampling: - A random sampling is the one where each item in the universe has an equal or known opportunity of being selected. A random sample is a sample selected in such a way that every item in the population has an equal chance of being included. Random sampling is more suitable in more homogeneous and comparatively larger groups.

Selection of Random sample

Four methods are generally used for drawing out a sample on random basis. They are:-

- i. **Lottery Method:** Under this method, the various units of the universe are numbered on small and identical slips of paper, which are folded and mixed together in a drum

thoroughly. A blindfold selection is then made of the number of slips required to constitute the desired size of sample.

ii. Use of table of random numbers: The most practical and economical method of selecting a random sample consists of the use of table of random numbers. These tables have been constructed by L.H.G. Tippet (1927), Fisher and Yates (1963), Kendall and B. Smith (1939). These numbers are very widely used in all the sampling techniques and have proved to be quite reliable as regards to accuracy and representativeness. The procedure for a sample using a table of random numbers is as follows:

1. Identify the total numbers of elements in the study population. It can have one or more digits .
2. Number each element starting from 1.
3. If the table of random numbers is in more than one page, choose the starting page by a random procedure. Select also the rows or columns by same procedure to determine you starting point and proceed in a predetermined direction (horizontally, vertically, or diagonally).
4. Randomly select number of rows or columns, in a random table, corresponding to the number of digits of the population
5. Decide on your sample size
6. Select the required number of your sample. If same numbers are selected, discard it and go to the next.

Example: Suppose a random sample of 15 is to be selected from AMU population of 500 Management students. A table extracted from a standardized random table is illustrated as under. Here, you have to choose three columns or rows randomly (say column 1, 4, and 8 respectively). Starting at the top of the left first column, the selected elements (samples) include 104, 373, 84, 92,121,312, 125,148,289,357, 327,418,162, 154, and 296.

Table 5.1. A sample of random numbers

104	092	273	125	233	476	352	501
373	054	320	148	405	764	489	947
084	142	226	289	153	419	364	650
092	001	590	625	729	709	837	667
121	080	479	599	970	180	015	573
660	506	057	747	217	434	107	327
312	106	301	508	705	945	257	418
851	626	897	976	702	502	305	162
633	057	233	721	635	805	932	154
734	379	264	357	553	703	652	296

iii. Grid system: This technique is used for selecting a sample of area. According to this method, a map of entire area is prepared. Then a screen with squares is placed upon the map and the areas falling within the selected squares are taken as samples.

iv. Coin Flipping method: If a small number of populations are considered, a coin may be flipped to select a sample by assigning head or tail to a unit. But you have to decide ahead of time whether the unit which gets a tail or head needs to be included in the sample.

Precaution in drawing a random sample

Following precautions may be observed in drawing a random sample:-

- Population to be sampled and the unit must be clearly defined
- Different units should be approximately of equal size
- The units must be independent of each other
- Method of selection should be completely independent
- Every unit should be accessible; units once selected should not be ignored or replaced by any other unit of the universe.

Merits of random sample method

This method has several advantages, which may be summarized as below:

- ⇒ It is more scientific method of taking out samples from the universe since its elements are free from personal bias

- ⇒ No advance knowledge of the characteristics of the population is necessary under this method
- ⇒ The sample drawn under this method is true representative of the universe
- ⇒ It is possible to ascertain the efficiency of the estimates by considering the standard errors of their sample distribution
- ⇒ It is very simple and easily practicable procedure of selecting sample
- ⇒ This method provides us most reliable and maximum information at the least cost which saves time, money and labor.

Demerits of Random Sample Method

Random sampling method has certain practical difficulties and inertness limitations, which are as follows.

- ⇒ The sampling method requires complete list of the universe. But such up-to-date list is not available in many enquiries which restrict the use of this method.
- ⇒ The sample may not be a true representative of the universe if its size is too small
- ⇒ Sometimes this method gives such results whose probability is very small
- ⇒ For a given degree of accuracy, this method usually requires larger sample as compared to stratified sampling.

?

Dear learner! What are the various methods employed to select a random sample? _____

- 2) **Stratified Random Sampling:** Where the *population is heterogeneous* with respect to the variables or characteristics under study, then the technique of stratified sampling is used to obtain more efficient and accurate results. Stratification means division of the universe into non-overlapping groups according to geographical, sociological or economic characteristics.

Process of Stratifying

1. Stratified random sampling involves the following steps The universe is first divided in to sub-groups based on the principal variables under study (for example, based on age, income, attitude, etc) and the required units are selected at random from each sub-group

2. The stratification should be conducted in such a way that the items in one stratum should be similar to each other but they should differ significantly from units of other strata
3. Each and every unit in the population must belong to one and only one stratum. In other words, various strata must be non-overlapping
4. The size of each stratum in the universe must be large enough to provide selection of items on random basis
5. Size of the sample from each stratum can either be proportional or disproportional to the size of each stratum.

Categories of stratified sampling

a. Proportionate stratified sampling- This sampling involves drawing a sample from each stratum in proportion to the share of the stratum in the total population. For example, the second year BA students of Faculty of Business and Economics of Arba Minch University consist of the following specialization groups:

Table 5.2. Enrollment in Arba Minch University, 2009

Departments	Number of students	Proportion of each stream
Management	340	0.35
Accounting	300	0.31
Economics	320	0.33
	960	1.00

Suppose the researcher decides to take a sample of 150. Then the strata sample size will be:

Table 5.3. Sample from the total population of Enrollment in AMU, 2009

Strata		Sample size
Management	$0.35 \times 150 =$	53
Accounting	$0.31 \times 150 =$	47
Economics	$0.33 \times 150 =$	50
		150

b. Disproportionate stratified random sampling- This method does not give proportionate representation to strata. It gives overrepresentation to some and under

representation to others. All strata may be given equal weight even though their shares in the total population can vary.

Merits of stratified random sampling

This method has several advantages which may be summarized as below:-

- If a correct stratification has been made, even a small number of units will form a representative sample
- Under stratified random sampling, no significant group is left underrepresented
- Stratified random sampling is more precise and to a great extent avoids bias. It also saves time and cost of data collection since the sample size can be less in this method
- It is the only sampling plan which enables us to achieve different degrees of accuracy for different segments of the population. Replacement of case is easy in this method if the original case is not accessible to study. If a person refuses to cooperate with the investigator he may also be replaced by another person from the same sub-group.
- Stratification is of great advantage if the distribution of the universe is skewed.

Demerits of stratified random sampling

This method has some demerits which are listed below:-

- It is a very difficult task to divide the universe into homogeneous strata
- If the strata are over-lapping, unsuitable or disproportionate, the selection of samples may not be representative
- If stratification is faulty, the results obtained may be biased. Such errors cannot be compensated even by taking large samples

3) Systematic Sampling: Under this method, a sample is taken from a list prepared on a systematic arrangement either on the basis of alphabetic order or on house number or any other method. In this method, only the first sample unit is selected at random and the remaining units are automatically selected in a definite sequence at equal spacing from one another.

Steps involved in systematic sampling

1. First of all, the population is arranged in serial numbers from 1 to N, and the size of sample is determined
2. The sample interval is determined by dividing the population by the size of sample
i.e. $N/n = K$
Where K = sample interval
N = Size of population
n = Sample size
3. Any number is selected at random from the first sampling interval. The subsequent samples are selected at equal or regular intervals

 Merits of Systematic Sampling

- ⇒ Systematic sampling is very easy to operate and checking can also be done quickly
- ⇒ Randomness and probability features are present in this method which makes sample representative.

 Demerits of systematic sampling

- ⇒ Systematic sampling works well only if the complete and up-to-date frame is available and if the units are randomly arranged
- ⇒ Any hidden periodicity in the list will adversely affect the representatives of the sample

4) Cluster Sampling: This method is adopted when the population elements are scattered and large, and the list of the elements is not readily available. Under this method, the total population is divided into some recognizable subdivisions which are termed as clusters and a simple random sample of these clusters, as the sampling unit, is drawn and then the survey of each and every unit in the selected cluster is made. Unlike the stratified random sampling technique, there is heterogeneity within the subgroups (clusters) and homogeneity among the subgroups.

Suppose a researcher wants to select a random sample of 600 households out of 10,000 estimated households in one rural *Kebele* for survey. Here the list of the individuals may

not exist and because of this it is difficult to select a direct sample of individual households. Instead, the researcher can select a random sample of a few villages/*Gotes*. If, for example, the average number of households per village/*Gotes* is 50, then 12 villages/*Gotes* comprise the sample. Since the number of households per block varies the actual sample size depends on the village which happens to be selected.

Principles of cluster sampling

The principles that are basic to the cluster sampling method are:

- Cluster should be as small as possible with the cost and limitations of the survey.
- The number of sampling units in each cluster should be approximately same

Merits of cluster sampling

- ⇒ This method provides significant cost gain
- ⇒ It is easier and more practical method which facilitates the field work

Demerits of cluster sampling

- ⇒ Probability and the representatives of the sample is sometimes affected if the number of the clusters is very large
- ⇒ The results obtained under this method are likely to be less accurate if the number of sampling units in each cluster are not approximately the same

?

Hello learner! Suppose you want to estimate proportion of machine-parts in an inventory which is defective. Also assume that there are 20,000 machine parts in the inventory at a given point of time, stored in 400 cases of 50 each. How many clusters can be formed here?

5) Area sampling: If clusters happen to be some geographical sub-divisions (such as regions, *woredas*, *kebeles*, and etc), in that case cluster sampling is better known as area sampling. In other words, cluster designs, where the primary sampling unit represents a cluster of units based on geographical area, distinguished as area sampling. All points of cluster sampling are also applicable to *area sampling*.

6) Multi-Stage Sampling: Multi-stage sampling is a further development of the principle of cluster sampling. The method is generally used in selecting a sample from a very large

area. As the name suggests, multi-stage sampling refers to a sampling technique which is carried out in two or more stages. Here the population is regarded as made of a number of first stage sampling units (e.g. regions), each of which is further composed of a number of second stage sampling units (e.g. zones) which is further composed of third stage sampling units (e.g. *woredas*) and so on till we ultimately reach the desired sampling unit in which we are interested. In each stage, the sampling units are clusters. At each stage, there is a *random selection* and the size of sample may be proportional or disproportional. Thus, the area of investigation is scientifically restricted to a small number of ultimate units which are representative of the whole.

Example: Suppose you want to investigate the opinion of merchants towards the “free market economy” in Ethiopia. You can draw the sample of merchants as follows. You can select a sample of four regions from the country. From the 4 regions, you should list all zones and you can select a random sample of 25 zones. From the 25 zones, you should list all *Woredas* and you can select 100 *Woredas* randomly as your sampling unit. From the 100 *Woredas*, you should list all merchants and you can randomly select 600 merchants to finally contact for survey.

Merits of Multi-stage sampling

- ⇒ Multi-stage sampling is more flexible in comparison to the other methods of sampling. It is simple to carry out and results in administrative convenience by allowing the field work to be concentrated in compact small areas and yet covering a large area
- ⇒ This technique is of great significance in surveys of underdeveloped areas where no up-to-date and accurate frame is generally available
- ⇒ It is reliable and satisfactory technique. Under this method surveys can be conducted with considerable speed

Demerits of multi-stage sampling

- ⇒ Errors are likely to be large in this method in comparison to any other method
- ⇒ It is usually less efficient than a suitable single stage sampling of the same

- ⇒ It involves considerable amount of listing of first stage units, second stage units etc. though complete listing of units may not be necessary

B. Non-probability sampling

Non-probability sampling or judgment sampling is based on the personal judgment. Under this method, a desired number of sample units are selected *deliberately or purposely* depending upon the object of the enquiry so that only the important items representing the true characteristics of the population are included in the sample. Purposive sampling, quota sampling and convenience sampling are considered as non-probability sampling.

1) Purposive sampling: This is also called “*deliberate sampling*” or *judgment sampling*”. When the researcher deliberately selects certain units for study from the universe, it is known as *purposive sampling*. Thus under this method, there is a deliberate selection of certain units on the judgment of the researcher and nothing is left to the chance. But it must be kept in mind that the units selected must be representative of the universe.

Merits of Purposive sampling

- ⇒ Under proper controls and safeguards, this method can prove very useful. It is more economical and less time consuming
- ⇒ This method ensures proper representation of a cross-section of various strata of the universe if the researcher has full knowledge of the composition of the universe
- ⇒ This method is very useful specially when some of the units are very important and their inclusion in the study is necessary
- ⇒ It is a practical method where randomization is not possible

Demerits of Purposive sampling

- ⇒ Under this method, considerable prior knowledge of the universe is necessary which in most cases is not possible
- ⇒ Control and safeguards adopted under this method are sometimes not effective and there is very possibility of the selection of biased samples
- ⇒ Under this method, the calculation of sample errors is not possible. Therefore, the hypothesis framed cannot be tested

2) Quota Sampling: This is a special type of stratified sampling. First of all, in this method, the population is stratified on some basis, preferably on the basis of the characteristics of the population under study. After this, the number of sample units to be selected from each stratum is decided by the researcher in advance (*i.e. selection within the strata is non-random*). This number is known as *quota*, which may be fixed according to some specific characteristics such as income groups, sex, occupation, political or religious affiliations, and etc. The choice of the particular units for investigation is left to the investigators themselves. The investigators try to complete the quota assigned to them from each stratum. The investigators usually apply their judgments in the choice of the sample and try to get required information quickly. In case of non-response due to unco-operative nature of the respondents, the investigator selects some fresh units himself to complete his quota. To show a model of assignment given to an interviewer is shown below:

Table 5.4. Assignment of quota

Sex		Age		Social class	
Male	11	20-40	5	Higher	3
Female	9	41-50	8	Middle	10
Total	20	51-60	4	Lower	7
		61 and above	3	Total	20
		Total	20		

Merits of Quota sampling method

- ⇒ Quota sampling is the combination of stratified and purposive sampling and thus enjoys the benefits of both methods. It makes the best use of stratification economically. Thus, it is a practical as well as convenient method.
- ⇒ If proper controls or checks are imposed on the investigator, quota sampling is likely to give accurate results
- ⇒ It is useful method when no sample frame is available

Demerits of Quota Sampling

- ⇒ This method suffers from the limitations of both stratified and purposive sampling
- ⇒ Control over fieldwork is a very difficult task. Here, the results may be biased because of the personal beliefs and prejudices of the investigator in the selection of the units under study
- ⇒ Since quota sampling is not based on random sampling, the sampling, error as well as standard error can not be estimated
- ⇒ Since the samples are not randomly selected, the sample selected under this technique may not be true representative of the universe.
- ⇒ The bias may also occur due to substitution of unlike sample units.

3) Convenience sampling: It is known as unsystematic, careless, accidental or opportunistic sampling. Under this method, a sample is selected according to the convenience of the investigator. This convenience may be in respect of availability of data, accessibility of the units, etc. Moreover, the method is all about selecting whatever sampling units are available, e.g., a producer/seller may select the customer coming to his shop. This method may be used in the following cases:-

1. When the universe is not clearly defined
2. When sampling units are not clear

Thus the names may be selected from a Telephone Directory, Automobile Registration Records, Industrial or stock Exchange Directories and contracting any one who may have chance to be easily available etc.

4) Snowball sampling: This is a colorful name for technique of building up a *list* or a *sample of a special population* by using an *initial* set of its members *as informants*. For example, if you want to study the problems faced by Ethiopians living in some country, say, you may identify an initial group of Ethiopians through some source like Ethiopian Embassy. Then you can ask each one of them to supply names of other Ethiopians known to them, and continue until you get an exhaustive list from which you can draw a sample or make a census survey.

Merits of Snowball sampling

- ⇒ It is very useful in studying social groups, informal groups in a formal organization, and diffusion of information among professional of various kinds.
- ⇒ It is useful for smaller populations for which no frame is readily available

Demerits of Snowball sampling

- ⇒ It does not allow the use of the probability statistical method. Elements included are dependent on the subjective choice of the original selected respondents.
- ⇒ It is difficult to apply it when the population is very large
- ⇒ It does not ensure the inclusion of all elements in the list.

?

Hello learner! Answer the following two questions.

1. What is the difference between probability sampling and non-probability sampling?

2. List the sub-categories of probability sampling and non-probability sampling methods

7. How a sample should be selected: Steps in sampling design

The sample design is the blueprint for the method of selection of the sample; still there are certain fundamental factors which are to be considered. These are:-

- i. Type of the universe.** What is the nature of the universe from which the sample will be taken? Define the universe whether it is finite or infinite. In finite population, the number of items is certain while in the infinite population; the number of items in it is infinite. E.g. the population of a city, the number of workers in a factory, and etc are examples of a finite population, where as the number of stars in the sky, listeners of a specific radio program, throwing of a dies and etc are the examples that explain an infinite population.
- ii. Sampling unit.** The sampling unit is the individual population object or elements or a group of population object or element of sample. A sampling unit may be a geographical: state, district, village, etc; or a construction unit such as house, flat, etc, or it may be social unit such as family, club, school, etc, or it may be an individual.
- iii. Source list.** This is also known as sampling frame. It contains the names of all items of a universe (in a finite population) from which a sample is drawn. If source list is not available, the researcher has to prepare it. The sampling frame has to be the

representative of the population. Additionally, the source list should have the following qualities:

- a) It should be exhaustive so as to include all items of the universe
- b) It should be accurate and duly verified
- c) It should contain all relevant information about the items, so that if strata are to be made, it should be possible to do so
- d) No item should be repeated
- e) The source list should be in terms of those units in which investigation is to be conducted. For example, for a study of families, this should be in terms of families, not in houses or persons
- f) It should be reliable
- g) It should be available to the investigator

iv. Size of the sample. This is the number of the items to be selected from the universe to constitute a sample. It should be neither excessively large nor too small. It should be optimal. The size of the population should also be taken in to account as it limits the sample size. Budgetary constraints should also be taken in account. Detail discussions on the issues related to sample size are discussed in the new sub-title that follows.

v. Parameters of interest. The specific population parameters including the population mean, proportion, or population standard deviation and/ or others of interest should be taken in to account.

vi. Budgetary constraints. Such constraints have big impact both on the sample size and the sample type. This fact can even lead to the use of a non-probability sample.

vii. Sampling procedure. Finally, the researcher has to be able to decide about the technique used to select the sample. In other words, this is all about the selection of the sampling design. It must be the one which has a smaller sampling error.

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Dear learner! While developing a sample design, what points should be taken in to attention? _____

8. Sample Size

This means the number of sampling units selected from the population for investigation. The size of sample is directly related to standards of accuracy, time, cost and administration of the investigation. The larger the size of sample, greater will be the representation of the items of the investigation or universe in it, but it brings difficulties of managing it. In large-sized samples, intensive study of the units becomes difficult. If the size of the sample is kept small, it will not represent the universe and the degree of accuracy in the results will be limited. Hence, *the size of a sample should be neither too big nor too small*. It should be “optimum”. An optimum sample survey is one which fulfills the requirements of efficiency, representativeness, reliability and flexibility. The size of the sample should ensure minimum of cost and minimum of standard error. A small sample will have a large standard error compared with a large sample of the same characteristics, while the cost will be less in the first as compared to the latter. In determining the size of your sample, you have to be able to answer the following questions.

- At what level of confidence do you want to test your results, findings or hypotheses?
- With what degree of accuracy do you wish to estimate the population parameters?
- What is the estimated level of variation (standard deviation), with respect to the main variables you are studying, in the study population?

The size of the sample which is required in order to make a valid inference about a population is dependent upon several factors such as:-

1. **Homogeneity or heterogeneity of the universe:** - In a universe consisting of homogeneous units, a small sample is suitable, while in a universe consisting of heterogeneity units; a large sized sample is inevitable, for yielding good results.
2. **Number of classes proposed:** - If it is necessary to classify data in a large number of classes, a large size sample should be taken to facilitate analysis of the data. If the size of the sample is small and classes are many, then there will be some classes which may not get due representation in the sample. If the proposed number of classes is more, the size of the sample should be big.
3. **Nature of Study:**-Nature of study also affects the size of a sample. For an intensive and continuous study, small sample will be suitable because such study

in large-sized sample will require more resources. For general survey, the size of the sample should be larger, but for technical studies the size of the sample may be kept small.

4. **Practical Considerations:** - The availability of finance, time and trained personnel are other practical considerations which affect the size of the sample
5. **Standard of accuracy:** - Though it is believed that larger is the size of the sample, great will be the degree of accuracy, yet this is not always so. If a sample is selected by experts through scientific method, a small sample can produce more accurate results.
6. **Type of Sampling:** - In random sampling, greater accuracy in results will be achieved only in a large sample. In a properly drawn stratified sampling, a small sample can give better results.
7. **Nature of units:** - where it is expected that a large number of units will not respond, and then a large sample should be taken.
8. **Size of questionnaire:** - If the size of the questionnaire is large and it contains difficult questions, the size of the sample should be kept small.

Moreover, when deciding upon sample size, the researcher must note the following points:-

1. **The size of the population to be sampled:** - Even if it is difficult to make general rules with out the knowledge of the population, researchers recommend that a sample size of around 30 cases seems to be the bare minimum for studies which require statistical data analysis and which do not involve many sub divisions of data. Although some studies (like laboratory experiments, qualitative studies, etc) use fewer than 30 cases, experienced researchers advise sample of 100 or more.
2. **The actual number of cases from which data are collected:** - In some instances, the data actually gathered might be substantially fewer because of respondents who can not be located, refuse to be interviewed, or return illegible questionnaires. Furthermore, the actual number of cases varies from question to question depending on the percentage of respondents who fail to answer each particular question. For example, respondents who are in hurry to complete the questionnaire may inadvertently omit some items. Other respondents may deliberately leave blank questions that they feel are an invasion of privacy. A question that requires much

thought before it can be answered properly may also be left unanswered. Other questions may be ambiguous or have unclear answer categories, causing respondents to refuse to answer. Moreover, some questions are not applicable to all respondents, then lowering the sample size for that question.

3. The number of cells the sample may have to be subdivided during data analysis:

Every researcher is required to estimate the number of cells in which the sample has to be shared during data analysis of variables and ensure an adequate sample size for each subdivision. The cross tabulation analysis of variables requires the following criteria to be met with:

- a. Each category of an independent variable included in a cross tabulation should contain at least 50 cases, for percentage calculated on the basis of fewer than 50 cases tend to be unreliable. The minimum sample size required to ensure at least 50 cases in each category of the independent variable is obtained by dividing 50 by the proportions(Ps) of total cases you expect in the smallest category of the variables, i.e., $n=50/Ps$.
- b. The expected number of cases in each cell of a table should be at least five. The sample required to ensure a minimum expected cell size of five is obtained by dividing five by the product of the proportions falling in the smallest categories of the variables in the cross tabulation table.

To illustrate those points, consider the sample size required in analyzing a cross tabulation of advertisement-sensitivity of customers and their educational attainment.

Table5.5. Sample size determination in a cross tabulation (advertisement-sensitivity and educational attainment of customers)

Advertisement	Educational attainment			
	College (5%)	Secondary school (15%)	Primary school (40%)	No schooling (40%)
Highly sensitive (40%)				
Poorly sensitive (35%)				
Not sensitive (25%)				

In this particular example, education is considered as the independent variable. Therefore, in order to have a sample large enough to ensure that at least 50 cases will

fall in to the smallest category of the variable “educational attainment” (i.e. in to the category of “College” which accounts for only 5% of the total number of cases)), we need to divide 50 by 5% (i.e., $n=50/P_s= 50/.05=1000$ cases required).

To find the minimum sample size needed to ensure an expected cell frequency of at least five, we must divide 5 by the product of the proportions falling in the smallest categories of the two variables (0.05 for the “College” and 0.25 for “No sensitive”) so that the number of cases will be 500, i.e. $n= 5/(0.05 \times 0.25)= 500$ cases.

4. **Degree of accuracy that is needed for the study:** - If you seek a higher degree of accuracy (such as 0.01), your sample will need to be much larger than if you seek a relatively low degree of accuracy (such as 0.05). You can calculate the sample size needed to measure a given proposition with a given degree of accuracy at a given level of statistical significance by using a simple formula, provided that the total population size is greater than 10,000.

$$n = (z \times p \times q) / (d^2)$$

where:

n =the desired sample size (when population greater than 10,000)

z =the standard normal deviate, usually set at 1.96 (or simply 2.0), which corresponds to the 95% confidence interval

d =degree of accuracy desired, usually set at 0.05 or occasionally at 0.01.

p =the proportion in the target population estimated to have a particular characteristic.

$q = N - p$

For example, if the proportion of a target population with a certain characteristic is 0.50, the Z-statistic is 1.96, and the desired accuracy at the 0.05 level, then the sample size is

$$n = zpq/d^2 = (1.96) (0.5) (0.5)/(0.05^2) = 384$$

Note that if N (the entire population size) is less than 10,000, the required sample size will be smaller. In such cases, calculate a final sample estimate (nf) by using the following formula.

$$nf = n / (1 + (n/N)) \text{ where}$$

nf =the desired sample size (when the population is *less than* 10,000)

n = the desired sample size (when the population is *greater than* 10,000)

N = the estimate of the population size.

9. Characteristics of a good sampling design

- ✚ Sample design must result in a truly representative sample
- ✚ Sample design must be the one which results in a small sampling error
- ✚ Sample design must be viable in the context of funds available for the research study
- ✚ Sample design must be the one in which systematic bias can be controlled in a better way
- ✚ Sample design should be such that the results of the sample study can be applied, in general, for the universe with a reasonable level of confidence.

10. Sampling and Non-Sampling Errors

In the study of sample surveys, it is necessary to keep in mind sampling and non-sampling errors. Since a sample survey implies the study of small proportion of the total universe/population and drawing inference about the population, there would naturally be a certain amount of inaccuracy or errors.

Let us now see the two types of errors in the following part I and II as follows.

I. Sampling Errors

These are the errors created because of chance only. The errors occur because only a partial observation of the universe is made. If a census is taken, sampling errors could be expected to disappear. Thus, *the difference between a sample result (sample statistic such as sample mean and standard deviation) and the result of a complete census (parameter such as population mean and standard deviation) taken under the same circumstances is called sampling error. It is thus the measure of the divergence between statistic and the parameter values (i.e., Sampling error = statistic - parameter).* Thus, such errors reveal themselves in a partial distortion of the statistical derivations such as mean, standard deviation or correlation based on sample observations from those based on complete enumeration.

Sampling error can be measured for a given sample design and size as above. The measurement of a sampling error is also referred to by the term precision of sample plan/results. The difference is also called the accuracy of the sample result. If we can increase the sample size, the precision of the sample result can be improved. However, it should be

noticed that increasing the sample size does have its own limitations (in terms of cost and surfacing of the sampling bias).

The errors which arise due to the use of sampling surveys are known as the sampling errors. Thus, if two or more samples are taken and further if the required information is collected without any mistake, whatsoever, the results obtained from the samples would usually differ to some extent as compared to one obtained from a complete enumeration. These differences constitute the errors due to sampling. For example, the mean of the sample may be different from the population mean by *chance alone*. In other words, even when a sample is random one, it may not be exactly representative of the population from which it is chosen. Samples may be taken from the same population four times. All samples may not be alike. This also gives rise to sampling errors. Thus they are due to the fact that samples are used and to the particular method used in selecting the items from the population. The sampling error decreases with the increase in the size of the sample, and it happens to be a smaller magnitude in case of homogeneous population. If the size of the sample is large, there would be the tendency of the movements to neutralize one another making the data more stable and thus more reliable. Such error is, therefore, also known as non-cumulative or compensating error as distinguished from biased errors or non-compensating or cumulative errors.

Example: Suppose an individual student has scored the following grades in 6 subjects: 70, 75, 80, 85, 90, and 95. If three samples are selected at random (say 75, 85 and 90), the mean of this sample will be $(75 + 85 + 90) / (3) = 83.3$. The mean of another three randomly selected sample (say 80, 85, and 90) will be $(80 + 85 + 90) / (3) = 85$. But the mean of the population is computed to be 82.5, i.e., $[(70 + 75 + 80 + 85 + 90 + 95) / (6)] = 82.5$. From these, we can observe the sampling errors of 0.8 (i.e., $83.3 - 82.5$) and 2.5 (i.e., $85 - 82.5$).

II. Non-sampling Error (Systematic Bias)

Systematic bias results from errors in the sampling procedures, and it *can not be reduced or eliminated by increasing the sample size*. Rather, the causes for these errors can be detected and corrected. Such error occurs because of the human mistakes and not by chance.

Non-sampling errors include bias and mistakes at the stages of ascertainment and processing of data, i.e., clerical errors, computation of errors, or an incorrect answer to an unclear question. Naturally, this type of error can occur in any survey, whether it is a complete enumeration or sampling. The bias/sampling error can be **caused** by the following factors.

i. Faulty Selection of the Sample:- Faulty selection of the sample may give rise to bias in a number of ways, such as:-

- a. *Purposive Sampling*: - Purposive and/or deliberate sampling method may be adopted in place of a simple random sampling method. As a result of such subjective selection, some errors are bound to arise, and they are known as biased errors. Such errors are also known as *cumulative errors or non-compensating errors*. Errors due to bias increase with an increase in the size of the sample. Moreover, the investigator's desire or motive to obtain a certain result from the sample survey may influence his/her selection of the sample, consciously or unconsciously.
- b. If the selection of the sample is *haphazard*, the chance of bias error is great.
- c. *Substitution*: That is, substitution of the selected item in the sample by another. Due to non-availability of the person chosen in the sample, another may be interviewed, who may not have the same characteristics as the original one. This will introduce substitution bias in the sample and invalidate the result of the enquiry.
- d. *Incomplete investigation or non-response*, i.e. Failure to cover the whole of the sample. If the selected person is not visited for the first or for the second time, or, if necessary, the third time, to elicit information, bias may arise.
- e. *An appeal to the vanity of the person interviewed* may give rise to yet another kind of bias. For example, the question, 'have you read that book?' is such that most of the students would succumb to vanity and answer 'yes'.

Example on inappropriate sampling: Suppose a researcher plans to conduct his study on the attitude of adults towards education in Ethiopia. If a certain political party membership lists are used as sampling frame, the sample to be selected can not represent the general adult population in Ethiopia, as it has been selected from a biased source.

ii. Bias due to faulty collection of data: During the process of collecting the actual information in a survey (whether sample or census), certain inaccuracies and mistakes may creep in. These may arise due to the following reasons:-

- Negligence or prejudice of the person collecting information either in asking the questions or in recording the answers
- Negligence or prejudice or lack of knowledge or forgetfulness on the part of the person furnishing information
- Poorly designed questionnaire
- Unorganized collection procedure, faulty editing or coding of response

The danger of such error is, however, likely to be greater in sampling work, since the units measured are often smaller.

iii. Accessibility bias: -In a considerable number of research studies, researchers tend to select respondents who are most accessible to them. This can create bias.

iv. Defective measuring device: - In case, for example, the questions are not well phrased; they can not be fully understandable by the respondents. Accordingly, the answers obtained can not be accurate. Moreover, some items can be miss-measured due to the errors in the procedures of observation, interviewing, coding and etc.

Example: If a test is conducted on a sample of students for general mental ability, and if the researcher unintentionally allows an extra time of 5 minutes for the test, there will probably be a systematic tendency for the sample statistic to be higher than it should be, therefore, there will be a bias in the measurement of mental ability.

v. Bias in Analysis: Faulty methods of analysis of data may also introduce bias.

Now let us see the mechanisms how to avoid a bias.

Avoidance of bias

If possibilities of bias exist, the conclusions drawn from the sample cannot be regarded as fully objective. The first essential of any sampling or census procedure must, therefore, be the elimination of all sources of bias. To avoid bias in the selection process is to draw the

sample either entirely at random. However, in certain cases systematic sampling may be adopted.

Bias arising from substitution should not be allowed to enter the survey. The survey once selected should be strictly adhered to throughout the enquiry. The investigator should try to investigate the entire sample.

Bias arising from faulty collection of data may be removed in a number of ways. For instance, the information relating to age, income, language etc may be wrongly entered either by the respondent or by the enumerator. The statistician should be able to corroborate such information to put a check on such wrong reporting.

Chapter Summary

Dear learner! Now let us summarize what you have already gone through in chapter five. In this chapter, you learnt about sampling design and population, and accordingly about census and sampling methods. While census method deals with the investigation of the entire elements of the population, sampling enables you to learn about the population on the basis of a sample drawn from it. Here, population implies the aggregate number of the units pertaining to a study. But Sample refers to part of a population from which you obtain the information about the population.

Broadly, speaking, sampling technique is divided in to two: Probability sampling and non-probability sampling techniques. Under probability sampling technique, each unit of a sample has some definite pre-assigned probability of being chosen in the sample. But in non-probability sampling technique, a desired number of sample units are selected *deliberately or purposely* to incorporate only the important items representing the true characteristics of the population.

The probability sampling technique can take on different categories such as random sampling, stratified random sampling, systematic sampling, cluster sampling, area sampling, and multi-stage sampling techniques. In non-probability sampling; purposive sampling, quota sampling, convenience sampling, and snowball sampling techniques are employed. However, in the study of sample surveys, sampling and non-sampling errors can possibly surface. Sampling errors are created because of chance only. However, in systematic bias errors result from such factors as human mistakes including faulty selection of the sample, faulty collection of data , accessibility of bias, using defective measuring device, bias arising in analysis.

The sampling error can be reduced by increasing the size of the sample. However, the non-sampling error *can not be reduced or eliminated by increasing the sample size*. Rather, such techniques as selection of the sample at random, investigating the sample in its entirety, and checking for wrong reporting can be employed.

Exercises**Part I. Multiple Choice Questions (The answers are provided after the last question.)**

1. The process of drawing a sample from a population is known as _____.
 - a. Sampling
 - b. Census
 - c. Survey research
 - d. None of the above
2. A number calculated with complete population data and quantifies a characteristic of the population is called which of the following?
 - a. A datum
 - b. A statistic
 - c. A parameter
 - d. A population
3. A sampling frame is:
 - a. A summary of the various stages involved in designing a survey
 - b. An outline view of all the main clusters of units in a sample
 - c. A list of all the units in the population from which a sample will be selected
 - d. A wooden frame used to display tables of random numbers
4. A simple random sample is one in which:
 - a. From a random starting point, every n^{th} unit from the sampling frame is selected
 - b. A non-probability strategy is used, making the results difficult to generalize
 - c. The researcher has a certain quota of respondents to fill for various social groups
 - d. Every unit of the population has an equal chance of being selected
5. When each member of a population has an equal chance of being selected, this is called:
 - a. A nonrandom sampling method
 - b. A quota sample
 - c. A snowball sample
 - d. An Equal probability selection method
6. Which of the following techniques yields a simple random sample?
 - a. Choosing volunteers from an introductory management class to participate
 - b. Listing the individuals by ethnic group and choosing a proportion from within each ethnic group at random.
 - c. Numbering all the elements of a sampling frame and then using a random number table to pick cases from the table.
 - d. Randomly selecting schools, and then sampling everyone within the school.
8. Which of the following statements are true?
 - a. The larger the sample size, the greater the sampling error
 - b. The more categories or breakdowns you want to make in your data analysis, the larger the sample needed

- c. The fewer categories or breakdowns you want to make in your data analysis, the larger the sample needed
 - d. As sample size decreases, so does the size of the confidence interval
9. Which of the following sampling techniques is an equal probability selection method (EPSEM) in which every item in the population has equal chance of being selected?
- a. Simple random sampling
 - b. Systematic sampling
 - c. Proportional stratified sampling
 - d. Cluster sampling
 - e. All of the above are EPSEM
10. Which of the following is not a form of nonrandom sampling?
- a. Snowball sampling
 - b. Convenience sampling
 - c. Quota sampling
 - d. Purposive sampling
 - e. All
13. How often does the Ethiopian Census Bureau (the CSA) take a complete population count?
- a. Every year
 - b. Every five years
 - c. Every ten years
 - d. Twice a year
14. People who are available, volunteer, or can be easily recruited are used in the sampling method called _____.
- a. Simple random sampling
 - b. Cluster sampling
 - c. Systematic sampling
 - d. Convenience sampling
16. In which of the following nonrandom sampling techniques does the researcher ask the research participants to identify other potential research participants?
- a. Snowball
 - b. Convenience
 - c. Purposive
 - d. Quota
20. It is recommended to use the whole population rather than a sample when the population size is of what size?
- a. 500 or less
 - b. 100 or less
 - c. 1000 or less
 - d. you should always use a sample
21. Which of the following sampling methods is the best way to select a group of people for a study if you are interested in making statements about the larger population?
- a. Convenience sampling
 - b. Quota sampling
 - c. Purposive sampling
 - d. Random sampling

22. _____ is a set of elements taken from a larger population according to certain rules.
- a. Sample b. Population c. Statistic d. Element
23. Determining the sample interval (represented by k), randomly selecting a number between 1 and k , and including each k^{th} element in your sample are the steps for which form of sampling?
- a. Simple Random Sampling c. Systematic Sampling
b. Stratified Random Sampling d. Cluster sampling
- 24 The nonrandom sampling type that involves selecting a convenience sample from a population with a specific set of characteristics for your research study is called _____.
- a. Convenience sampling c. Purposive sampling
b. Quota sampling d. Snowball sampling
25. The sampling error is a statistical measure of:
- a. The normal distribution of scores around the sample mean
b. The extent to which a sample mean is likely to differ from the population mean
c. The clustering of scores at each end of a survey scale
d. The degree to which a sample has been accurately stratified
26. What effect does increasing the sample size have upon the sampling error?
- a)** It reduces the sampling error **c)** It has no effect on the sampling error
b) It increases the sampling error **d)** None of the above
27. Which of the following is *not* a type of non-probability sampling?
- a)** Snowball sampling **c)** Quota sampling
b) Stratified random sampling **d)** Convenience sampling
28. Snowball sampling can help the researcher to:
- a)** Access deviant or hidden populations
b) Overcome the problem of not having an accessible sampling frame
c) a & b are the answers

Chapter Six

Data Collection

6.1. Introduction

Dear learner! In the preceding chapter, you learnt about sampling and sampling design. The next activity in the research process is data collection and methods. Accordingly, in this particular chapter, you will learn about data types and the rationale for each, preliminaries to data collection, sources of data, and finally the methods of data collection with their pros and cons will be dealt in-depth.

6.2. Lesson objectives

Up on the successful completion of the chapter, you will be able to:

- ✓ Make a distinction between primary data and secondary data
- ✓ Understand when and why to use primary data and/or secondary data
- ✓ Confidently explain the preliminaries to the collection of data
- ✓ Differentiate the types of enquiry
- ✓ Determine the requirements of a statistical unit
- ✓ Identify the types of statistical units
- ✓ Identify the various sources of data
- ✓ Investigate the sources of data and methods of data collection with their pros and cons.

6.3. Primary Data and Secondary Data

Data are facts, figures and other relevant materials serving as bases for a study and analysis. Data are of two types: primary and secondary data. Primary data are original observations collected by a researcher *for the first time* for any investigation and used by him/her in the statistical analysis whereas secondary data are *collected by others and used by others*. The difference between primary and secondary data is only the degree of detachment with the original source. The data, which are primary one time, may be secondary at another. Once the primary data have been used, it loses its original character and becomes secondary. Such secondary data are mostly published in newspapers, periodicals and journals.

6.3.1. Choice between Primary and Secondary Data

The choice between primary and secondary data mainly depends up on the nature, objectives and scope of inquiry, availability of time and money, degree of accuracy desired and the status of the investigator. The primary data are more reliable in the **face**, but the secondary data can be relied only by examining the source from which they have been obtained, their true significance, completeness and methods of collection. Sometimes in a certain investigation, both primary and secondary data are used as supplement to one another. It may be pointed out that, today, in a large number of statistical inquires secondary data are generally used because fairly reliable published data on a large number of diverse fields are now available. In fact, primary data are collected only if there do not exist any secondary data suited to the investigation under study.

The following points are noteworthy for the preference of primary source in comparison to the secondary source.

1. The primary source gives data in greater details compared to secondary source. The secondary source often omits part of the information.
2. In the secondary source, there is possibility of mistakes due to errors in transcription made then the figures were copied from the primary source.
3. The primary source includes definitions of terms and units used. It is essential that the investigator understands the meaning of units in which data are recorded.
4. The primary source also includes a copy of the schedule used in data collection together with the prescription of the procedure used in selecting the sample and the size of the sample.
5. The availability of finance also influences the method to be adopted in the collection of data. For example, research scholars mostly depend upon secondary data due to shortage of money.
6. The availability of trained investigators also affects the choice of the method to be employed in data collection. If trained investigators are available, primary method should be adopted for the collection of data. In case of non-availability of trained investigators, it is recommended to adopt secondary method of data collection.
7. The objective and the scope of the enquiry also determine the method to be adopted in the collection of data. The secondary method must suit the objective and the scope of the enquiry.

6. 4. Preliminaries to the Collection of Data

The collection of data is by no means the first step for any statistical investigation. Before we embark up on the collection of data for a given statistical enquiry, it is imperative to examine carefully the following points as preliminaries to data collection.

1. Objectives and Scope of the Inquiry

The first and foremost step in organizing any statistical enquiry is to define, in clear and concrete terms, the objectives of the enquiry. This is very essential for determining the nature of the data to be collected and also the statistical techniques to be employed for the analysis of the data. The objectives of the enquiry would help in eliminating the collection of irrelevant information, which is never used subsequently and also reflect up on the uses to which such information can be put. In the absence of the purpose of the enquiry being explicitly specified, we are bound to collect irrelevant information and also omit some important information, which will ultimately lead to fallacious conclusions and wastage of resources.

Furthermore, the scope of the enquiry will also have a great bearing up on the data to be collected and also the techniques to be used for its collection and analysis. Scope of the enquiry relates to the coverage with respect to the type of information, subject matter and geographical area. For instance, if we want to study the cost of living index numbers, it must be specified if they relate to particular city or state or whole of Ethiopia.

2. Types of Inquiry

Statistical enquiry may be classified under the headings which are discussed below:-

a. Open or confidential: A statistical enquiry is termed open when its findings are public.

On the contrary, when findings of an enquiry are kept secret, it is called confidential.

b. Original or Repetitive: An enquiry, which is conducted for the first time, is called original. On the other hand, an enquiry which is conducted in continuation is called repetitive.

c. Official or Non-official: An enquiry which is conducted on behalf of the government is named official enquiry whereas an enquiry conducted by a private authority or an institution is called non-official.

d. Census or Sample: In a census type enquiry, data is collected about each and every individual item in the population whereas in a sample enquiry, only a few selected items in the population are studied. On the basis of information gathered from these items, conclusions are drawn about the nature of the population.

Now a question crops up, which type of enquiry is to be used in data collection? Here there is no hard and fast rule which may be laid down for selecting the type of enquiry.

3. Statistical units to be used

A well-defined and identifiable object or a group of objects with which the measurements of count in any statistical investigation are associated is known as *statistical unit*. For example, in a socio-economic survey the unit may be an individual *person, family, a household or a block of locality*. A very important step before the collection of data begins is to define clearly the statistical units on which the data are to be collected. In a number of situations, the units are conventionally fixed like the physical units of measurements such as meters, kilometers, quintals, hours, days, week, etc. which are well defined and do not need any elaboration or explanation.

However, in many statistical investigations, particularly relating to socio-economic studies, arbitrary units are used which must be clearly defined. This is imperative since in the absence of clear-cut and precise definitions of the statistical units, serious errors in the data collection may be committed. This will ultimately lead to fallacious conclusion.

?

Dear learner! 1. What do you understand by statistical unit? _____

2. Briefly discuss the preliminaries required to collect data? _____

Requirements of a Statistical Unit

The following are the main requirements which a statistical unit should possess:-

- 1) *It must be specific and unmistakable:* - A statistical unit employed in the collection of data must be specific and unmistakable. Such a unit will be interpreted in same sense by different persons and the collected data could be **homogeneous** in nature. If the unit is ill-defined, different persons would interpret differently to suit their own way of thinking and the collected data would be **heterogeneous** in nature and unfit for comparison.
- 2) *It should be stable:* - The second requirement of a good statistical unit is stability. If there are continuous changes in the definition of unit, the data collected at different times would become unfit for comparison. The utility of such data is lost. Therefore, the unit must be stable for meaningful comparison.

3) *It should be homogeneous*:-A statistical unit employed in the collection of data must be homogenous throughout the enquiry. It must not imply different characteristics at different times and at different places. If a single unit is not applicable to all cases under study, then this problem can be solved by sub-dividing data in uniform groups.

4) *It must be appropriate to enquiry and capable of correct ascertainment*: - The statistical unit selected for the collection of data should be appropriate to the enquiry and capable of correct ascertainment. For example, when compiling labor statistics, it is necessary to select the unit appropriate to the information required. E.g. workers engaged directly on production, those employed in indirect factory services, those in administrative office or those in sales department.

Types of Statistical Units

There are two types of statistical units:

1) Units of Collection: Such units are concerned with the collection of data. They take the form of counts and measurements. Examples of unit of measurement are production of wheat in quintals, height in inches or weight in kilograms and examples of units of counting are the number of persons in a family, number of rooms in a house or number of houses. Units of collection may further be divided in to two categories, namely:-

1.1 . *Simple unit*: A simple unit is one which describes single determining characteristics such as wage, weight, ton, yard, etc. Such units are not difficult to define and are in common use. But care must be taken in interpreting some of them. For example, a wage may be defined as money wage or kind wage.

1.2 . *Composite unit*:-A composite unit is one, which is formed by the addition of some qualifying conditions to a simple unit. For example, electric power is measured in units of kilowatt-hours while railway transport is expressed in ton-miles are equal to the number of tons multiplied by the number of miles carried. Other examples of composite units are educated-unemployed, factory- building, man-hours or passenger miles.

2) Units of Analysis: Statistical data are collected for analyzing the behavior of the variable under study. Therefore, units of analysis are those with which statistical data are analyzed and interpreted. These units are employed for making comparisons between two or more series of variables. Suppose we are told that the sales of firm “A” has increased by

8% and firm “B” by 15% in 1994. Here we can easily conclude that firm “B” made more progress than firm “A”.

?

Dear learner! What is the difference between unit of collection and units of analysis? ____

6.5. Sources of Data

There are two sources of data: Primary and secondary sources.

1. Primary sources are original sources and the data collected are first-hand or primary data
3. Secondary sources contain data which have been collected by some other agency for other purpose; the data to be used from such sources is *Second-hand or secondary data*.

Now the question crops up, which type of data -primary or secondary- should be employed for studying the behavior of variables? This decision cannot be undertaken independently but depends upon the purpose and scope of the enquiry. In most of the enquiries relating to current problems, it is recommended to use primary-data than secondary data.

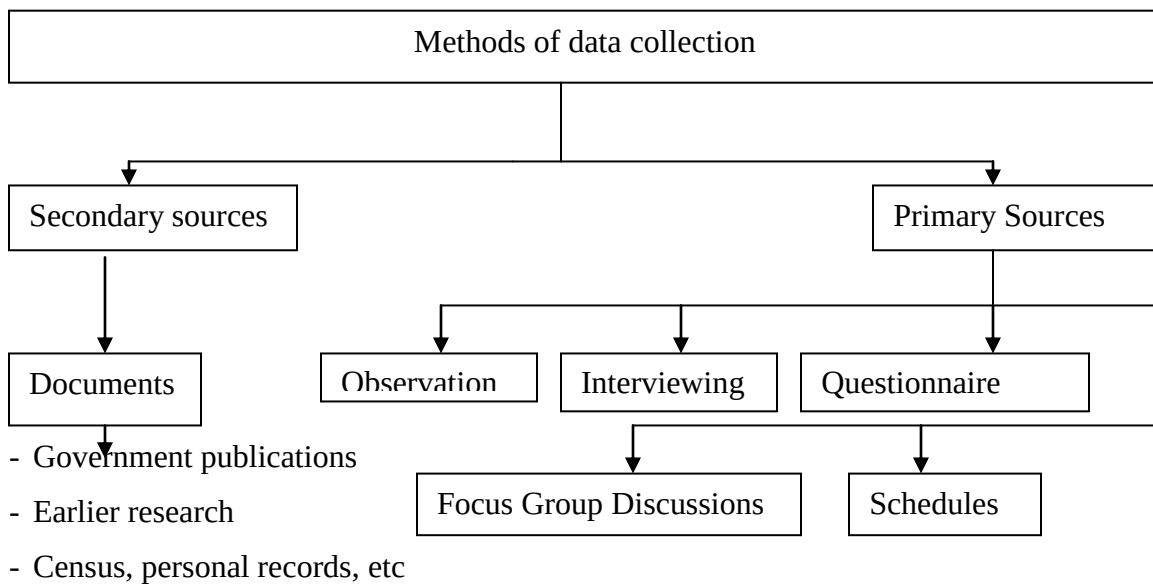
Moreover, it is always necessary to examine the following questions.

- i. Who collected the data?
- ii. How the data were collected?
- iii. Was the correct method used for the data collection?
- iv. Whether the collected data are sufficient for the present enquiry or not
- V. Whether the purpose and the scope of the past enquiry were the same as that of the present enquiry.

6.6. Methods of Data Collection

If primary data are to be collected, a decision has to be taken whether (i) census method or, (ii) sample technique is to be used for data collection. A choice between the census method and sample method is to be made depending up on the objectives and scope of the survey, the limitations of the resources in terms of time, money, manpower, etc and the degree of accuracy desired. In case of sample method, the size of the sample and the technique of sampling, like simple random sampling, stratified random sampling, systematic sampling, and etc are to be decided. Broadly, data collection methods are presented as follows.

Fig 6.1. Data collection methods



6.6.1. Methods of Collecting Primary Data

When the preliminary stages relating to a statistical inquiry have been satisfactorily arranged and the sources of data are listed, the next stage is the actual collection of the data. The choice of the primary data collection method depends upon the purpose of the study, the resources available and the skill of the researcher. The socio-economic-demographic characteristics of the study population play an important role in selecting the method of data collection. Whatsoever method is used, the way the purpose of the study is explained to the respondents also determines the quality of the data collected.

The following methods of collecting primary data can be distinguished:-

I. Observation

Observation is a purposeful, systematic, and selective way of watching and listing to an interaction or phenomenon as it takes place. There are many situations in which observation is the most important tool of data collection. For example, when you want to learn about the interaction in a group, study of dietary patterns of a population, ascertain the functions performed by a worker, or study the behavior or personality traits than the perception of an individual. When we can not elicit accurate information through questioning because the respondents may not co-operate or because they are unaware of the answer, observation method is still important. Or when subjects are so involved in the interaction that they are unable to provide objective information about it, observation is the best approach to collect the required information.

Types of Observation

There are two types of observation:

- ❖ Participant observation, and
- ❖ Non-participant observation

Participant observation is when a researcher participates in the activities of the group being observed in the same manner as its members, with or without their knowing that they are being observed.

Example: Suppose you want to examine the reactions of the general population towards people in wheel chairs. To study their reactions, you can sit in a wheel chair yourself.

Or else, if you want to study the life of prisoners, pretend to be a prisoner to observe.

Non-participant observation, on the other hand, is when the researcher does not get involved in the activities of the group but remains a *passive observer*, watching and listening to its activities and drawing conclusions from this.

Example: If you want to investigate the functions carried out by nurses in a hospital; you have to thoroughly watch, follow, and record the activities as they are carried out. Ultimately, you can reach a certain conclusion regarding the functions.

Problems with using observation as a method of data collection

- When the groups or individual become aware that they are being observed, they may change their behavior either positively or negatively. The change in the behavior of the individuals or groups because they recognized that they are being observed is known as the *Hawthorne Effect*. In this situation, what is observed does not represent their normal behavior.
- There is always the possibility of the observer bias.
- The interpretations drawn from observations may vary from observer to observer.
- There is a chance of incomplete observation and/or recording, which varies with the method of recording. An observer may devotedly watch but at the expense of detailed recording or the vice versa.

Situations in which observations can be made

Observations can be made under two conditions:

1. Natural, and
2. Controlled

Observing a group in its natural operation rather than intervening in its activities is classified as observation under natural conditions. Introducing a stimulus to the group for it to react to and observing the reactions is called controlled observation.

The Recording of Observation

There are many ways of recording observation, the selection of which depends upon the purpose of the observation, the complexity of the interaction and the type of population being observed. The methods of recording used include the following:

a. Narrative:-In this method, the observer records the description of the interaction in his/her words. Usually, he/she makes brief notes while observing and then makes detail notes soon after the observation. Besides, some researchers may interpret the interactions and arrive at a certain conclusion.

 **Advantage of narrative method:** - It provides a deeper insight in to the interaction.

 **Disadvantage of narrative method:-**

- ⇒ An observer may be biased in his/her observation. As a result, the interpretations and the conclusions drawn will be biased.
- ⇒ If his/her attention is on observation, a part of recording may be missed and when recording a part of the interaction may be missed.
- ⇒ With different observers, the comparability of narrative recording can be a problem.

b. Scales:-Some observers may prefer to develop a scale in order to rate various aspects of the interaction or phenomenon. The recording is done on the scale. The scale can have many directions depending on the purpose of the observation.

Problems/disadvantages of Scales:

- ⇒ it does not give detail information about the interaction
- ⇒ The observer may avoid the extreme positions on the scale, using mostly the central part. The error this tendency creates is termed as *error of central tendency*.
- ⇒ Some observers may prefer certain sections of the scale in recording the interaction. This phenomenon is known as the elevation effect.
- ⇒ When the way an observer rates an individual on one aspect of the interaction influences the way he/she rates that individual on another aspect of the interaction, an error is introduced. This type of effect is called *halo effect*.

Illustration: For more understanding of the method, look at the following figure in which the scale is designed to record the nature of interaction with in a group. (Here, there are three directions: positive, negative, and neutral).

Fig. 6.2. Example on Scale method of recording observation

Aspects of Interaction	Positive ←					Neutral	→ Negative				
Participation	5	4	3	2	1	0	1	2	3	4	5
Rapport	5	4	3	2	1	0	1	2	3	4	5
Confidence	5	4	3	2	1	0	1	2	3	4	5
Aggressiveness	5	4	3	2	1	0	1	2	3	4	5
Withdrawnness	5	4	3	2	1	0	1	2	3	4	5
Friendliness	5	4	3	2	1	0	1	2	3	4	5
Aloofness	5	4	3	2	1	0	1	2	3	4	5

c. Categorical recording: sometimes an observer may decide to use categories to record his/her observations. For example: passive/active (two categories); introvert/extrovert (two categories); always/sometimes/never (three categories); strongly agree/agree/uncertain/disagree/strongly disagree (five categories).

Disadvantage:-the method has the same problem as those associated with scales.

d. Recording on Mechanical devices: - This includes use of video and audio tapes.

Advantage: - the observer can see the interaction in a number of times before drawing any conclusion and even can invite other professional to reach a more objective conclusion.

Disadvantage: Some may feel uncomfortable or may behave differently before a camera. So the interaction may not be a true reflection of the situation.

?

Dear learner! Identify and briefly discuss the ways used to record what observed is. _____

II. Interview

Interviewing is a commonly used method of collecting data through communication or interaction in which the subject or interviewee gives the needed data verbally in a face-to-face situation or through telephone calls.

Interview Designs

- i. Structured Interview:** - A structured interview is one in which the procedure to be followed is standardized and is determined in advance of interview. It involves the use of a set of predetermined questions and of highly standardized recording techniques. In other words, in this method of data collection, the same types of questions are presented in the same order to each interviewee, and wording of instructions to the interviewees is specified. The interviewer has no freedom to rephrase questions, and extra ones, or change the order in which the questions have to be presented. Thus, the interviews in a structured interview follows a rigid procedure laid down, asking questions in a form and order prescribed.
- ii. Unstructured Interview:-** This is characterized by a flexibility of approach to questioning. Although the series of questions to be asked and the procedure to be followed are decided upon before hand, the interviewer is largely free to arrange the form, order and timing of the questions. He can thus rephrase the questions, modify them, and add some new

questions to his list. But this sort of flexibility results in lack of comparability of one interview with another and the analysis of the unstructured responses becomes much more difficult and time consuming than that of the structured responses obtained in case of structured interviews. The unstructured interview also demands deep knowledge/skill of the interviewer. Unstructured interviews are usually labeled as “focused”, “depth”, and “non-directive”. The focused interview aims at some particular event or experience rather than on general lines of inquiry about an event. The depth interview is searching and giving emphasis to psychological and social factors. The non-directive interview permits much freedom to the interviewees to talk about the problem under investigation.

Unstructured interview is found to be very important technique of data collection in case of exploratory/formulative studies. But in case of descriptive studies, we quite often use structured interview technique because of its being economical, providing a safe basis for generalization and requiring relatively less skill (knowledge) on the part of the interviewer.

Methods of Interview

The main techniques of interview are as follows:

- i. Personal Interviews
- ii. Telephone Surveys

i. Personal Interviews

An interview is called personal when the interviewer asks the questions face-to-face with the interviewee. It is a two way communication initiated by an interviewer to obtain data from the respondent. The respondent is asked to provide information with little hope of any immediate or direct benefit from this co-operation. Personal interviews can take place in the home, at a shopping mall, on the street, outside a movie theater or polling place, and so on.

Advantages: high in-depth and detail information, the interviewer has more control, they can set up control question, and the interviewer can make adjustment (i.e., flexibility) to the language of the interviewee.

Limitations: costly both in terms of time and money, biases in the form of sampling, non response and response errors may creep into the process.

ii. Telephone Interviews

This method is a non-personal method used to collect data by contacting respondents on telephone. Though it is not widely used, it plays important part in industrial surveys,

especially in developed nations.

Advantages:

- ⇒ Moderate cost i.e., cost in administration and travel can be minimized.
- ⇒ Faster completion of the study
- ⇒ Responses can be directly entered to the computer
- ⇒ Fewer interviewer biases.
- ⇒ Respondents not willing to reveal their identity would be encouraged with this method.

Disadvantages: Respondents must be having a telephone facility, the length of interview is short or limited, the adjustments and advantages which are available with the face to face kind of interview is lacking in this method.

 Techniques of Interviewing

The techniques include preparation for interview, interview conduction, and data recording.

i. *Preparation for Interview:* - It is quite necessary to plan for the interview carefully to make it effective. Preparation focuses on recruitment of interviewers, the type of data collected, and the procedures used for data collection. Large gaps (in terms of social or physical characteristics) between an interviewer and interviewee can adversely affect the quality of data collected. Therefore, as much as possible, the interviewers need to be from the same tribe or race, of equivalent age, sex, and of the same social status as that of the respondents. They should also dress neutrally so as not bias the respondents' answers and dress unobtrusively so that the respondents' emphasis will be on the interview rather than the interviewers' appearance.

It is also advisable to try out the interview on another person before using it for actual study to reveal the deficiencies that should be corrected before the actual interview.

ii. *Conducting the interview:* - During the execution of the interview, a harmonious relationship between the interviewer and the interviewee is essential as a good rapport helps the interviewee to feel at ease and express him/herself willingly and freely. To establish a good rapport, an interviewer should note the following rules:-

- ✓ Tell the respondent who the respondent is and whom s/he represents, including showing the identification card or letters of assignment.

- ✓ Tell the respondent what the interviewer is doing, and what the study is about to stimulate his/her interest.
 - ✓ Tell how the respondent was chosen, ensuring that he/she was not chosen personally for harassment or intimidation but was chosen in impersonal way merely because a cross section of the population is needed.
 - ✓ Use a positive approach. Assume that the respondent will not be too busy for the interview. Say, “I would like to come in and talk with you about this” rather than, “May I come in?” or “Should I come back later?” or “Do you have time now?” or any other approach that gives the respondent a chance to say no.
 - ✓ Ask non-threatening questions at the beginning
 - ✓ Ask only one question at a time
 - ✓ Repeat a question if necessary
 - ✓ Try to make sure that the subject understands the question
 - ✓ Listen carefully to the subject’s answer
 - ✓ Observe the subject’s facial expressions, gestures, and tone of voice
 - ✓ Allow the subject sufficient time to answer the question, but do not let the interview drag
 - ✓ Avoid suggesting answers to the questions
 - ✓ Do not show signs of surprise, shock, anger, or other emotions if unexpected answers are given.
 - ✓ Have neutral attitude with respect to controversial issues during the interview process.
 - ✓ Take note of answers that seem to be vague, ambiguous, or evasive and ask for their clarification (if necessary).
 - ✓ In unstructured interview, ask additional questions to follow up clues or to obtain additional information
 - ✓ Use tact and skill in getting the subject back to an area of inquiry when he has stayed too far from the original question. You must wind up the interview before the respondent becomes tired.
- iv. *Recording of the interview*:-In this case, the interviewer may make use of a schedule, a structured format, rating scale, or a tape recorder to record the responses of the interviewee. The use of tape recorder during the conduct of the interview eliminates the omissions, distortions, elaborations and other modifications of data usually found in written interview, and it provides an objective basis for evaluating the adequacy of the

interview data in relation to the performances of the interviewer. It also allows the interviewer to devote his attention to the interviewee and to save his time utilized for recording during and after the interview. The conscious and unconscious selection of the interview data by the researcher can also be avoided by using tape recording mechanism. If tape-recorder is not available, the interviewer can use note taking.

NB: Note the differences of the following three terminologies.

a) Interview- A method of data collection

b) Interviewer- the person who asks respondents to respond to the interview questions

c) Interviewee/subject- A respondent who gives responses to the interview questions.

?

Dear learner! Briefly explain the various methods of conducting an interview_____

III. Questionnaire

A questionnaire is a written list of questions, the answers to which are recorded by respondents. In a questionnaire, the respondents read the questions, interpret what is expected and write down the answers. The only difference between a questionnaire and an interview schedule is that it is the interviewer who asks questions (and if necessary, explains it), and records the respondents' responses on an interview schedule. But in the case of a questionnaire, no one explains the meaning of the questions to respondents. So it is important that the questions should be clear and understandable. The layout of a questionnaire has to be easy to read and attractive to the eye and the sequence of questions should be easy to follow. It has to be developed in an interactive style, i.e., the respondents should feel as if some one is talking to them. A sensitive question or a question the respondents feel hesitant about answering should be prefaced by an interactive statement explaining the relevance of the question. Use of a different font for such statements is also advisable.

Methods of Administration and Data Collection through Questionnaire

1. Mailed questionnaire to be filled in by the respondent: - The collection of data through mailed questionnaire is one of the most popular methods used these days. A

questionnaire contains many questions pertaining to the field of inquiry and provides space for answers. It is sent to the informant by post. The informant sends back the questionnaire duly filled in within the stipulated time mentioned in the covering letter sent with the questionnaire.

➤ **Advantages of Mailed questionnaire**

- ⇒ Mail surveys are among the least expensive.
- ⇒ Persons who might otherwise be inaccessible can be contacted
- ⇒ The questionnaire can include pictures to add value.
- ⇒ Mail surveys allow the respondent to answer at their leisure. For this reason, they are not considered as invasive as other kinds of interviews.

➤ **Disadvantages of Mailed questionnaire:**

- ⇒ This method is used if you have the names and addresses of the target population
- ⇒ Respondents can take more time to collect facts.
- ⇒ Non-response error is expected
- ⇒ Large amount of information may not be acquired.
- ⇒ This method has the least response rate. For example, in populations of lower educational and literacy levels, response rates to mail surveys are often too small to be useful. Even in well-educated populations, response rates vary from as low as 3% up to 90%. But as a rule of thumb, the best response levels are achieved from highly-educated people and people with a particular interest in the subject (which, depending on your target population, could lead to a biased sample).

2. Questionnaires in charge of enumerators:-Sometimes the questionnaire is filled by the enumerator or the interviewer in which case it is known as *schedule*. In the case of a schedule, the enumerator fills the answers to the questions asked by him/her. This eliminates to a large extent the problem of non-response present in the case of mailed questionnaire method.

3. Collective administration: This method involves obtaining a captive audience such as students in a class room, people attending a meeting, participants of a program, etc. As you have personal contact with the group/gathering, you can explain the purpose of the study, its relevance and importance, and can clarify the questions that respondents may have. Also, there will be high response rate with the method.

3. *Administration in a public place* including shopping center, health center, hospital, school, or pub. Apart from being time consuming, this method has same advantage with *collective administration* method.

Designing a questionnaire

There is no one generally accepted method for designing a questionnaire. Various research texts suggest procedures ranging from 4 to 14 sequential steps. Questionnaire design, no matter how formalized, requires a measure of art with a good dose of humility. In designing a questionnaire, presumption must be set aside. Although for simplicity of format, the sequence for developing a questionnaire is given on a step-by-step basis rarely is a questionnaire constructed in such a routine way. Quite often it is necessary to skip from one step to another and to look back through a previous series of steps. The following steps represent a sequential procedure that need to be considered for the development of an effective questionnaire.

1. Determine the specific information needed to achieve the research objectives.
2. Identify the sources of the required information
3. Choose the method of administration that suits the data required and sources
4. Determine the types of questions to be used and form of response.
5. Develop the specific questions to be asked
6. Determine the sequence of the questions and the length of the questionnaire
7. Predetermine coding
8. Pretest the questionnaire.
9. Review and revise the questionnaire.

A brief discussion of each of the questionnaire development steps is made as follows:

Step1. Determine the specific information needed

This is the initial step required to determine the specific information needed to achieve the research objectives – which would answer the research questions and test the hypothesis. One of the most common and costly errors of research is the omission of an important question on the data gathering instrument. Once the questionnaire is filled, it is too late to go back for additional data without significant delay and additional cost. The researcher must determine all of the information required before the questionnaire is developed. Sometimes the articulation of research objectives into specific questions by setting up dummy tables to be used when analyzing the results will trigger a new idea of research question that should be included in the survey. Every effort needs to be marshaled at this point of the design process to ensure relevant results for the analysis phase of the research process. In some cases, it is advisable to

conduct exploratory research to ensure that all of the relevant variables are identified.

In short, first clear definition and individual listing of all the specific objectives, research questions, or hypothesis to be tested should be made. To answer each list, associated questions need to be listed.

Step 2. Identify the sources

This step involves the important aspect of identifying the sources of the information needed determined in step one. The sophistication, intelligence, frame of reference, location, ethnic background, and other characteristics of the potential respondents are vital to determine the type of questionnaire, the wording of questions, the means of administration, and other aspects of the questionnaire. E.g. Consider the difference in the design of a questionnaire if the questions are asked for ten-year old children or twenty-year old College students.

Step 3. Choose the method of administration

In this step, the decision is made on whether to use a method of personal interview, telephone survey, or mail questionnaire. Issues such as whether a stimulus is required, the length of the questionnaire, the complexity of the questions, and the nature of the issue being researched must be considered in choosing which type of survey is best. These decisions should be made with respect to the information required and the nature of the sources of the information. Other considerations that affect this decision are the cost and time constraints placed on the research project.

Step 4. Determine the type of questions

To decide up on the type of questions used, a researcher must look both backward and forward. The researcher looks back to review the information required and the nature of the respondents. The researcher must look forward to the analysis stage of the research process to ensure that the right form of data is obtained to accommodate the proper analytical techniques. There are two basic types/forms of questions that can be used in a questionnaire:

- a) Open-ended,
- b) Closed-ended
 - Dichotomous,
 - Multi-chotomous, and
 - Scales.

Most questionnaires have more than one type of questions.

a. Open-ended questions: open-ended (unstructured free response) questions such as "What did you like about the product?" or "What is your favorite television program?" provides an unlimited range of replies the respondent can make.

Advantages of Open-ended questions

- ⇒ Permit an unlimited number of possible answers.
- ⇒ Respondents can answer in detail and can qualify and clarify responses
- ⇒ Unanticipated findings can be discovered.
- ⇒ Permit adequate answers to complete questions.
- ⇒ Permit creativity, self-expression, and richness of detail
- ⇒ Reveal a respondent's logic, thinking process and frame of reference.
- ⇒ Provide respondents with maximum flexibility in answering questions

Disadvantages of Open-ended questions

- ⇒ Different respondents give different degrees of detail in answers - responses may not be consistent.
- ⇒ Responses may be irrelevant or buried in useless detail.
- ⇒ Comparison and statistical analysis become very difficult.
- ⇒ Coding responses is difficult.
- ⇒ Articulate and highly literate respondents have an advantage.
- ⇒ Responses are written verbatim, which is difficult for interviewers.
- ⇒ Greater amount of respondent time, thought and effort is necessary.

b. Closed-ended (structured, fixed response) questions: A closed ended questions ask questions and give the respondent fixed response from which to choose. Most answers are variables having two or more discrete categories. Category style depends at least up on whether the variable is (i) nominal, (ii) ordinal, (iii) interval or (iv) ratio.

e.g. Is the president doing (a) very good, (b) good, (c) fair, (d) poor job in your opinion?

Advantages of closed-ended questions

- ⇒ Easier and quicker for respondents to answer.
- ⇒ Easier to compare the answers of different respondents.
- ⇒ Easier to code and statistically analyze.
- ⇒ Response choices can clarify questions for respondents
- ⇒ Less articulate or less literate respondents are not at a disadvantage

✚ **Disadvantages of closed-ended questions**

- ⇒ Can suggest ideas that the respondents would not otherwise have.
- ⇒ Respondents with no opinion or no knowledge can answer anyway.
- ⇒ Respondents can be confused because of too many choices.
- ⇒ Misinterpretation of a questions can go unnoticed
- ⇒ Force people to make choices they want of not make in the real world.

The closed-ended questions can further be seen from the following sub-categories:

- i. **Dichotomous questions:** Dichotomous questions give two choices: either “Yes” or “No”, or a choice of two opposite alternatives. They are preferred for starting interviews, are easy to tabulate, and can be used as a lead-in for more specific questions to a particular group.

✚ **Weaknesses of dichotomous questions:**

- ⇒ They can force a choice where none exists or lead to bias. E.g. "Do you still cheat on your taxes?" Yes__ No__
- ⇒ Only few questions can be framed in terms of a dichotomy; more complex questions cannot be done with this method.

- ii. **Multi-chotomous questions:** Such questions provide several alternatives to respondents. In some cases, they may be asked to select only one from a list or asked to select as many as are applicable. The **advantages** of the multi-chotomous question are: ease of administration, ease of tabulation, and flexibility for factual as well as attitudinal responses. Some **disadvantages** exist in the length of the list of alternative responses; response alternatives may not be mutually exclusive; or reading the list may lead the respondent to a particular response.

- iii. **Scales:** Although a scale can be considered a multi-chotomous question, it deserves separate consideration. The benefits of a scale are that it permits the objective measurement of attitude and feelings. This allows for the identification of segments that are more favorable inclined toward a product, service, or issue. Scaling is somewhat subject to various frames of reference differences and to the effect. Some debate exists concerning whether scales would be balanced or unbalanced. Balanced scales have the same number of responses on the positive side of the scale as on the negative side of the scale. Unbalanced scales have more possible responses on one side of the scale than the other, usually the positive side. Some debate also exists over whether to have a neutral response. Those who feel there should be a “neutral “ option believe so because they think that this is a legitimate response category and for the sake of accuracy the respondent should have that “ground to stand on”. People against having a neutral response category argue that most

individuals are unlikely to be authentically neutral on the issue. They consider the neutral as an easy way out or a “polite negative.” Consequently, they feel that it is much better to frame the issues that the respondent is “forced” to indicate a preference, however slight.

The “don’t know, or “no opinion” response should always be available to respondents. An honest expression of a lack of awareness, experience, or thought given to an issue should be allowed so that these responses are not lumped together with those people who have considered the issue and find their position is the neutral point on the scale.

Step 5. Develop the specific questions

This step deals with the actual writing of the specific questions that will be asked. The elements in the previous steps, to a large extent, control the content of the questions. Questions should be worded in ways that avoid leading, pressuring, or embarrassing the respondent. The questions should avoid ambiguous words.

Some basic rules and considerations for wording questions

i) Keep the questions short, simple and in everyday language, and to the point. Reasons for keeping the questions as brief as possible are that longer questions make the response task more difficult, are more subject to error on the part of both interviewers and respondents, cause a loss of focus, and are less likely to be clear. Moreover, your respondents may not be well-educated and even if they are, they may not be able to understand some jargon words. Ambiguous questions (i.e., a question that is interpreted differently by different respondents) should also be avoided.

Example: Let us consider the following two questions:

“Is any one in your family a *dipsomaniac*?”

“Is your work made difficult because you are expecting a baby?” Yes ☐ No ☐

From these, the first question suffers from lack of clarity because the word *dipsomaniac* is jargon. On the other hand, the second question suffers from its ambiguous nature because, for example, we do not know what type of work and the stage of pregnancy the question implies.

ii) Do not ask double-barreled questions: - This implies a question with a question.

Therefore, one question should not ask more than one thing at a time because we do not know which particular question the respondent has answered.

Example, “How often and how much time do you spend on each visit?” This question has two questions. Here, we can not identify for which a respondent gives his/her response.

iii) Avoid identifying the sponsor of the survey: In such cases as customer satisfaction surveys, the researcher may want to identify the sponsor of the survey. In most case, however, it is best to not identify the sponsor to collect more objective responses.

iv. Keep the questions as neutral as possible (avoid leading questions). A leading question is one which, by its structure, contents, or wording leads a respondent to answer in a certain direction (either positively or negatively).

Example, “Living cost is getting cheaper, is not it?”

“Equity on payment has a direct implication on employees turnover, is that not?”

One problem from these questions is that they are not questions rather they are statements that lead the respondents to say “Yes”. That means, these non-objective questions tend to be leading because they suggest an answer.

v. Do not ask unnecessary questions. Always ask yourself “Is this question necessary?”

Each question should relate or add meaning to a specific research objective. If the question falls in the “nice to know” but “not necessary” category, leave it out.

vi. Avoid asking questions the respondent either cannot answer or will not answer

Since respondents will answer most questions, it is important to determine if a respondent can be expected to know the information desired. On other occasions, respondents will have the information required may not be willing to divulge it. If a question appears to be sensitive, it may be best to leave it out. It is crucial to the research objective, and then it is best to ask the sensitive question later in the survey after a degree of trust has been established with the respondent.

vii. Avoid asking overly personal questions: Words or phrases that show bias are emotionally charged and will lead to inaccurate results. When sensitive questions must be asked, they should be asked after a warm-up period and/or hidden within a series of less personal questions.

viii. Do not ask questions that are based on presumptions. – In such questions, the researcher assumes that respondents fit in to a particular category and seeks information based up on the assumption.

Example, if these questions are asked, “How many bottles of beer do you drink per day?”

Or “What type of contraceptives do you use?”, they were asked without knowing whether the respondents drink beer or sexually active.

ix. Keep it readable.

x. Make all definitions, assumptions, and qualifiers clearly understood: For example, the terms like “adequate”, “minority”, “value”, “how much”, “usually”, “convenient”, “most”, “quality”, “procedures”, “diversity”, “not too often”, “acceptable”, “very often”, “now”, “you”, “never”, and “salary” are full of ambiguities. If you ask about salary, you have to specify whether it is gross or net and its rate (per day, per month, per year, etc). For numerical responses, specify the unit of measurement. Make also sure that the respondent has all the information necessary to answer the question.

xi. Avoid making significant memory demands: For example, the question “How many times did you visit the library last year?” requires the respondent to sit for long and recall. Rather, if the question is asked like this, “About how much time did you spend to visit library in the last year?” the respondent can accurately answer it because its memory and cognitive demand are not so much demanding.

xii. Avoid asking for very precise responses: Such specific answer demanding question as “What is your exact yearly income in Birr?” also become cognitive demanding.

xiii. Avoid asking for broad generalizations about attitudes or options: For example, instead of asking “Do you consider your self hard working business person?”, ask, “How many hours did you study yesterday?”, “How many hours per day do you usually study?”, and other questions to infer the opinion rather than simply ask opinion. In order not to make your questionnaire lengthy, try not to ask very specific questions as well.

xiv. Make the item easy and fast to be answered

ix. Keep it interesting: Do not make your items monotonous.

?

Dear learner! List the basic considerations that should be taken in wording questions?_____

Response-category format

i) Blank space: These are used especially for answers to open-ended questions. The researcher should be able to carefully estimate the amounts the respondents write and sufficient space.

ii) For closed-ended questions, response categories are written in different styles based on the type of variable. Responses can be checked on **blank spaces, in box, or they may be encircled.**

Example:

Check the appropriate blank

1- Sex: Male____ Female____

2- What type of soft drink do you drink? Coca-cola____ Sprite____ Fanta ____ Pepsi____

Check the appropriate box

1. Sex: Male ☐ Female ☐

Circle the appropriate number

1. Sex: Male 1 Female 2

iii) Inventory method: An inventory is list of items to be checked or marked by the respondent.

Example: check all the ways in which you have learned about product X in the last week.

- ___ Internet
- ___ Television
- ___ News papers
- ___ Magazine
- ___ Word of mouth
- ___ Other means (please specify)

iv. Grid format: These are boxed formats frequently used to evaluate each item in an inventory. It can also be a two way inventory.

Illustration: Look how a grid is designed to seek response from respondents on both the type of News and Media used.

	TV	Internet	Radio	News Paper	Magazine	Friends or colleagues
BBC news						
CNN news						
ERT news						

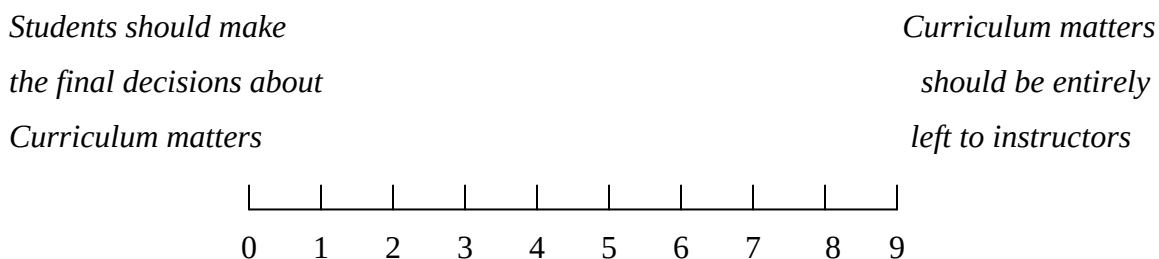
iv.Ordinal/ Likert Scale: - By this technique, you construct the same answer format for a whole series of questions. In other words, one set of category labels can serve for a whole set of categories if the labels are placed above the categories like titles.

Illustration: Suppose Arba Minch Tourist Hotel has made feedback surveys from its clients. It presents response-format in the following grid.

Please rate your satisfaction with each of the following:	Very Satisfied	Somewhat Satisfied	Neutral	Somewhat Dissatisfied	Very Dissatisfied
Overall satisfaction with this experience	☐	☐	☐	☐	☐
Receiving a warm and sincere greeting upon arrival	☐	☐	☐	☐	☐
Staff greeting you by name	☐	☐	☐	☐	☐
Staff remembering you as a regular guest	☐	☐	☐	☐	☐
Timeliness of check-in	☐	☐	☐	☐	☐
Ability of the staff to anticipate your needs	☐	☐	☐	☐	☐
Cleanliness of the guest room furnishings	☐	☐	☐	☐	☐
Receiving a fond farewell when you checked out	☐	☐	☐	☐	☐
Did you experience any problems during your stay? Yes ☐ No ☐					
If you reported any problems, how satisfied are you with the resolution	☐	☐	☐	☐	☐
Comments:					

v) Semantic differential responses: In this method, a list of responses as a continuum with labels only at the extremes can be used.

Illustration: To what extent do you feel Students should have a voice in determining course content and degree requirements?



vi. Ranked Responses: Sometimes it is useful for the respondent to rank a set of options by numbering them in order from 1 to the maximum number you are interested in.

For example: “Place in order of importance to you the following reasons of employee turnover in Arba Minch University” (Indicate by numbering from 1-4 in order where 1 is the most important factor)

	<u>The hot Climate</u>	
	<u>Poor Fringe Benefits</u>	
	<u>Shortage of Basic facilities</u>	
	<u>Administrative problems</u>	
	<u>Employees relations problems</u>	

Note that each option will need to be coded as a separate variable and in the above case five variables are required (even though only four ranks are to be identified). This approach can generate a lot of data and so the number of options used should not be excessive. Apart from this, respondents find it difficult to discriminate meaningfully between lots of options.

vi. Interval Scales: In this case, the questions have the responses in interval (range) format.

Example:

1. What is your age (in years)? 10-20 ☐ 21-30 ☐ 31-40 ☐ Above 40 ☐
2. What is your daily income in Br? Below 25 ☐ 251-50 ☐ 501 100 ☐ Above 100 ☐

Comprehensive example for the preceding steps: - Let us consider the study to evaluate community responsiveness in a health program. Consider the following table:

Table 6.1. Sample of questionnaire developed based on certain research questions or objectives

Objective/question/hypothesis	Main and associated research questions	Information required	Questions
1.To find out the understanding of the concept “community responsiveness” among health administrators, managers, service providers and consumer of health services	What is community responsiveness? What does it mean when people use the term “community responsiveness?”	1.Perception of community responsiveness	Q1.1. When you use the term “community responsiveness”, what do you mean by it? Q 1.2. What comes to your mind when you use the term “community responsiveness” in the delivery of health services? Q1.3. What, in your opinion, are the characteristics of “community responsiveness”? Q1.4. What, in your opinion, is the difference between “community development and community responsiveness”?
2. To identify the strategies needed to foster community responsiveness in the delivery of health services	What are the differences in the perception of community responsiveness among health administrators, planners, managers, service providers and consumers of health services?	2.Occupational status, i.e. health administrator, planner, service provider or consumer	Q2.1. Would you categorize your job as? Health administrator/planner Service provider Health consumer
3. To develop a set of indicators to evaluate the effectiveness of strategies to foster community responsiveness		3. Age, gender, education	Q 3.1. How old are you?_____ Q3.2. Are you ___Female ____Male? Q3.3. What is the highest level of educational achievement you have attained?

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Dear learner! On writing your questions of a questionnaire, what types of formats for response can you employ?_____

Step 6. Determine question sequence and length of the questionnaire

This step six can be extremely important in the completion rate of the survey. Excessive length may deter some respondents from completing the survey. This step can be equally important to the quality of the results acquired. The sequence of the questions should follow a logical pattern so it follows smoothly and effortlessly from one section to another. Another important consideration is to ensure that questions building on previous questions are placed properly in the questionnaire. Generally, questions will flow from the general to the specific. This warms up the respondents and lets them reflect appropriately before being asked specific questions that require good recall and some degree of detail. Skip patterns of branching questions need to be carefully designed. Quite often, a respondent must be directed to a specific location in the questionnaire based on a previous question .For example, “Have you baked brownies in the past three months? “Yes” respondents go on to answer specific questions about the baking occasion while “No” respondents skip to a different part of the survey. Personal, telephone, and Internet surveys lend themselves more to branching than do mail questionnaires.

Most questionnaires have three basic sections

- (a) Introduction
- (b) Body/content, and
- (c) Classification section.

Now let us discuss each in detail as follows:

(a) Introduction: The introduction tells the respondent who the researcher is, why he is requesting the respondent’s information, and explains what is expected of the respondent. The introductions should explain the purpose of the questionnaire and enlist the co-operation of the respondent. In case of a mail survey, this may take place in an attached cover letter. On the phone, this is accomplished with a positive tone without sounding like a polished sales pitch. Most people want to express their opinion as long as they know it is not a sales device. The introduction should promise to keep the respondents identity anonymous and

his/her information will be kept confidential. Often the introduction will qualify the respondent in special way to make sure that the interviewer is talking only to the right respondents. Qualification questions may include brand usage, age/sex categories, security issues such as whether the respondent or any member of his her households for a competitor of the client, an advertising agency or a marketing research company. (For more information on this theme, please see discussions given under the section *Organizing Questionnaire*).

(b) **Body/Content:** The body or main content of the questionnaire provides basic information required by the research objectives. This is usually the most substantial portion of the questionnaire. If the respondent has not been qualified in the introduction, the first question of this section should identify the proper respondents and have instructions to end the survey for all others. *This section should begin with easy, direct question. Place questions that are personal or reflect a sensitive issue well toward the end of the body.* Once respondents have become interested in the survey, they are more likely to respond to personal or sensitive issues.

(c) **Classification section:** The final section is designed to obtain cross-classification information such as sex, income, educational level, occupation, marital status, and age. The demographic data allows for comparisons among different types of respondents. Certain market segments may emerge through the use of this cross-classification data.

Step7. Predetermine coding

Questionnaires should be pre-coded so that any difficulties encountered in entering data in to computer tabulation and analysis programs may be solved before the data are gathered. Once the questionnaires are completed and returned, the responses can be quickly entered in to the database and results generated.

Step 8. Pretest the questionnaire

This is an essential step that should not be ignored. Following the maxim, "The proof is in the pudding," a questionnaire should be pre-tested before it is administered. A questionnaire may be too long, ambiguous, incomplete, unclear, or biased in some way. A thorough pretest help overcome these problems and it will help refine any procedural problems a questionnaire might have. Some of the procedural problems might be improper skip patterns and misunderstanding the interviewer. A pretest will evaluate and fine-tune the

questionnaire, estimate the time required for completion of the interview, and allow for setup of coding refinements for tabulations. The pretest should be administered under actual field conditions to get an accurate response. If significant changes result from an original pretest, it is advisable to conduct a second pretest after making appropriate revisions. A relatively small number of interviews are sufficient to pretest a normal questionnaire. Once they have been completed, all interviewers who participated should report their findings. They will be able to determine whether the questions work and be able to make suggestions for revisions.

Step 9. Review and revise the questionnaire

Based on the pretest and a thorough review of all the previous steps, the questionnaire should be revised. The bottom line for an effective data-gathering instrument is the accuracy of the data collected. Everything that can possibly be done in advance should be done to ensure the accuracy of the instrument.

The steps used to design an effective data-gathering instrument need to be viewed as a dynamic outline and not a sequential step-by step process. Each step should be applied with a sharp eye on the research objectives and the data needs they require. The finished questionnaire is the result of rearticulating the study objectives in to a set of effective questions for extracting the required information from the pertinent set of respondents.

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Dear learner! Write the steps required in developing a questionnaire_____



General rules of Question order in a questionnaire

- i. *Put sensitive questions, open-ended questions late in the questionnaire.* Since open-end require more thought and writing, we should not place them first. Additionally, if sensitive questions (e.g. such as sexual behavior, income, religion, and etc) are placed first, respondents may react to them negatively or they refuse to answer.
- ii. *Ask easy-to answer questions first:* - The opening questions should require little thought. Moreover, they should be interesting to stimulate respondents to complete the questionnaire. Usually, opening questions are about background information such as age, gender, educational status, marital status, etc.

iii. *Place questions and answer Choice in logical order:* Answer choice order can also make individual questions easier or more difficult to answer. Whenever there is a logical or natural order to answer choices, use it. Always present agree-disagree choices in that order. Presenting them in disagree-agree order will seem odd. For the same reason, positive to negative and excellent to poor scales should be presented in those orders. When using numeric rating scales higher numbers should mean a more positive or more agreeing answer.

The questions have to be arranged in some general order, such as time sequence and sub-topics of the study. For example, if fertility survey of a given family is designed, it should include job history, contraceptive practices, birth rates, and etc. Therefore, time and care must be taken against mixing topics or frames of reference as well as against jumping abruptly from one frame of reference to another.

Grouping together questions on the same topic makes the questionnaire easier to answer.

Question order can affect the results in two ways. One is that mentioning something (an idea, an issue, a brand) in one question can make people think of it while they answer a later question, when they might not have thought of it if it had not been previously mentioned. In some cases you may be able to reduce this problem by randomizing the order of related questions. Separating related questions with unrelated ones can also reduce this problem, though neither technique will eliminate it.

The other way question order can affect results is **habituation**. This problem applies to a series of questions that all have the same answer choices. It means that some people will usually start giving the same answer, without really considering it, after being asked a series of similar questions.

If you are using telephone, computer direct or Internet interviewing, good software can help with this problem. Software should allow you to present a series of questions in a random order in each interview. This technique will not eliminate habituation, but will ensure that it applies equally to all questions in a series, not just to particular questions near the end of a series.

Another way to reduce this problem is to ask only a short series of similar questions at a particular point in the questionnaire. Then ask one or more different kinds of questions, and then another short series if needed.

A third way to reduce habituation is to change the “positive” answer. This applies mainly to level-of-agreement questions. You can word some statements so that a high level of agreement means satisfaction (e.g., “My supervisor gives me positive feedback”) and others so that a high level of agreement means dissatisfaction (e.g., “My supervisor usually ignores my suggestions”). This technique forces the respondent to think more about each question. One negative aspect of this technique is that you may have to modify some of the data after the results are entered, because having the higher levels of agreement always mean a positive (or negative) answer makes the analysis much easier. However, the few minutes extra work may be a worthwhile price to pay to get more accurate data.

The order in which the answer choices are presented can also affect the answers given. People tend to pick the choices nearest the start of a list when they read the list themselves on paper or a computer screen. People tend to pick the most recent answer when they hear a list of choices read to them.

iv. *Avoid establishing a response set:* A response set is a tendency to reply to items in particular way, regardless of the questions content or the correct answer. The response set is may be created due to social desirability of statements, tendency of the respondents to answer yes rather than no, a problem of question sequence (e.g. sequence of questions asked in a similar format can lead to stereotyped responses), and etc.

v. *Separate reliability-check question pairs:* Pairs of questions, one stated positively and the other negatively, are often used to check reliability.

Example, “Abortion should be legalized (agree/disagree)” at one point in the questionnaire and “abortion should not be legalized (agree/disagree)” at other point.

vi. *Vary questions by length and type:* This may help keep the respondent interested.

vii. *Determine whether funnel technique is applicable:* In this technique, broad, general or easy to answer open-ended/ closed-ended questions are asked first, and then asking more

specific questions narrows the funnel. This method puts the respondents at ease. By using the technique, in conjunction with “filter” questions, the researcher can determine whether specific questions apply to the respondent, and thus avoid asking questions that do not. A filter or screening determines whether future questions are applicable to the respondent. For example, instead of asking the respondent how many bottles of beer he/she smokes or what brand, the researcher first asks whether the respondent drinks beer. If the answer is no, the specific drinking questions are skipped.

?

Dear learner! Write the rules required for a proper order of Questions in a questionnaire_____

Organizing questionnaire

1. Introducing a questionnaire: All surveys, irrespective of how they are administered, require an introduction. In case of a mail survey, this may take place in an attached cover letter. An introduction for an online survey should be no longer than one computer screen (about three paragraphs). For a written survey, it should be no longer than one page.

In introducing a questionnaire, start with a Title (e.g., Banking service-market survey). Always include a short introduction - who you are and why you do the survey. If you are asking about different services, it is often a good idea to give the name of the research company (e.g., XYZ Research Agency) rather than the client (say, the service producer) in order to avoid possible bias. However, in some cases, to mention the client may be important. For instance, if the organization asks opinions of its members on how it can best meet their needs, the members may be more likely to respond. The same could be true when you are surveying users of a particular service.

Reassure your respondent that his/her responses will not be revealed to your client, but only combined with many others to learn about overall attitudes.

A cover letter is the introduction to a mail survey and should be friendly and persuasive. A good cover letter will increase the response rate. In a cover letter,

- ✓ Explain how the data will be collected, stored, analyzed, and be used

- ✓ Brief how the survey recipients were selected
- ✓ Point out who will have access to the data and who will have access to the results
- ✓ Identify who the sponsor of the survey is
- ✓ Write how a person is contacted for questions
- ✓ Specify how long it takes the respondent to complete the survey
- ✓ Indicate how should the respondent submit the survey?
- ✓ Mention the incentive (if any).
- ✓ Describe how to return the questionnaire.
- ✓ Include the name and telephone number of someone the respondent can call if they have any questions.
- ✓ Include instructions on how to complete the survey itself.
- ✓ Ask the recipient to take the survey.
- ✓ Explain why taking it will improve some aspect of the recipient's life (it will help improve a product, make an organization better meet their needs, make their opinions heard).
- ✓ Appeal to the recipient's sense of altruism ("please help"). Ask the recipient again to take the survey.
- ✓ Specifically, number mail questionnaires on each page and include the return address on the questionnaire itself, because pages and envelopes can be separated from each other. Envelopes should have return postage prepaid. Using a postage stamp often increases response rates, but is expensive, since you must stamp every envelope - not just the returned ones.

Additionally, you may want to leave a space for the respondent to add their name and title. Some people will put in their names, making it possible for you to re-contact them for clarification or follow-up questions. Indicate that filling in their name is optional. If the questions are sensitive in nature, do not have a space for a name. Some people would become suspicious and not complete the survey.

If the survey contains commercially sensitive material, ask a "security" question up front to find whether the respondent or any member of his/her family, household or any close friend works in the industry being surveyed. If so, terminate the interview immediately.

They may work for the company that commissioned the survey - or for a competitor. In either case, they are not representative and should be eliminated. If they work for a competitor, the nature of the questions may betray valuable secrets. The best way to ask security questions is in reverse (e.g. if you are surveying for a pharmaceutical product, phrase the question as "We want to interview people in certain industries; do you or any member of your household work in the pharmaceutical industry?") If the answer is "Yes", thank the respondent and terminate the interview. Similarly, it is best to eliminate people working in the advertising, market research or media industries, since they may work with competing companies.

After the security question, start with general questions. If you want to limit the survey to users of a particular product, you may want to disguise the qualifying product. As a rule, start from general attitudes to the class of products, through brand awareness, purchase patterns, specific product usage to questions on specific problems (i.e., work from "What types of coffee have you bought in the last three months" to "Do you recall seeing a special offer on your last purchase of Brand X coffee?"). If possible, put the most important questions into the first half of the survey. If a person gives up half way through, at least you have the most important information.

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Dear learner! What important elements should be included on a cover letter to send a mailed survey? _____

2. Questionnaire layout: Always consider the layout of your questionnaire. This is especially important on paper, computer direct and Internet surveys. You want to make it attractive, easy to understand and easy to complete. If you are creating a paper survey, you also want to make it easy for your data entry personnel.

Try to keep your answer spaces in a straight line, either horizontally or vertically. A single answer choice on each line is best. Eye tracking studies show the best place to use for answer spaces is the right hand edge of the page. It is much easier for a field worker or respondent to follow a logical flow across or down a page. Using the right edge is also easiest for data entry.

The Survey System lets you create a Questionnaire Form with the answer choices in two columns. Creating the form that way can save a lot of paper or screen space, but you should recognize doing so makes the questionnaire a little harder to complete. It also slows the data entry process when working with paper questionnaires.

Questions and answer choice grids, as in the second of the following examples, are popular with many researchers. They can look attractive and save paper, or computer screen space. They also can avoid a long series of very repetitive question and answer choice lists. Unfortunately, they also are a bit harder than the repeated lists for some people to understand. As always, consider whom you are studying when you create your questionnaire.

Hello student! Look at the following layouts and decide which you would prefer to use:

Do you agree, disagree or have no opinion that this company has-

A good vacation policy?	- Agree/not sure/disagree.
Good management feedback?	- Agree/not sure/disagree.
Good medical insurance?	- Agree/not sure/disagree.
High wages?	- Agree/not sure/disagree.

An alternative layout is:

Do you agree, disagree or are not sure that this company has:

	Agree	Not Sure	Disagree
A good vacation policy?	3	2	1
Good management feedback?	3	2	1
Good medical insurance?	3	2	1
High wages?	3	2	1

The second example shows the answer choices in neat columns and has more space between the lines. It is easier to read. The numbers in the second example will also speed data entry, if you are using a paper questionnaire.

Pre-testing/ piloting questionnaire

The last step in questionnaire design is to test a questionnaire with a small number of interviews before conducting your main interviews. Ideally, you should test the survey on the same kinds of people you will include in the main study. If that is not possible, at least have a few people, other than the question writer, try the questionnaire. This kind of test run can reveal unanticipated problems with question wording, instructions to skip questions, etc. It can help see if the interviewees understand your questions and giving useful answers.

If you change any questions after a pre-test, you should not combine the results from the pre-test with the results of post-test interviews. The Survey System will invariably provide you with mathematically correct answers to your questions, but choosing sensible questions and administering surveys with sensitivity and common sense will improve the quality of your results dramatically.

Incentives to improve response for a questionnaire

One of the most important aspects of designing a questionnaire is improving your Response Rate, which requires providing the respondent with the motivation to complete the questionnaire. So consider:

- Ways to increase perceived reward
 - Show positive regard
 - Say *thank you*
 - Ask them for “advice”
 - Give social validation
 - Give a tangible reward
 - Make the questionnaire interesting
 - State an upcoming deadline

- Ways to decrease perceived cost
 - Avoid subordinating language
 - Do not embarrass the respondent
 - Do not inconvenience the respondent
 - Make questions appear short and easy
 - Avoid asking personal information (if it must be asked, use 'soft' wording and inform about confidentiality)
 - Keep subsequent requests similar
- Ways to establish trust
 - Provide a token of appreciation
 - Provide a sense of legitimate authority
 - Make completing the questionnaire seem important
 - Remind respondent of previous relationship with sponsor (if applicable)

As a general rule, with only a few exceptions, long questionnaires get less response than short questionnaires. So keep your questionnaire short. In fact, the shorter is the better. Response rate is the single most important indicator of how much confidence you can place in the results. A low response rate can be devastating to a study. Therefore, you must do everything possible to maximize the response rate. One of the most effective methods of maximizing response is to shorten the questionnaire.

Remember **KISS** - *Keep It Short and Simple*. If you present a 20-page questionnaire, most potential respondents will give-up in horror before even starting. Ask yourself what you will do with the information from each question. If you cannot give yourself a satisfactory answer, leave it out. Avoid the temptation to add a few more questions just because you are doing a questionnaire anyway. If necessary, place your questions into three groups: *must know*, *useful to know* and *nice to know*. Discard the last group, unless the previous two groups are very short.

Many researchers have examined the effect of providing a variety of non-monetary incentives to subjects or respondents. These include token gifts such as small packages of coffee, ball-point pens, postage stamps, key rings, trading stamps, participation in a raffle

or lottery, or a donation to a charity in the respondent's name. Generally (although not consistently), non-monetary incentives have resulted in an increased response. A meta-analysis of 38 studies that used some form of an incentive revealed that monetary and non-monetary incentives were effective only when enclosed with the survey. The promise of an incentive for a returned questionnaire was not effective in increasing response. The average increase in response rate for monetary and non-monetary incentives was 19.1 percent and 7.9 percent, respectively.

Most researchers have found that higher monetary incentives generally work better than smaller ones. One researcher proposed a diminishing return model, where increasing the amount of the incentive would have a decreasing effect on response rate. A meta-analysis of 15 studies showed that an incentive of 25¢ increased the response rate by an average of 16 percent, and \$1 increased the response by 31 percent.

Allow a “Don't Know” or “Not Applicable” response to all questions, except to those in which you are certain that all respondents will have a clear answer. In most cases, these are wasted answers as far as the researcher is concerned, but are necessary alternatives to avoid frustrated respondents. Sometimes “Don't Know” or “Not Applicable” will really represent some respondents' most honest answers to some of your questions. Respondents who feel they are being coerced into giving answers do not want to give and often do not complete the questionnaire. For example, many people will abandon a questionnaire that asks them to specify their income, without offering a "decline to state" choice.

For the same reason, include “Other” or “None” whenever either of these is a logically possible answer. When the answer choices are a list of possible opinions, preferences, or behaviors, you should usually allow these answers.

The response rate in questionnaires and especially in mail surveys can also be improved through quality printing, preparing a cover letter in a pleasant style, providing advance information, making a follow-up , and drawing a larger sample size.

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Dear learner! What techniques of incentives do you employ to increase response for your questionnaire survey? _____

IV. Focus Group Discussion /FGD/

A Focus Group Discussion (FGD) is a group discussion of approximately 6 - 12 persons guided by a facilitator, during which group members talk freely and spontaneously about a certain topic. An FGD is a qualitative method. Its purpose is to obtain in-depth information on concepts, perceptions and ideas of a group. An FGD aims to be more than a question-answer interaction. The idea is that group members discuss the topic among themselves, with guidance from the facilitator.

FGD techniques can be used to:

1.Focus research and develop relevant research hypotheses by exploring in greater depth the problem to be investigated and its possible causes.

Example: a district health officer had noticed that there were an unusually large number of cases of malnutrition of children under 5 reported from one area in her district. Because she had little idea of why there might be more malnutrition in that area she decided to organize three focus group discussions (one with leaders, one with mothers from the area and one with health staff from the area). She hoped to identify potential causes of the problem through the FGDs and then develop a more intensive study, if necessary.

2. Formulate appropriate questions for more structured, larger scale surveys.

Example: In planning a study of the incidence of childhood diarrhea and feeding practices, a focus group discussion showed that in the community under study, children below the age of 1 year were not perceived as having 'bouts of diarrhea' but merely 'having loose stools' that were associated with milestones such as sitting up, crawling,

and teething. In the questionnaire that was developed after the FGD, the concept ‘diarrhea’ was therefore carefully described using the community’s notions and terms.

3. Help understand and solve unexpected problems in interventions.

Example: In District X, the recent national (polio) immunization days (NID) showed widely different coverage’s per village (50-90%) and in a number of villages a marked decrease in coverage was observed compared to last year. Eight FGD were held with mothers, two in town, three in rural villages with a marked decrease in NID coverage and three in villages with a high coverage throughout. It appeared that overall, the concept NID had raised confusion. Most people believed that this mass campaign strengthened the children’s immunity against any (childhood) disease, including malaria and Respiratory Tract Infections. In the villages with a low NID coverage there had been a high incidence of malaria in children immediately after the previous NID campaign and several children died. Mothers therefore believed that the NID campaign was useless.

4. Explore controversial topics.

Example: Sexual behavior is a controversial topic in the sense that males and females judge sexual relations and sexuality often from very different perspectives. Sexual education has to take this difference into account. Through FGDs, first with females, then with males, and then with a mixed group to confront both sexes with the different outcomes of the separate discussions (listed on flip charts) it becomes easier to bring these differences in the open. Especially for teenagers, who may have many stereotypes about the other sex or be reluctant to discuss the topic openly (particularly girls), such a ‘multi-stage’ approach is useful.

Strengths and limitations of FGD

Implementation of FGDs is an iterative process; each focus group discussion builds on the previous one, with a slightly elaborated or better-focused set of themes for discussion. Provided that the groups have been well chosen in terms of composition and number (see

below), FGDs can be a powerful research tool which provides valuable spontaneous information in a short period of time and at relatively low cost.

FGD should not be used for quantitative purposes, such as the testing of hypotheses or the generalization of findings for larger areas, which would require more elaborate surveys.

However, FGDs can profitably complement such surveys or other, qualitative techniques. Depending on the topic, it may be **risky** to use FGDs as a **single tool**. In group discussions, people tend to centre their opinions on the most common ones, on ‘social norms’. In reality, opinions and behavior may be more diverse. Therefore, it is advisable to combine FGDs with at least some key informant and in-depth interviews. Explicitly soliciting other views during FGDs should be routine as well.

In case of very **sensitive topics**, such as sexual behavior or coping with HIV/AIDS, FGDs may also have their limitations, as group members may hesitate to air their feelings and experiences freely. One possible remedy is the selection of participants who do not know each other (e.g., selection of children from different schools in FGDs about adolescent sexual behavior), while assuring absolute confidentiality.

It may also help to alternate the FGD with other methods, for example, to precede it by a self-developed role play on sexual behavior, or to administer a written questionnaire immediately after the FGD with open questions on sexual behavior in which the participants can anonymously state all their questions and problems.

Another way to ensure confidentiality in a FGD on a sensitive topic is giving participants an option to introduce themselves under any name they would like to use (not necessarily their own). Further, before the discussion, it should be stressed that they may bring up experiences of friends and brothers/sisters as well as their own, and that it is not necessary to bring painful personal experiences in the open.

How to Conduct a Focus Group Discussion

i. Determine the purpose: A FGD can be regarded as a mini-study. It, therefore, requires one or two clear objectives. These objectives will guide the research team in the formulation of discussion questions.

ii. Situation analysis: Any FGD requires good knowledge of local conditions. Communities are seldom or never homogeneous. There are always differences between community members, for example in education, political power, gender, economic status and ethnic group. These differences will be reflected in their perceptions of the problems they suffer from and possible solutions. A researcher must be aware of these differences, otherwise (s)he may miss important groups of participants or obtain a hotchpotch of information. Similarly, (s)he must know which key persons or organizations could be good entry points for the selection of participants in the FGDs (e.g.: women's groups, parent associations, youth clubs, etc.). If a FGD forms part of a bigger study, or project, it may be easy to define target groups for the discussions. Otherwise, the first task of the researcher(s) will be to explore the area and identify possible target groups. Interviews with some key informants and a rudimentary situation analysis are then indispensable. The situation analysis should preferably be carried out in a participatory way, with representatives of the study population on which the FGD focuses.

Points to be considered when preparing the FGD

i. Recruitment of participants: Participants should be roughly of the same socio-economic group or have a similar background in relation to the issue under investigation. The age and sexual composition of the group should facilitate free discussion. Often you, therefore, need to obtain information on a topic from several different categories of informants who are likely to discuss it from different perspectives in separate FGDs, though in a later stage groups may be joined.

Always ensure confidentiality of opinions. Ask co-operation from the group members as well. If group members present very personal problems and need advice or help, this should be followed up after the FGD.

Participants should be invited at least a day or two in advance, and the general purpose and procedures of the FGD should be explained, in order to obtain their **consent to join**.

ii. Selection of participants: If you are an outsider in the research area, you may have to rely on your key informants for the first selection of participants in FGDs. Your key informants to whom you have explained thoroughly the purpose and the process of the FGD might each suggest some individuals who could be invited to a focus group discussion.

Note that the key informants may select persons similar to themselves so that you do not get an adequate variety of views in your discussion group. So in your explanations be sure to emphasize that you want a group of people that can express a range of views, to be able to have a proper discussion. Participants in a first FGD may assist to find relevant participants for other groups.

Another way of getting participants is to conveniently select individuals in a systematic way, to try and ensure a range of views. You might, for example, ask every third or fourth person you find. This method might be more suitable in urban areas.

iii. Physical arrangements: Communication and interaction during the FGD should be encouraged in every way possible. Arrange the chairs in a circle. Make sure that there will be no disturbances, sufficient quietness, adequate lighting, etc. Try to hold the FGD in a neutral setting which encourages participants to freely express their views.

iv. Preparation of a discussion guide: There should be a **written** list of topics to be covered. It can be formulated as a series of open-ended questions. Guides for different groups gathered to discuss the same subject may vary slightly, depending on their knowledge or attitudes and how the subject should first be explored with them.

v. Conducting the session: One of the members of the research team should act as 'facilitator' or 'moderator' for the focus group discussion. The other one should serve as 'recorder'. The facilitator should preferably be as close as possible to the participants in their characteristics (same sex, roughly same age, etc.).

Functions of the facilitator in an FGD

The facilitator should NOT act as an expert on the topic. His or her role is to stimulate and support discussion. Dear learner! If you are a facilitator on FGD, your functions include the following.

1) Introduce the session: Introduce yourself as facilitator and introduce the recorder. Let participants introduce themselves with whatever names they wish to use. Put the participants at ease and explain the purpose of the FGD, the kind of information needed, and how the information will be used (for the planning of a health program, an education program, business development program, etc). Ask permission to use a tape-recorder, let people hear their own voices before the session starts. You might offer drinks and allow some informal discussion before the actual session starts.

2) Encourage discussion: Be enthusiastic, lively, and humorous and show your interest in the groups' ideas. Formulate questions and encourage as many participants as possible to express their views. Remember there are **no** 'right' or 'wrong' answers. **React neutrally** to both verbal and non-verbal responses.

3) Encourage involvement: Avoid a question-and-answer session. Some useful techniques include:

- a. Asking for clarification: E.g. 'Can you tell me more about. . .?'
- b. Reorienting the discussion when it goes 'off the track':
 Saying: 'Wait, how does this relate to. . .?'
 Saying: 'Interesting point, but how about. . .?'
- c. Using one participant's remark to direct a question to another, for example, 'Mrs. X said. . ., but how about you, Mrs. Y?'
- d. When dealing with a dominant participant, avoiding eye contact or turning slightly away to discourage the person from speaking, or thanking the person and changing the subject.
- e. When dealing with a reluctant participant, using the person's name, requesting his/her opinion, making more frequent eye contact to encourage his/her participation.

4) Deal correctly with sensitive issues: If you notice that the discussion stops when dealing with a sensitive topic, you could ask participants (if literate) to anonymously write down their responses or opinions on the topic. Alternatively, you could summarize for the group some of the opinions from previous focus group discussions, focusing on one or two major contrasting opinions. Still another strategy is to form sub-groups, and to get a member of the sub-group to summarize and present the opinions of their sub-group members after which the whole group can still discuss these opinions.

5) Build rapport, empathize: Observe non-verbal communication. Ask yourself, ‘What are they saying? What does it mean to them?’ Be aware of your own tone of voice, facial expressions, body language, and those of the participants.

6) Avoid being placed in the role of expert: When asked for **your** ideas or views by a respondent, remember that you are not there to educate or inform. Direct the questions back to the group by saying: ‘What do you think?’, ‘What would you do?’ Set aside time, if necessary, after the session to give participants the information they have asked for.

Do not try to comment on everything that is being said. Don’t feel you have to say something during every pause in the discussion. Wait a little and see what happens.

7) Control the rhythm of the meeting, but in an unobtrusive way: Listen carefully, and move the discussion from topic to topic. Subtly control the time allocated to various topics so as to maintain interest. If participants spontaneously jump from one topic to another, let the discussion continue for a while since useful more information may surface; then summarize the points brought up and reorient the discussion.

8) Take time at the end of the meeting to summarize, check for agreement and thank the participants. Summarize the main issues brought up, check whether all agree and ask for additional comments. Thank the participants and let them know that their ideas have been a valuable contribution and will be used for planning the proposed research, intervention, or health education materials.

9) Listen for additional comments and spontaneous discussions which occur after the meeting has been closed.

?

Dear learner! 1. Define an FGD _____

2. List the function of a moderator/facilitator in an FGD _____

Functions of a recorder

The recorder should keep a record of the content of the discussion as well as emotional reactions and important aspects of group interaction. Assessment of the emotional tone of the meeting and the group process will enable you to judge the validity of the information collected during the FGD.

Items to be recorded include:

- ❖ Date, time, place
- ❖ Names and characteristics of participants
- ❖ General description of the group dynamics (level of participation, presence of a dominant participant, level of interest)
- ❖ Opinions of participants, recorded as much as possible in their own words, especially for key statements
- ❖ Emotional aspects (e.g., reluctance, strong feelings attached to certain opinions)
- ❖ Vocabulary used - particularly in FGDs that are intended to assist in developing questionnaires or health education materials
- ❖ Spontaneous relevant discussions during breaks or after the meeting has been closed

It is highly recommended that a tape-recorder be used to assist in capturing information. Even if a tape-recorder is used, notes should be taken as well, in case the machine malfunctions and so that information will be available immediately after the session for discussion.

If there is no reliable tape-recorder available, it is advisable to have **two recorders**.

A **supplementary role** for the recorder could be to assist the facilitator (if necessary) by drawing his or her attention to:

- ✓ missed comments from participants
- ✓ missed topics (the recorder should have a copy of a discussion guide during the FGD)
- ✓ If found, the recorder could also help resolve conflict situations within the group that the facilitator finds difficult to handle on her own.

Number and duration of sessions

1) Number of sessions: The number of focus group sessions to be conducted depends upon project needs, resources, and whether new information is still coming from the sessions, (that is, whether contrasting views within and between various groups in the community are still emerging). If not, you may stop.

One should plan to conduct at least two FGDs for each sub-group (for example, two for males and two for females). Otherwise you have no way of assessing whether the information you get from the first FGD is representative for that group.

2) Duration: A focus group session typically lasts up to an hour and half. Generally, the first session with a particular type of group is longer than the following ones because all of the information is new. Thereafter, if it becomes clear that all the groups have a similar opinion on particular topics, the facilitator may be able to move the discussion along more quickly to other topics which still elicit new points of view.

Processing and Analysis of Results

1) After each focus group session, the facilitator and recorder should meet to review and complete the notes taken during the meeting. This is the right moment to evaluate how the focus group went and what changes might be made in the topics when facilitating the next focus group. Immediately afterwards, a full report of the discussion should be prepared which reflects the discussion as completely as possible, using the participants'

own words. List the key statements, ideas, and attitudes expressed for each topic of discussion.

- 2) After the transcript of the discussion is prepared, **code-** following your topics- the participants' statements right away, using the left margin. Make finer sub-codes. **Write comments** (your first interpretation of the data) in the right margin. Formulate additional questions if certain issues are still unclear or controversial and include them in the next FGD. Further, categorize the statements for each topic, if required. (See Example 2 below)
- 3) When you have all the data, **summarize** it in a **compilation sheet** organizing the findings per topic for each. Number the FGD interviews and use key words to summarize group statements in the compilation sheet so that you can always go back to the full statement. If you have different categories of informants, e.g., male and female, you can summarize the information from the male and female groups on two **separate** compilation sheets.
- 4) Then you should do a **systematic comparison** between groups on all topics. Use your **objectives** and problem analysis diagram as a framework for analysis and comparison.
- 5) The next step could be to put the major findings for different study populations on one sheet. You may want to use some of these sheets in your research report.
- 6) Sometimes you may also wish to use diagrams when summarizing the causes or components of the problem under study.
- 7) Only now can you report the major findings of the FGDs in a narrative.

Report Writing on FGD

Start with a description of the purpose of the FGDs, the selection and composition of the groups of FGD participants and a commentary on the group process, so the reader can assess the validity of the reported findings.

Present your findings, following your list of topics and guided by the objective(s) of your FGD. Include quotations whenever possible as illustrations, particularly for key statements.

Example 1. FGD on reasons of nurses' turn over; discussion guide

- 1) When did the leaving of nurses become a problem to you? When did it start?
- 2) What, do you think, were (are?) the major reasons why they left the profession?
- 3) What problems do you experience due to this turnover of nurses?
- 4) How did the Ministry of Health react?
- 5) Did you take any initiatives to alert the Ministry to your problems?
- 6) Is it only because so many nurses have left that you experience problems in your work, or are other factors playing a role as well?
- 7) Do you have any concrete suggestions to the Ministry to improve your working situation?

Example 2 (*on next page*): FGD on reasons why nurses leave the profession; transcribed text from tape-recorder (Group of four nurses). **NB:** *This example was adopted from internet source.*

(codes)

(5), (4)

(5), (4)

Blaming
representa-
tives
that
notBalanced
view,
own resp.Action
not? (4)

(5)

(2), (4)
Blaming
MDH(2), (6),
Low
salaries

Nurse A (angry): We have been complaining about how things are in this Ministry but nobody listens to us. Every time we complain, we are told that things are being looked at and that soon things will work out. Nothing!

Nurse B: Yes! During our last strike on conditions of service, we were told that a committee had been set up to look at how our working conditions can be improved. Strange thing is that we had two representatives. I think we ourselves are also not very active in pushing our needs – how can we have representatives in the committee and nothing happens in two years?

Nurse C: Representatives in the committee, I think, they are promised better things for themselves and forget about everybody else's needs. I tell you, we ourselves are killing the nursing profession in this country.

Nurse A: I agree with you. We have asked these colleagues who are representing us to meet with us and tell us what has happened. We met them, but you don't hear what they are saying in the meetings. I really think that they are not expressing our views. No wonder they do not tell us what they are saying.

Nurse D (an elderly nurse who has not said anything yet): That is not fair to the representatives. I think they are trying their best. We all know how difficult it is to negotiate for these things. They are up against the whole Ministry. After all, most of the changes we want will take a long time. Of course we are suffering now but it is clear that the Ministry is trying to change things. We were told by the representatives that our salaries may be reviewed but that nothing can be done now. It will have to be at the beginning of the new fiscal year. I personally think we should not blame our representatives. If they are not performing, it is our fault for selecting the wrong people.

Facilitator: What then do you think needs to be done to ensure that nurses do not leave and who should do this?

Nurse A: I think we have done all we can. I told you, we have begged, asked, struck, and still ..., nothing.

Facilitator: What do you think are the major reasons why nurses are leaving their jobs?

Nurse A: I really think nurses are leaving because the Ministry is not interested in the welfare of nurses in this Ministry. Everything about nurses is wrong. We could write books about this! Even the President stated in his New Years speech that the nurses in our community are not performing as they should.

(Nurses start to discuss the New Years speech and go slightly off the track.)

Facilitator: What do you mean by welfare of nurses?

Nurse B: You know we get very low salaries here. I have still the same salary since I came to this Ministry five years ago. How can I pay for transport, the school fees for my children, food and rent – I don't even have any subsidised rent. You buy food and pay for transport and the money is finished.

(own
remarks)NB. *
Interview
representa-
tives !!*
Get overview
of salary
scales and
changes over
past 5 yrsMDH blames
nurses
instead of
looking for
structural
causes of
problem*Get News-
paper with
New Year's
speech!

Salaries

Nurse A: I think the amount of salary we get shows the Ministry does not value our work. I have had increments- but what can you do with an annual increment of R20.00? It's nothing. It is a joke on our efforts.

Salaries

Nurse B: Look at what X is getting since she left for Australia. At least three times as much as what we have, and things are expensive here, not much difference with Australia.

(6)
Working
Conditions

Nurse C: But we all know, the problem is not only salaries. Our working conditions are horrible. No consideration is given to our working conditions. Sometimes you are called back on your free days because a nurse is sick and there is no one to replace her.

Nurse B: Yes, even our welfare, nobody cares. I can give examples of many people who have big complaints about their welfare here.

Facilitator: I am still not so sure what you mean by welfare. Could you explain? (Asks nurse B to continue)

(6)
welfare/
working
conditions

Nurse B: Actually what we are discussing now is the welfare. You have no life and there is no consideration of your life outside your performance. We are sometimes transferred to places where there are no schools, no shops – nothing. You are not even given transport to come home. I was transferred to a very rural clinic and stayed there for five years – five years!! – they had completely forgotten about me.

Money not
first problem?
Working cond.
and attitude
RDH more
important?

6.6.2. Secondary Data and Sources

Secondary data are those which have been collected by someone else and which have already been passed through the statistical processes. Thus, studies made by others for another purpose represent secondary data.

Advantages of secondary data

1. It is found more quickly and cheaply
2. Helps much on past events and distant places, where the researcher cannot afford to collect the data right now.

Disadvantages of Secondary data

1. The information does not meet one's specific needs, since it is collected by others for their own purpose, definitions would differ, units of measurements would differ, and different time periods may be involved.
2. It is difficult to assess the accuracy of the information because one knows little about the research design or the conditions under which the research took place.
3. It is often out of date: a study made five years ago may not be relevant today.
4. This type of data may not be free from personal bias and prejudices.
5. It may not be adequate.

Classifications of Secondary Data Sources

1) Internal sources: Internal data come from internal sources related with the functioning of an organization or a firm where records regarding purchase, production, sales, profits, etc are kept on a regular basis. However, the internal data can be either insufficient or inappropriate for the statistical enquiry into a phenomenon. In that situation, we need external data. Organizational sources include department reports, producer summaries, financial and accounting reports, marketing and sales studies.

2) External sources of information: The external data are collected and published by external agencies. It includes the sources outside the organization. E.g. books, periodicals, government documents, reference books, encyclopedia, journals, magazines, government and non-government reports.

Chapter Summary

Dear learner! In this chapter, you learnt about data collection. Data refer to facts, figures and other relevant materials serving as bases for a study and analysis. You saw two types of data: Primary Data and Secondary Data. Primary sources are original sources for first-hand or primary data. Second-hand or secondary data in turn are sourced from secondary sources which contain data which have been collected by some other agency for other purpose.

For the collection of primary or secondary data, a number of Methods apply. For primary data collection, some of the basic methods used include observation, interviews, questionnaire, Schedules, and Focus Group Discussions. Mainly, there are two sources of secondary data: Internal sources and External sources. Whereas internal secondary data are obtained from sources related with the functioning of an organization or firm, the external data are collected and published by external agencies or outsider organization.

Exercises

Multiple Choice Questions

1. The data which are collected afresh are_____.
 - a. Primary data
 - b. Secondary data
 - c. Second hand data
 - d. band c
2. Of the following, one statement does not contain a quality a statistical unit should have
 - a. A statistical unit must be specific and unmistakable.
 - b. A statistical unit must be stable for meaningful comparison
 - c. A statistical unit must not give varying characteristics at different times and places.
 - d. None of the above
3. An interview in which the interviewer asks the questions face-to-face with the interviewee is called_____.
 - a. Personal Interviews
 - b. Telephone Interviews
 - c. Computer Direct Interviews
 - d. E-mail Surveys
4. Mailed questionnaire is characterized by one of the following
 - a. Mail surveys allow the respondent to answer at their leisure.
 - b. Mailed questionnaires are both very economical and very fast.
 - c. The respondents must have access to a computer
 - d. Response rates are usually higher in Mailed surveys
5. In relation to open-ended questionnaires, closed-ended questionnaires
 - a. Permit an unlimited number of possible answers
 - b. Easier to code and statistically analyze
 - c. Permit creativity, self-expression, and richness of detail
 - d. Consume greater amount of respondent time, thought and effort
6. Which of the following is not a method of data collection?
 - a. Questionnaires
 - b. Interviews
 - c. Experiments
 - d. Observations
7. Secondary/existing data may include which of the following?
 - a. Official documents
 - b. Personal documents
 - c. Archived research data
 - d. All of the above

8. Which of the following terms best describes data that were originally collected at an earlier time by a different person for a different purpose?
- a. Primary data
 - b. Secondary data
 - c. Experimental data
 - d. Field notes
9. Open-ended questions provide primarily _____ data.
- a. Confirmatory data
 - b. Qualitative data
 - c. Predictive data
 - d. None of the above
11. Introducing a stimulus to the group for it to react to and observing the reactions is called _____ observation.
- a. Controlled
 - b. Naturalistic
 - c. Complete
 - d. Probed
12. When constructing a questionnaire, it is important to do each of the following, except
- a. Use "leading" or "loaded" questions
 - b. Use natural language
 - c. Understand your research participants
 - d. Pilot your test questionnaire
13. Another name for a *Likert* Scale is a(n):
- a. Interview protocol
 - b. Event sampling
 - c. Summated rating scale
 - d. Ranking
14. The type of interview in which the specific topics are decided in advance but the sequence and wording can be modified during the interview is called:
- a. The interview guide approach
 - b. The informal conversational interview
 - c. A closed quantitative interview
 - d. The standardized open-ended interview
15. What is the role of the moderator in a focus group?
- a)** To stimulate discussion and keep the conversation on track
 - b)** To ask leading questions and dominate the discussion
 - c)** To sit away from the group and observe their behavior
 - d)** To evaluate the group's performance on a particular task
16. What should the moderator say in their introductory remarks?
- a)** Thank the participants for coming
 - b)** Who they are and what the research is about
 - c)** How the focus group will proceed
 - d)** All of the above

17. The researcher has secretly placed him or herself (as a member) in the group that is being studied. This researcher may be which of the following?

- | | |
|---------------------------|-----------------------|
| a. A complete participant | c. A partial observer |
| b. An interviewer | d. None of the above |

Chapter Seven

Data Processing and Analysis

7.1. Introduction

Dear learner! In the preceding chapter, you learned about data and methods of data collection. But in this chapter, you will see the next step in a research process, i.e. how to process and make sense of the data collected in the form of written text. As discussed in previous chapter, this text may be derived from observations, interviews, case studies or other sources. Now we will discuss how to categorize, synthesize and analyze this data.

7.2. Lesson objectives

Up on the successful completion of the chapter, you will be able to:

- ✓ Familiarize yourself with the crucial stages in statistical processing of data
- ✓ Understand the various concepts of editing, coding, classification, and tabulation;
- ✓ Clearly define what data analysis is
- ✓ Identify the various techniques of data analysis
- ✓ Understand the roles of statistics in research

7.3. Data processing

Technically speaking processing implies editing, coding, classification and tabulation of collected data. Data collected during the research is processed with a view to reducing them to manageable dimensions. A careful and systematic processing will highlight the important characteristics of the data, facilitates comparisons and render it suitable for further statistical analysis and interpretations. In other words, data processing is an intermediate stage between the collection of data and their analysis and interpretation. Once data are processed, the data presentation exercise begins. A detail discussion of some broad areas of work for statistical processing and analysis is given below.

1. Editing the Primary Data

After the data have been collected by the primary method, the next step is to edit the filled schedules or questionnaires. Editing is always done in the office, far away from the

place of data collection. In many cases, it is not possible to check all schedules. In such a case, a sample of required size is sufficient to ascertain the degree of accuracy of filled schedules. The duty of editing the schedules must be entrusted to honest, efficient and sincere editors who are committed to the job. The editors are required to check the filled schedules in the following respects.

Consistency
Completeness
Accuracy
Uniformity
Computations

- **Editing for consistency:** The schedules are to be edited by editors for consistency. They should see whether answers to questions supplied by informants are consistent or not. If the answers are of contradictory nature then the editors are required to send such schedules to respective supervisors for recording the correct answers from concerned informant. For illustration, an informant might enter an “unmarried” in one column. While in the second column, he may enter “two children”. Such contradictory replies are to be corrected. Same is the case with age date of birth information, which may not agree.
- **Editing for completeness:** It is the duty of the editor to examine whether all questions in the schedule or questionnaire are answered or not. In many cases, important questions are not answered and the analysis of the problem is not possible due to such unanswered questions. The editor should send such schedules to concerned investigators for recording the answers from informants. In case the editor has not got the needed data in spite of his best efforts, he should enter N.R. (Not Reported), where the entries are missing. If important questions remain unanswered, then such schedules should not be for analysis.
- **Editing for accuracy:** The job of the editor is highly specialized; therefore, he must know the nature of information which is to be supplied by the informants. The accuracy and the reliability of the findings depend upon accurate information, therefore, the editor has to examine whether all questions are answered correctly or not. In case, any doubt

develops about any question in the mind of the editor, he must seek clarification from the supervisor or from the informants.

- **Editing for uniformity:** The problem under study can be rationally analyzed when there is uniformity in the answers of the questions supplied by different informants. Uniformity of answers means whether all questions are interpreted in the same sense by all informants or not. If the questions are interpreted in different ways by different informants, then the data supplied becomes heterogeneous. Such data should not be employed for analysis.
- **Editing to make computations:-** It is advised not to ask the investigators or the respondents supplying information to make any computation. This additional work may give risk to errors. Computations, if necessary, should always be made by editors.

?

Dear learner! As editor, against what important elements should you edit the filled survey before analysis? _____

Levels of Editing

The editing can be done at two levels: on the field or centrally in the office.

a) Field level editing (where the data is collected): during the time of data collection, the interviewer often uses ad hoc abbreviations, special symbols and the like. As soon as possible, after an interview, field workers should review their reporting forms, complete what was abbreviated, translate personal shorthand and re-write illegible entries. But here attention must be given that investigator must not correct errors of omission simply by guessing what the respondent would have said if the question had been asked.

b) Central Editing: at this stage, the research form or schedule should get a thorough editing and this takes place when all forms and all schedules have been completed and returned to the office. One editor or team of editors may correct obvious errors such as entry in wrong place, recordings in wrong units, etc. In case of inappropriate or missing replies, the editor can sometimes determine the proper answer by reviewing the other information in the schedule. At times, the respondent can be contacted for clarification.

2. Coding

After editing of the collected data, the next step to follow is coding. Coding refers to assigning of number, digits or letters or both to various responses so as to enable tabulation of information easy. The purpose of coding is to classify the answers in to meaningful categories, which is essential for tabulation. Coding is, therefore, necessary to carry out the subsequent operations of tabulation and analyzing data.

Coding consists of assigning a number or symbols to each answer, which falls in a predetermined class. Coding means an operation by which data are organized into classes and number or symbol is given to each item according to the class in which it falls. For example, a researcher may code Male as 0 and Female as 1.

The process of coding involves two distinct steps. The first is to decide on *the categories* to be used, the second *to allocate individual answers to them*. The set of categories to be used is referred to as the *coding frame*. The set of coding frames covering all the information to be abstracted from the questionnaires is commonly known as the *codebook*. The set of rules stating that certain numbers are assigned to variable attributes is called *coding procedure*.

3. Classification

Once the data is collected and edited, the first task of the researcher or statistician is the organization of the figures in such a form that their significance, for the purpose in hand, may be appreciated that comparison with masses of similar data may be facilitated, and that farther analysis may be possible. This is done through classification and tabulation. But before tabulating the data into different homogeneous classes, it is necessary to sort out the relevant and significant features from the irrelevant and insignificant ones. *The process of arranging the data into groups or classes according to their resemblances and similarities is technically called classification.*

Classification is the process of arranging data groups according to their common characteristics, or separating them in to different but related parts.

Example: The number of students registered in Arba Minch University during the academic year 2000-2008 may be classified on the basis of any of the following criterion.

- Sex
- Age
- The region to which they belong
- Religion
- Different faculties like FBE, Social sciences and Humanities, Natural and Computational science, Engineering and Technology, Agriculture, and Health.
- Heights or weights, etc

In short, classification can be in the form of either *attribute* (literacy, sex, honesty, etc) or *class intervals* (like income, population, age, weight etc).

a. Classification according to attributes: data are classified according to some common characteristics which can be either descriptive (literacy, sex, honesty, etc) or numerical (such as weight, height, income, etc). Descriptive characteristics refer to qualitative phenomena which cannot be measured quantitatively. In this case, we classify the data only by noticing the presence of these characteristics. Data obtained in this way on the basis of certain attributes are known as statistics of attributes and their classification is said to be classification of attributes.

b. Classification according to class intervals: this refers to statistical data or quantitative phenomena like income, population, age, weight, etc which can be classified on the basis of class intervals. For instance, persons whose incomes are within Birr 101 to 200 can form one group where as those whose incomes are within Birr 201 to 300 forms another group. In this way, the entire data may be divided in to a number of groups or classes or what are usually called class intervals. Each group or class interval has an upper limit as well as a lower limit known as class limits. The difference between the two class limits is known as class magnitude. The number of items that fall in a given class is known as the frequency of the given class. All the classes or groups, with their respective frequencies taken together and put in the form of table, are described as group frequency distribution or simply frequency distribution. Classification according to class intervals usually involves the

following problems:

i. How many classes should be there? What should be their magnitude? There is no specific answer to the first question and the decision is left to the skills and experiences of the researcher. However, we may have 5 to 15 classes. With regard to the second question, to the extent possible, class magnitudes should be of equal magnitudes, but in some cases, unequal magnitudes may result in better classification. Hence the researcher's objective judgment plays an important part in this connection. Multiples 2, 5, and 10 are generally preferred while determining class magnitudes. Some statisticians adopt the following formula:

$i = R / (1 + 3.3 \log N)$ Where i = size of class interval

R = Range (i.e. the difference between the values of the largest item and smallest item among the given items)

N = Number of items to be grouped

ii. How to choose the class limits? In this case, the researcher must consider the criteria that the mid point (generally worked out first by taking the sum of the upper limit and the lower limit of a class and then divide this sum by 2) of a class interval and the actual average of items of that class interval should remain as close to each other as possible. Consistent with this, the class limits should be located at multiples of 2, 5, 10, 20, 100, and such other figures. Class limits may generally be stated in any of the following forms:

◇ *Exclusive type class intervals are usually stated as follows:*

10-20

20-30

30-40

40-50

The above interval should be read as follows:

10 and under 20

20 and under 30

30 and under 40

40 and under 50

◇ Inclusive type class intervals are usually stated as follows:

11-20

21-30

31-40

41-50

In inclusive class interval, the upper limit of a class interval is also included in the concerning class interval. Thus, the item whose value is 20 will be put in 11-20 class intervals. The real upper limit in this case is 20.999999 and as such 11-20 class interval really means 11 and under 21.

iii. How to determine the frequency of each class? This can be done either by tally sheets or by mechanical aids. Under the tally sheet technique, the class-groups are written on a sheet of paper known as tally sheet and for each item; a stroke or (a small vertical line) is marked against the class group in which it falls. The general practice is that after every four small vertical lines in a class group, the fifth line for the item falling in the same group, is indicated as horizontal line through the said four lines on the resulting flower (///) represents five items. See the following example.

Table 7.1. An illustrative tally sheet determining the number of 73 employees in different income categories

Income group (in Br.)	Tally mark	Number of employees (class frequency)
100-300	/// /// ///	15
301-500	/// /// /// ///	18
501-600	/// /// /// ///	20
Above 600	/// /// /// ///	20
Total		73

Objects of classification

The chief objectives of classification are:-

1. *To present the facts in a simple form:* - classification process eliminates unnecessary details and makes the mass of complex data simple, brief, logical and understandable. For example, the data collected in a population census is so huge and fragmented that

it is not possible to draw any conclusion from them. When these massive figures are classified according to sex, education, marital status, occupation, etc. then the structure and nature of the population can easily be understood.

2. *To bring out clearly points of similarity and dissimilarity:* - Classification brings out clearly the points of similarity and dissimilarity of the data so that they can be easily grasped. Facts having similar characteristics are placed in a class; such as educated, uneducated, employed, unemployed, etc.

3. *To facilitate comparison:* - classification of data enables one to make comparison, draw inference and locate facts. This is not possible in an unorganized and unclassified data. If marks obtained by students in two colleges are given; no comparison can be made of their intelligence level. But classification of students into first, second, third and failure classes on the basis of marks obtained by them will make such comparison easy.

4. *To bring out relationship:* - Classification helps in finding out cause-effect relationship, if there is any in the data. For example, data of small-pox patients can help in finding out whether small-box cases occurred more in vaccinated or unvaccinated population.

5. The process of classification *enables one to form a mental picture of objects of perception and conception.* Summarized data can easily be understood and remembered.

6. *To prepare the basis for tabulation:* - Classification prepares the basis for tabulation and statistical analysis of the data. Unclassified data cannot be presented in tables.

Though many advantages are derived from classification, some amount of details is lost in the process of summarization. The greater is the extent of summarization, the more is the loss of details. The statistician will have to weight and balance the advantages he would derive from summarization, the loss of details he will have to suffer, and decide on the extent of summarization he will have.

Types of classification

The raw data are not digestible; therefore, it must be classified for understanding the salient features of the data. There are limitless ways of classifying the data but the main ones are:-

1. Classification based upon differences in kind
2. Classification based upon differences of degree of a given characteristic
3. Geographical classification
4. Chronological classification
5. Alphabetical classification

?

Dear learner! What do we mean by classification? _____

4. Tabulation

Tabulation involves the orderly and systematic presentation of numerical data in a form designed to elucidate the problem under consideration.

Objects of Tabulation

Tabulation is a process, which helps, in understanding complex numerical facts. The purpose of table is to summarize a mass of numerical information and to present it in the simplest possible form consistent with the purpose for which it is to be used. Tabulation has the following objectives to fulfill:-

1. *To clarify the object of investigation:* - The function of tabulation in the general scheme of statistical investigation is to arrange in easily accessible form the answer with which the investigation is concerned. The presentation of data in tables makes the problem under study simple and clear and answers many problems that are found there.
2. *To clarify the characteristic of data:* - A table presents facts clearly and concisely, eliminating the need for wordy explanation. It brings out the chief features of data.
3. *To present facts in the minimum of space:* - A table presents facts in minimum of space and communicates information in a far better way than textual material.
4. *To facilitate statistical process:* - It simplifies reference to data and facilitates comparative analysis and interpretation of the facts.

Advantages of Tabulation

The tabular presentation of data has several distinct advantages. They are:-

1. *It simplifies facts:* - Tabulation is helpful in presenting mass of complex data clearly, so that their features can easily be understood. After tabulation, data get a definite form

and structure, and can be studied from up to below and left to right easily. It reveals patterns within the figures which can not be seen in narrative form.

2. **Economy**:- It occupies less space than textual reports, hence saving of paper and time.

Required figures can be located more quickly.

3. **Helpful in comparison**:- Tabulation makes comparison easy. Comparable figures are placed in juxtaposed columns; hence a comparative study can be made.

4. **Facilitates computation**:- It helps in computation of different statistical measures. For calculating average, dispersion, correlation, etc. data must be presented in tables.

5. **Helps classification**:- Tabulation is helpful in classification. After classifying data, the characteristics of different classes are not clear till they are placed in a suitable table.

6. **Helps in Reference**:- Tabulated data are good for reference purposes

7. **Helps in interpretation and presentation**:- It is convenient to interpret tabulated data. Tabulated data make it easy to present the facts in a graphical and diagrammatic form.

Tabulation may be done manually by hand or mechanically or electronically using computers. In the case of computer tabulation, computer programs such as SPSS, LOTUS, EXCEL, STATA, etc could be used.

In general, tabulation refers to the orderly arrangement of data in a table or other summary format. At the very beginning, we have to develop a master sheet, and transform all information from each questionnaire on this master sheet. Counting the number of responses to a question and putting them in a frequency distribution is tabulation.

Example 1: Consider the following hypothetical question and possible way of tabulation:

“Do you have Computer? a. Yes b. No”

Table 7.2. Illustration on the tabulation on possession of Computer

Responses	Frequency	Percentages (%)
Yes	27	54
No	23	46
Total	50	100

i. **Frequency**- refers to the number of times same responses are repeated or it simply shows the number of respondents under specific response

ii. **Percentage**- refers to the relative weight of some frequency in terms of hundredth scale of the total elements under survey. For simplicity and comparison purpose frequency distribution has to be followed by percentages.

Example 2: Let's further consider the following technique of cross-sectional tabulation:

Cross – tabulation: a technique of organizing data by group category or classes.

Do you have job? Yes----- No-----.

If “yes”, in what category are you employed? A) Self B) Government C) NGO

Do you have Computer? Yes-----No-----.

Now by cross-tabulation, we consider computer possession by occupation.

Table 7.3. Computer possession by occupation

Occupation	Response					
	Yes	%	No	%	Total	%
Self employed	8	29	20	71	28	100
Government	19	27	51	73	70	100
NGO	10	45	12	55	22	100
Total	37	31	83	69	120	100

Difference between Classification and Tabulation

In statistical investigation, both classification and tabulation are important processes by which collected data are summarized and put in a systematic order. Still, there are some differences between these three processes. They are:-

1. Both the processes have to be done in a sequence. One is not the substitute for the other. First, data classified, and then they are presented in tables. Hence, classification forms the basis for tabulation.
2. In classification process, data are classified into different classes according to their similarities and dissimilarities. In tabulation, the classified data are placed in columns and rows. Thus, tabulation is a mechanical function of the classification
3. Classification is a process of statistical analysis; tabulations a process of presenting data in suitable structure. In brief, in classification process, data are classified in to classes

and sub-classes, while tabulation presents those classified data permanently under headings or sub-headings. Tables may also contain derivatives like percentages, coefficients, proportions, etc to facilitate comparison.

Rules for tabulation

Rules for tabulation may be divided into two groups

- a) Rules relating to table structure
- b) General rules

What do we mean by each? Let us see them as follows.

a) The structure of a table has the following parts.

1) Number- Each table should be numbered so that it may be easily identified. The number of the table should be given at the top above the title of the table so that it may easily be noticed.

2) Title- A table should have a complete title as it helps in finding the information wanted. A title should tell in concise language.

- ✓ The nature of data
- ✓ The data source
- ✓ What time period the data covers
- ✓ How the data are classified

The titles of the table should be short because long titles are not easy to read. The wording of the title should be carefully planned so that it may give only one interpretation.

3) Stub: The designations of the horizontal rows or the data in the table are called stub or stub items. They are listed along the Y-axis (the left hand column of a table). The complete column of this designation is known as stub. The stub items should be complete and clear. It is always advisable to condense the stub items so that they may be written in one line.

4) Caption: The heading of the column is called caption. Caption should be carefully worded and written in the center at the top of the column. If different columns are expressed in different units, the definition of the units should be included in the caption.

5) Body: The body of the table contains figures that the table is designed to present to readers. The data is the body corresponding to classification under caption and stub.

6) Foot-notes: The footnote is a phrase or a statement, which makes clear some specific items. If some figure in a column or a row is exceptional and does not belong under caption or stub, then foot-note is used to describe the exact situation. A letter or some other non-numerical symbol should be employed to key a foot note. The foot-note is always given at the bottom of the table and above the source.

7) Source: The source of the data, embodied in the table should be written so that whosoever uses the data may trace the data to the source without any difficulty. The data source should give information about the place from which data were obtained.

Specimen of a Table

Table Number ____ ____ ____ ____

Title ____ ____ ____ ____

Head note ____ ____ ____ ____

Stub head	Caption	
	Column head	Column head
Sub entries	Body of the table	

Foot-note ____ ____ ____

Source: ____ ____ ____

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Dear learner! 1. Define tabulation _____
2. What important parts does the structure of a table include? _____

7.4. Data Analysis

7.4.1. Meaning of Analysis

As you remember from the previous section, you learned about editing, coding, classifying and tabulation of data. Once data are edited, coded, classified and tabulated the data presentation exercise begins. Large volume of raw statistical information need to be reduced to more manageable dimension if one has to see the meaningful relationships in it.

Data analysis is the computation of certain indices or measures. It involves the computation of certain measures along with searching for patterns of relationships that exists among data group. Analysis of data involves a number of closely related operations that are performed with the purpose of summarizing the collected data and organizing these in such a manner that they will yield answer to the research questions or suggest hypothesis or question.

Analysis of data refers to seeing the data in the light of hypothesis or research questions and the prevailing theories and drawing conclusion that are as amenable to theory formation as possible.

Analysis of data is the most skilled task of all the stages of the research. It is a task calling for the researcher's own judgment and skill. Proper analysis requires a familiarity with the background of the survey and with all its stages. The steps envisaged in the analysis of data vary depending on the types of the study. If there exists a set of clearly formulated hypotheses to start the study with, then each hypotheses can be seen as a work prescribing a certain action to be taken vis-à-vis the data. The more specific the hypothesis, the more specific the action will be. In such a study, the analysis is almost completely a mechanical procedure. The task of analysis is reduced just to getting the appropriate combinations of data and reading them off against the instructions for verification and falsification of hypothesis.

Part of analysis is a matter of working-out *statistical distribution*, constructing of *diagrams* and calculating simple measures like *averages*, *measures of dispersion*, *percentages*, *correlation*, etc. Thus, statistical analysis forms part of survey analysis. Analysis means verification of hypothesis under ideal conditions of precision and simplicity; analysis presents very few problems since the statement of the hypothesis and the elaboration of the experimental design will automatically provide for the analysis of the data.

In the case of survey or experimental data, analysis involves *estimating the values of the unknown parameters of the population and testing hypothesis for drawing inferences*. Analysis may therefore be categorized as *descriptive analysis or inferential (statistical)*

analysis. With respect to the number of variables involved in the analysis, it can be also divided in to *uni-variate analysis and multi-variate analysis*. Uni-variate analysis refers to the analysis with respect to one variable. It is also called one dimensional analysis. Such analysis is the study of distribution of one variable. In this case, the process of data reduction is the preserve of descriptive statistics. The initial uni-variate analysis may be the presentation of descriptive analysis in the form of frequency distributions. The uni-variate analysis could either be presented in the form of statistical measures such as measures of central tendencies, measures of variation or dispersion or in the form of graphs. Graphical illustrations (such as histograms, line graphs, bar graphs, circular graphs or pie charts etc) could also be used to demonstrate the frequency distribution.

Secondly, the multi-variate analysis involves the consideration of two or more variables. If we have two variables, we call it a bi-variate analysis; but if we have more than two variables, we call it multi-variate analysis. This analysis can be undertaken in the construction of bi-variate tables, or multi-variate analysis such as multiple regressions, ANOVA, etc

The problems raised by the analysis of data are directly related to the complexity of the hypothesis or hypotheses.

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Dear learner! What does it mean by data analysis? _____

7.4.2. Scales of Measurement

In our daily life, we use various measuring instruments to determine weight, height, or some other features of a physical object. We also measure how well we like a song, a painting, a film, or personality of our colleagues. In short, we measure physical and abstract concepts. Measurement is relatively complex. Here, measurement is to mean the process of assigning numbers to objects or observations. The level of measurement is a function of the rules under which the numbers are assigned.

It is easy to assign numbers for some objects and difficult for some others. For example, whereas it is easy to assign numbers for properties like weight, height, mass, etc, it will be difficult to assign numbers (to measure) such properties as the attitude of individuals,

the motivation to succeed, comprehension of a student, etc. There is high accuracy of measurement in measuring the properties of a physical object. Yet when the concept is abstract and the measuring units are not standardized, we are less confident as to the accuracy of the results of the measurement.

Types of Scores/scales of measurement

The most widely used classifications of measurement scales are four:

- a. Nominal scale
- b. Ordinal scale
- c. Interval scale, and
- d. Ratio scale

a. Nominal scale of measurement:-Nominal scale is simply a system of assigning numbers or symbols to events in order to label them. For example, the assignment of numbers to identify gender (say, 1=Female and 0=Male). The numbers do not show an order and do not have any quantitative value. They are merely convenient labels for the particular class of events. One can not do much with the numbers involved: they can not be averaged and compared; no measure of dispersion is employed. The counting of members in each group is the only arithmetic operation when a nominal scale is employed. We are limited to use only Mode as a measure of central tendency. Chi-square test is the most common test of significance that can be utilized, and for the measure of correlation, the contingency coefficient can be worked out.

Nominal scale is the least powerful level of measurement. It indicates no order or distance relationship and has no arithmetic origin. A nominal scale simply describes differences between things by assigning them to categories.

b. Ordinal scale of measurement: - The ordinal scale places events in order or rank, but there no attempt to make the intervals of the scale equal in terms of some rules. The rank orders are usually used in research relating to qualitative phenomena. An example is the rank of Athletes on the track. For example, if Tirunesh's position in the race is 1st and Ejigayehu's position is 4th, it can not be said that Tirunesh's position on the track is 4 times as good as that of Ejigayehu's. Ordinal scales only permit the ranking of items

from highest to lowest. Ordinal scales have no absolute value, and the real differences between the adjacent ranks may not be equal. It can be said one person is higher or lower on the scale than another, but more precise comparisons can not be made, i.e., “greater than” or “less than” (or “equal to”) statements are acceptable but by how much greater or less can not be stated. The real difference between ranks 1 and 2 may not be similar to the difference between rank 6 and 7 as the numbers of this scale have only a rank meaning. The appropriate measure of central tendency used in this measuring scale is the median.

c. Interval scale of measurement: - In this case, the intervals are adjusted in terms of some rule that has been established as a basis for making the units equal. The units are accepted so far as one accepts the assumptions on which the rule is based. Interval scales can have an arbitrary zero, but it is not possible to determine for them what may be called an absolute zero or the unequal origin. The limitation of an interval scale is that it has no true zero value; it does not have the capacity to measure the complete absence of a trait or characteristic. The Celsius /Fahrenheit scale is an example of an interval scale and shows similarities in what one can or can not do with it. One can say that an increase in temperature from 10^0 to 20^0 involves the same increase in temperature as an increase from 70^0 to 80^0 , but one can not say that the temperature of 80^0 is four times as far as the temperature of 20^0 as both numbers are dependent on the fact that the zero on the scale is set arbitrarily at the temperature of the freezing point of water. The ratio of the two temperatures of 20^0 and 80^0 does mean nothing as zero is an arbitrary point.

Interval scales provide more powerful statistical measurement than ordinal scale for interval scale also incorporates the equality of interval. Mean is an appropriate measure of central tendency, while standard deviation is the most widely used measure of dispersion.

d. Ratio scale of measurement: - The ratio scales have an absolute or true zero of measurement. The term “absolute zero” is not as precise it was ones believed to be. Ratio scale represents the actual amount of variables. Measures of physical dimensions such as weight, height, distance, etc are examples. With ratio scale values, all statistical techniques can be used and manipulations carried out. Unlike in others, multiplications and divisions can be employed in the ratio scale. Geometric and harmonic means (as

measures of central tendency) and coefficient of variation (as a measure of dispersion) can be calculated from the values of the ratio scale.

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Dear learner! Mention the types of scales of measurement? _____

7.4.3. Types of Data Analysis

1. Secondary Data Analysis

There are two aspects of scientific research, however, which makes the occurrence of so ideal a situation relatively infrequent. The first of these is the appearance of an anomalous empirical regularity or the absence of an expected regularity. Such instances require analysis not anticipated by the original design. Similar to this is the use of data gathered for one purpose, for quite another problem. Both these represent what is called secondary data analysis. From the very ‘*pure*’ experimental point of view, such analyses are considered to yield answers which are ‘plausible’ but not capable of being stated in the customary ‘probability’ terms of science.

Problems of data analysis involve all the questions raised in research design, for secondary analysis to involve the designing and redesigning of substitutes for the controlled experiment. This indicates that there are two types of questions about analysis. One of them relates to the techniques of representing the data and the other to the methods of logically ordering them so that questions can be raised and answered.

2. Primary Data Analysis

The basic ingredient of any analysis is the scientific method and the testing of hypotheses. However, in case studies, hypothesis may some times be absent. In normal circumstances, the statement of a hypothesis and an elaboration of the experimental design would provide for a rational analysis of the data.

An analysis would necessitate first the tabulation and presentation of the data in the most intelligible, attractive, appropriate and comprehensible form. It would enable the researcher to identify relationship and to make inductions and deductions, wherever necessary. On the basis of statistical tables, it is possible to present the facts in the form

of diagrams, graphs, pictograms, and statistical and mathematical models. Representation and analysis would become easier with the help of these tools. The analysis of data would throw light on the various problem areas, enabling the researcher to identify ways and means of arriving at solution. The problems raised by the analysis of data are directly related to the complexity of the hypothesis or hypotheses.

Statistical and mathematical representations have attained prominence in research analysis in recent times. Frequency distribution, tabulation, coefficient of correlation, regression, standard deviation, econometric models and various other mathematical techniques are usually employed in research and managerial analysis. Percentages, ratios and proportions are also of very common use in any research analysis, which simplify the process of comparison.

7.4.4. Characteristics of Data Analysis

Following are the main characteristics of analysis of data:-

1. Analysis of data is one of the most important aspects of research. Since it is highly skilled and technical job, it would be carried out by the researcher him/herself or under his/her close supervision. The researcher should also possess judgment skill, ability of generalization and be familiar with the background objects and hypothesis of the study.
2. Data, fact and figures are silent and they never speak for themselves but they have complexities. It is only by organizing, analyzing and interpreting the research data that we can know their important features, inter-relationships and cause-effect relationships. The trends and sequences inherent in the phenomena are elaborated by means of generalization.
3. The data to be analyzed and interpreted should
 - (i) be reproducible
 - (ii) be readily disposed to quantitative treatment, and
 - (iii) have significance for some systematic theory, and can serve as a basis for broader generalization.
4. The task of analysis is incomplete without interpretation. In fact, analysis and interpretation of data are complementary to each other. The end product of analysis is the setting up of certain general conclusions while the interpretation deals with what these conclusions really mean.

5. Since analysis and interpretation of data are interwoven, the interpretation should more properly be conceived of as a special aspect of analysis rather than a distinct operation. Interpretation is the process of establishing relationship between variables, which are expressed in the findings and why such relationships exist.
6. For any successful study, the task of analysis and interpretation should be designed before the data are actually collected with the exception of formulative studies where the researcher has no idea as to what kind of answer he wants. Otherwise, there is always a danger of being too late and the chances of missing important relevant data.

7.4.5. The Place of Statistics in Social Research

(For detail knowledge on this issue, remember what you learned on your course "Statistics for Management")

Knowledge of basic statistical concepts and techniques is necessary for an intelligent understanding of the generality of life. Out of the welter of single events, social researchers seek for general trends; out of the vast and confusing variety of individual characters, they continually search for underlying group characters.

There are essentially two reasons why the people of statistics and the need to study statistics have grown enormously in the field of social sciences. One reason is that the huge amount of data collected by a researcher need simplification so as to render them capable of being commonly understood without much difficulty. The second and even more important reason is the increasingly quantitative approach currently being employed in social researches.

In fact, statistical considerations enter only into the analysis stage of the research process, after the data have been collected, and near the point in time when the initial plans for analysis are made and a sample is to be drawn. But this does not mean that a social researcher can plan and carryout his/her entire research without any knowledge of statistics and then hand over the data to the statistician for analysis. If he/she did this, the results of costly research project would probably be disappointing if not useless. Evidently, problems that will be encountered in analysis and interpretation have to be anticipated at every stage in the research process, and in this sense statistical methods may be involved throughout. This implies that statistics is useful tool for the social scientist. It is a much more useful tool for exploratory analyses than might possibly be

imagined. Most social researches are based on highly tentative theoretical ideas. The variables that should be controlled in the analysis or even the priorities and sequences of analysis, and the steps that should be followed are neither precise nor predetermined. Researchers are generally surprised at the complexity of data-analysis as soon as one introduces a set of variables. It is in these instances; especially that knowledge of the statistical methods becomes an invaluable tool enabling the social researchers to disentangle highly complex interrelationships.

Statistical methods are very useful for the research purposes in the following situations:

1. When a large number of quantitative data needs simplification, analysis and verification
2. Where estimates are needed for some unknown events on the basis of available information
3. When the data collected are very large in numbers, it becomes practically very difficult to examine every individual in detail. Under such circumstances, conclusions are derived with the help of sampling techniques, which now occupy a place of central significance among statistical techniques.
4. When the research problem is affected by multiplicity of causes and the use of experimental method is not possible, then in such a case, the only possible method is statistical method.
5. When the problem under study is a phenomenon of uncertainty, then the use of statistical method becomes necessary.

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Dear learner! In what situations do we employ statistics in researcher? _____

Types of statistical Methods

Statistical Methods are the principles employed for the description, analysis and interpretation of the data. They may be classified into two categories, namely:

- i. Descriptive statistical methods:** Those methods which are employed primarily to describe what has been observed are called descriptive statistical methods. Their sole purpose is to describe the behavior of a variable and the attempt is made to analyze either by tables or by diagrams to describe the behavior of the data.

ii. Analytical statistical methods: Those methods which are employed to analyze and interpret what has been observed are called *analytical statistical methods*.

The two categories of statistical methods are not mutually exclusive but analytical statistical methods are based upon and make use of descriptive methods for analyzing and interpreting data. For example, a research worker may be entrusted with the job of describing as well as analyzing and interpreting the problem under study. In some of the investigations, the sole purpose is to analyze and interpret what has been observed rather than describing it. For example, a research worker in a biological laboratory who is studying the effectiveness of two drugs in curing a particular disease in rats is interested in an analysis, comparison and interpretation of his/her data. In special studies conducted by government, such as population census, per acre-yield, etc the data are collected and reported in order to describe the situation and they are analyzed and interpreted to determine trends and possible future developments.

Uses of Statistical Methods

Statistical Methods are employed, almost, in every branch of knowledge. They are valuable tools in the hands of research workers and are used to cast light upon the behavior of a variable or variables under study. The following are the main uses of statistical methods:

1. *Statistical methods are employed to throw light upon the situation and to probe the unknown.* For example, in agriculture, these methods may be employed to study the effects of different kinds of fertilizers upon crop-yield. Had there been no statistical methods, it would not have been possible to establish any type of relationship between crop-yield and fertilizer. In the absence of yield estimates, it would have been difficult to persuade the farmers to apply more of fertilizer.
2. *Statistical methods enable us to reduce a mass figure to something that is of manageable size and can be easily understood and interpreted.* Suppose, we are given the monthly income of 1,000 workers of certain factory and our brain is not so fertile as to remember such a huge mass of data. With the aid of statistical methods, we may reduce a mass of data to a single figure, which is easy to remember and to compare.

3. *Statistical methods are employed for comparing two or more series.* Suppose we wish to compare wages in two firms. We would not gain much if we place before us two long lists relating to the actual earnings of each worker in two firms. But we can compare with relative ease and better understanding if we have average earnings of the workers of two firms.

4. *Statistical methods enable us to draw inferences or conclusions from the data about the behavior of a variable or variables under study.* For example, by conducting an experiment, we can conclude that medicine-A is more effective in curing a certain disease as compared to medicine-B.

Thus statistical methods are extremely useful for studying the behavior of variables.

Misuses of Statistical Methods

In spite of the great utility of statistical methods, the methodology of statistics is sufficiently misunderstood to give rise to a number of humorous comments about statistics and statisticians. For example, “A statistician is a person who draws a mathematically precise line from an unwarranted assumption to a foregone conclusion”. Some have a conclusion already decided upon and then use statistical methods to prove their findings. Due to the wrong applications of statistical methods, statisticians have been classified third in the hierarchy of liars. “There are liars, damned liars and statisticians.” Another criticism leveled against statistician and statistical methods is, “many people use statistics as drunken man uses a lamp-post as a means of support, rather than of illumination.”

Those comments refer to the misuse of statistical methods or misunderstanding of statistical methods. But the fault lies not with the statistical methods but with their applications. For example, if a man cuts his head with sword, the fault lies not with the sword but with the man.

The remedy lies in better understanding of statistical methods and by trained persons for describing, analyzing and interpreting the behavior of variables. To quote Croxton and Cowden, “Without an adequate understanding of the statistical methods, the investigator

in the social sciences may be like the blind man groping in a dark room for a black cat that is not there”

Basic Statistical Techniques

While analyzing the data, researchers usually make use of many simple statistical devices. Therefore, knowledge of basic statistical techniques used in statistical analysis to the phenomena under study is inevitable. In fact, a thorough knowledge of at least the fundamentals of statistics is an indispensable part of the equipment of the researcher in the field of social science. Various statistical methods used in research are seen below:

1. Collection, classification and tabulation of data: It helps in quantification and objective valuation of social phenomena. It presents facts in a proper form and reduces complexity of data. The first step in a statistical inquiry is the collection of data relating to the problem under study. When a mass of data has been gathered, it is necessary to arrange the material in some sort of concise and logical order. The procedure is referred to as classification and tabulation of data.

2. Diagrammatic and graphic representation: Another important convincing, appealing and easily understood method of presenting the statistical data is the use of diagrams and graphs. They give a bird’s eye view of a given set of numerical data. They register a meaningful impression in the mind almost before we think. They also save a lot of time as a very little effort is required to grasp them and draw meaningful inferences from them. They highlight the salient features of the collected data; facilitate comparisons among two or more sets of data and enable us to study the relationship between them more readily. Graphs reveal the trends, if any, present in the data more vividly than the tabulated numerical figures. For example, sales managers, stock analysts, hospital administrators, school directors, and other busy executives often need a quick picture of the trends in sales, stock prices, hospital costs, or students’ enrollments. These trends can be shown by use of charts or graphs. The charts that help portray a frequency distribution graphically include histograms, bar graphs, the frequency polygon, cumulative frequency polygon, line graphs, circular graphs or pie charts, and etc.

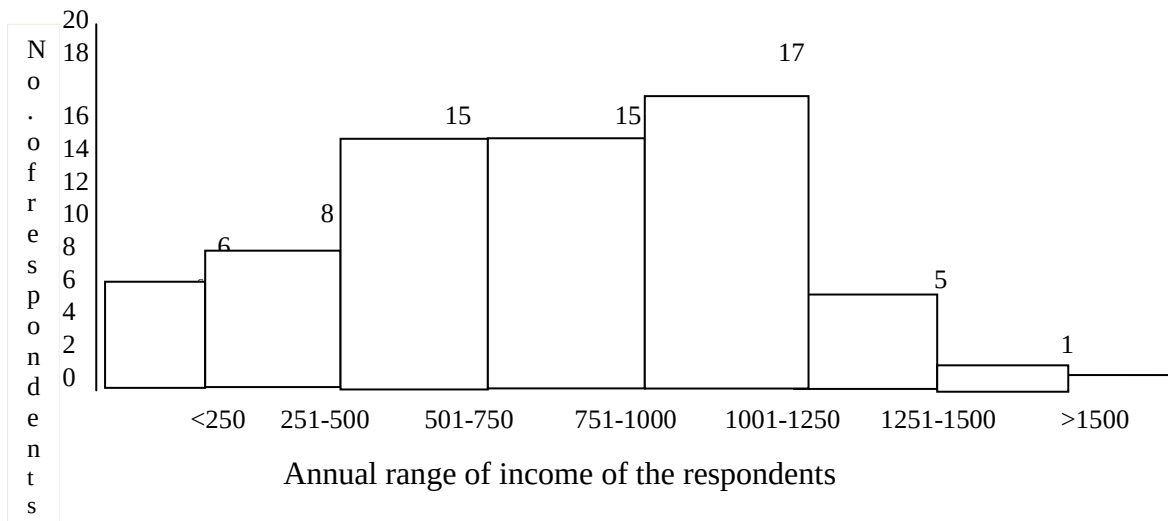
a. Histogram:-It is one of the most frequently used charts and one of the easiest to understand. A histogram describes a frequency distribution using a series of adjacent

rectangles, where the height of each rectangle is proportional to the frequency of the class it represents. For example, from the yearly income (in Birr) of individual partners in a certain business associations shown below, a histogram can be drawn.

Annual income (in Birr)	Number of partners
<250	6
251-500	8
501-750	15
751-1000	15
1001-1250	17
1251-1500	5
>1500	1

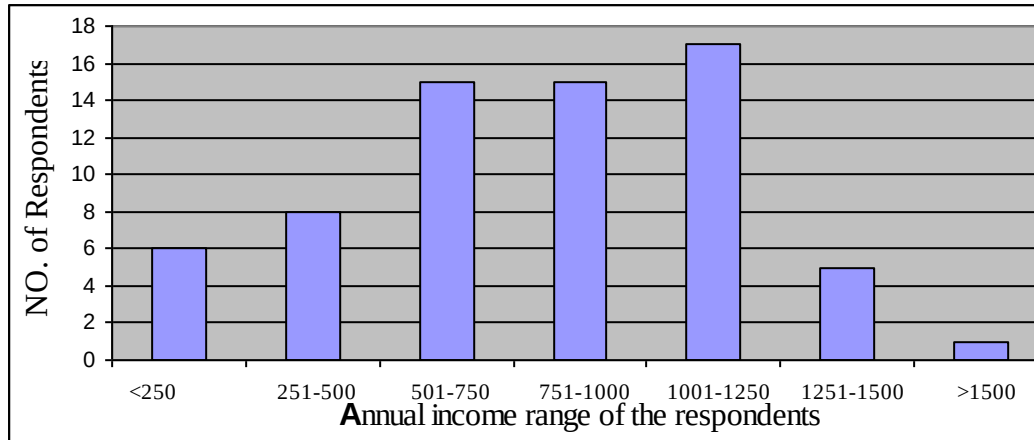
From this information, a histogram can be drawn as under by putting class frequencies on the Y-axis and the stated annual income ranges on the horizontal axis (X-axis).

Fig.7.1. Histogram



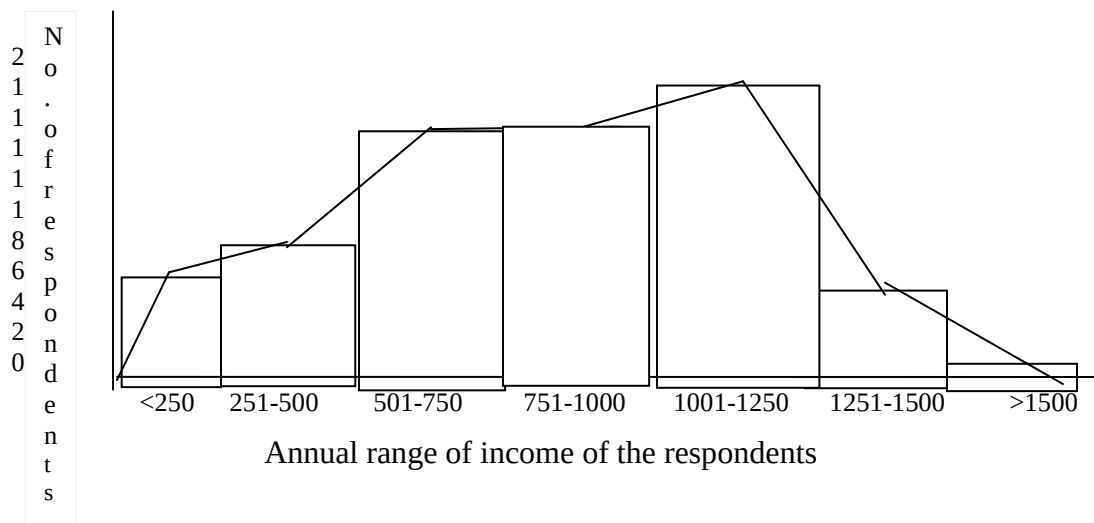
b. Bar Chart: - A bar chart or diagram is used for displaying categorical data. A bar chart is identical to a histogram except that in a bar chart the rectangles representing the various frequencies are spaced, thus indicating that the data are categorical. A bar chart is drawn as follows from the previous data.

Fig.7.2. Bar chart



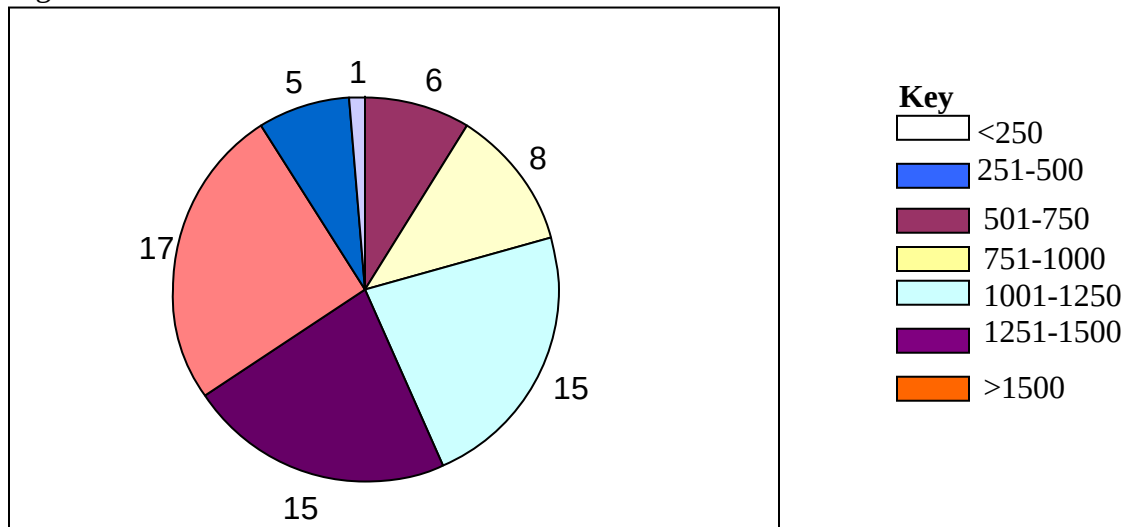
c. Frequency Polygon: - This is closely related to a histogram. It consists of line segments connecting the points formed by the intersection of the class mid point and the class frequency.

Fig.7.3. Frequency polygon



d. Pie-chart: - This is another way of depicting data graphically, this time as a circle. There are 360° in a circle, so the full circle can be used to represent 100 or the total population. The circle or pie is divided into sections in accordance with the magnitude of each sub-category, so each slice is in proportion to the size of each sub-category of a frequency distribution. Look at the following example.

Fig.7.4. Pie-chart



3. Measures of central tendency: This holds a very important place in all types of statistical work because they describe the inherent characteristics of a frequency distribution in a concise manner and help in comparative study of different distributions.

Measures of central tendency are different kinds but we will consider only five, namely, arithmetic mean, median, mode, geometric mean and harmonic mean.

a. **Arithmetic Mean-** This is the most widely used in statistical research because it is easy to understand and calculate. If the measure of each item in a series is known, the mean can be derived by adding the measures together and dividing by the number of items. In business and economics, it is used where all the items are of equal importance. It takes into consideration of all items from first to last and is considered to be more representative.

b. **Median-** This is another simple average or measure of central tendency. It is more frequently used where the extreme item is to be eliminated. Median is that value of variable, which divides the group into two equal parts, one part comprising all the values greater and the other all values less than median. Thus the median is the value of the variable, which exceeds and is exceeded by the same number of observations. It is very useful in the case of skewed distribution such as the distribution of incomes and wealth.

c. **Mode**-In a simple series, mode is the size of the measurement that occurs most frequently. In case of frequency distribution, it is the value of the variable corresponding to the maximum frequency. It is very easy to calculate and understand. It can also be estimated graphically from a histogram. It is not at all affected by extreme observations and as such is preferred to arithmetic mean while dealing with extreme observations. Whenever complete data is not available, mode is common form of averages to be used. It is not necessary to know the size of all units. Even if an approximate point of density is known, mode can be easily located. It is directly applicable to large number of items. In this respect, it is much superior to all other averages.

d. **Geometric mean**-Geometric mean is more mathematical and complicated than mean, median or mode. Geometric mean of different items of a series is the n^{th} root of the product of the n item values. In short, it is the root of the product of all the item values. It is mostly used in such cases where the data has to be put further mathematical analysis. It is especially suitable in those cases where less importance is to be given to large measurements. It is used in business and economics in finding the rate of growth of population, compound rate of interest, rate of depreciation of machinery and equipment, cost-benefit analysis, etc.

e. **Harmonic Mean**-It is reciprocal of the arithmetic mean or the reciprocal of the given observations. It is rigidly defined, based on all observations and is amenable to further algebraic treatment. It can be used in phenomena involving time, rate and price. It is the most suitable average when it is desired to give greater weight to smaller observations and less weight to the larger ones.

4. Index numbers: Index numbers are indicators which reflect the relative changes in the level of a certain phenomenon in any given period (or over a specified period of time) called the *current period* with respect to its value in some fixed period called the *base period* selected for comparison. It should be noted that index numbers measure only relative changes. Index numbers also help in forecasting the future trends in the level of phenomena. For example, index number of prices tells us about the changes in the general price level of a country and also predicts about future trends in prices. Cost of

living index numbers tells us about the change in the cost of living of different groups of people in a society.

5. Measures of Variability: Another important aspect in statistics is variability. The mean, median and mode give only one essential characteristics of a frequency distribution-its typical size of *central tendency*. Furthermore, it is possible for several distributions to have same average yet be markedly different in variability. In some distributions, the cases may be found very closely around the average and in other they may be widely scattered. It is, therefore, very important to determine the spread of the individual values on either side of their central tendency. Important measures of absolute variability are mean deviation and standard deviation and in relative variability, we calculate co-efficient of variation

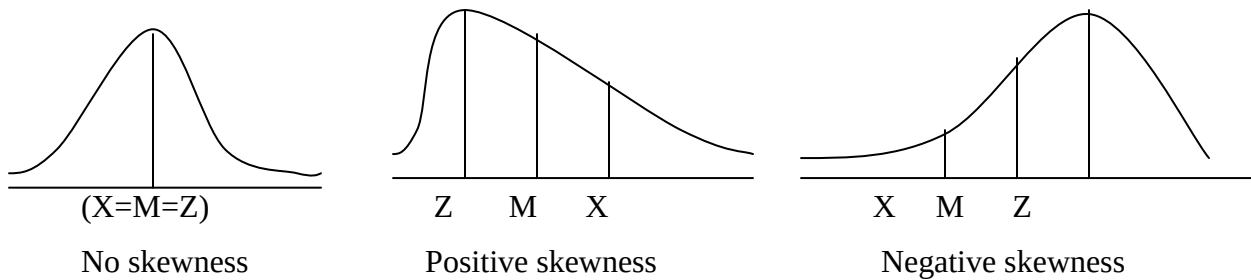
a. **Mean Deviation:** - Mean deviation is the arithmetic mean of the deviation of the individual values from the average of given data. Mean deviation is based on all the observations and is a good measure of dispersion. The averaging of the absolute deviations from the average removes the regularities in the distribution and thus mean deviation provides an accurate and true *measure of dispersion*.

b. **Standard Deviation:-**The standard deviation provides a more refined and statistically important measure of variability than the mean deviation. It is defined as the positive square root of the arithmetic mean of the squares of the deviations of the given observations from the arithmetic mean. It is least affected by fluctuations of sampling. It is always computed from the mean whereas the mean deviation may be compared from the mean, the median or sometimes the mode.

c. **Co-efficient of variation-** Co-efficient of variation is the relative variation in mean, whereas the standard deviation is the total variation in the mean. For comparing the variability, homogeneity, stability, uniformity and consistency of two series, the co-efficient of variation is calculated. The series having greater co-efficient of variation is said to be more variable than the other series having less co-efficient of variation is said to be more variable than the other series having less co-efficient of variations.

6. Measures of Asymmetry (Skewness):

Skewness is a measure that refers to the extent of symmetry or asymmetry in a distribution. It is used to describe the shape of a distribution, i.e. skewness shows the manner in which the items are clustered around the average.



Where X =Arithmetic mean, M =Median, and Z =Mode

Kurtosis: A measure of kurtosis indicates the degree to which a curve of a frequency distribution peaked or flat-topped.

Moments: In statistics, it is used to describe the peculiarities of a frequency distribution. Using moments, one can measure the central tendency of a set of observations, their scatteredness, i.e. dispersion, skewness, kurtosis of curve, etc.

7. Measures of association/relations

a. Correlation: When a change in one phenomenon is accompanied by a similar or opposite change in some other phenomena, it is called correlation. Correlation is a statistical technique used for analyzing the behavior of two or more variables. It measures the degree and the direction of sympathetic movements ranges between 0 and ± 1 . The direction of change is indicated by plus or minus signs. The former refers to the sympathetic movement in the same direction and the latter in opposite direction. According to Croxton and Cowden, “When the relationship is of a quantitative nature, the appropriate statistical tool for discovering and measuring the relationship and expressing it in brief formula is known as correlation.”

In economic theory and business studies, relationship between various variables is studied. The correlation analysis helps in deriving precisely the degree and direction of such relationships. The predication based on correlation analysis is more reliable and near to reality.

b. Association of Attributes: The problem of association is basically that of correlation but the only difference between them is in their quantitative and qualitative approach. The attributes are the qualitative characteristics as against the variables which refer to quantities or size or magnitudes of a phenomenon. Thus, the correlation deals with the variables, which may be numerically stated while the association of attributes is concerned with the attributes, which are not amenable to gradual measurement. Number of production units, wages of the workers and the like are the examples of quantitative characteristics of a variable while the classification of production into defective and non-defective are qualitative characteristics of a variable. Measures of association between attributes are limited to comparisons of predictability of a variable when association with another variable is taken into account as compared with the predictability when no association is utilized.

c. Regression Analysis: Regression analysis is the estimation or prediction of the unknown value of one variable from the known value of the other variable. In regression analysis, there are two types of variables. The variable whose value is influenced or is to be predicted is called dependent variable and the variable which influences the values or is used for prediction is called independent variable. The regression analysis for studying more than two variables at a time is known as multiple regression. The prediction is based on average of relationship derived statistically by regression analysis. In the words of M.M. Glair, “regression is the measure of the average relationship between two or more variables in terms of the original units of the data”

Regression is one of the very important statistical tools, which is extremely used in almost all sciences-natural, social and physical sciences. It is especially used in Business and Economics to study the relationship between two or more variables that are related causally, and for estimation of demand and supply curves, cost functions, production and consumption functions, etc. Regression is always very useful in model building.

8. Analysis of Time Series: A time series is an arrangement of statistical data in accordance with its time of occurrence. If the values of phenomena are observed at different periods of time, the values so obtained will show appreciable variations. Examples of time series are the series relating to prices, production and consumption of various commodities, agricultural and industrial production, national income, etc.

Analysis of time series is useful in administration, planning and evaluation of socio-economic progress as well as for research in various fields of science and humanities. It also helps in analysis of a phenomenon in terms of the effect of various technological, economic and other factors on its behavior over time. It makes more scientific comparison after considering the various components of the series to know how they have behaved over a period of time. This series also helps in evaluation of progress in any field of economic or business activities. It is also useful in forecasting a most likely value of a variable in near future.

9. Interpolation and extrapolation: Interpolation is a technique of estimating past figures whereas extrapolation means an estimation of a probable figure for future. Both are statistical techniques of arriving at the unknown facts from the known facts. With the help of interpolation intermediate figures, which have been destroyed on account of improper handling can also be estimated.

10. Probability:-Probability means the chance of occurrence of a particular event. This method helps not only in ascertaining the chance of occurrence of an event but also in finding out the total effect of any uncertain events if the consequences of various occurrences are known. It is extensively used in the quantitative analysis of economic or business problems. It is an essential tool in statistical inference and forms the basis of the decision theory, i.e., decision-making in the face of uncertainty with calculated risks. All statistical laws are based on probability. Therefore, statistics is called the science of estimates and probabilities. It enables a person to make guesses on matters where either full facts are not known or there is uncertainty about the outcome.

Some advanced statistical tools are also being employed in modern research, which have been discussed separately in details in the next chapters of statistical inference and testing of hypothesis whereas some have not been touched, as they do not play important part in social research.

7.4.6. Types of Statistical analysis

Statistical analysis is made for estimating the values of unknown characteristics (parameters) of the population and for testing hypothesis for drawing inferences. Therefore, analysis can be classified in to two:

1. Descriptive analysis, and
2. Inferential analysis

1. Descriptive analysis: This type of analysis describes the nature of an object or a phenomenon under study. This sort of analysis may describe data on one variable, two variables or more than two variables. Accordingly, it is called *univariate* analysis, *bivariate* analysis, and *multivariate* analysis, respectively. Multivariate analysis consists of

- of
- a) multiple regression analysis,
 - b) Multiple discriminant analysis,
 - c) Canonical analysis
 - d) Multivariate analysis of variance, and
 - e) Factor analysis

2. Inferential analysis: This is concerned with drawing inferences and conclusions from the findings of a study. There are two areas of statistical inferential:

- a) Statistical estimation, and
- b) the testing of hypothesis

Let us now see these areas in detail as below.

a) Statistical estimation involves estimation of the population parameters from the results of sample data analysis. To make this effective, it needs precisely defined population, adequate sample size, and the selection of representative sample.

b) Testing Hypothesis to an appropriate empirical scrutiny and statistical test to determine its validity. Hypothesis tests also address the uncertainty of the sample estimate. However, instead of providing an interval, a hypothesis test attempts to refute a specific claim about a **population parameter** based on the sample data. For example, the hypothesis might be one of the following:

- the population mean is equal to 10
- the population standard deviation is equal to 5
- the means from two populations are equal
- the standard deviations from 5 populations are equal

To reject a hypothesis is to conclude that it is false. However, to accept a hypothesis does not mean that it is true, only that we do not have evidence to believe otherwise. Thus

hypothesis tests are usually stated in terms of both *a condition that is doubted (null hypothesis)* and *a condition that is believed (alternative hypothesis)*.

A common format for a hypothesis test is:

1. H_0 : A statement of the null hypothesis, e.g., two population means are equal.
2. H_a : A statement of the alternative hypothesis, e.g., two population means are not equal.
3. Test Statistic: The test statistic is based on the specific hypothesis test.
4. Significance Level: The significance level, α , defines the sensitivity of the test. A value of $\alpha = 0.05$ means that we inadvertently reject the null hypothesis 5% of the time when it is in fact true. This is also called the type I error. The choice of α is somewhat arbitrary, although in practice values of 0.1, 0.05, and 0.01 are commonly used. The probability of rejecting the null hypothesis when it is in fact false is called the power of the test and is denoted by $1 - \beta$. Its complement, the probability of accepting the null hypothesis when the alternative hypothesis is, in fact, true (type II error), is called β and can only be computed for a specific alternative hypothesis.
5. Critical Region: The critical region encompasses those values of the test statistic that lead to a rejection of the null hypothesis. Based on the distribution of the test statistic and the significance level, a cut-off value for the test statistic is computed. Values either above or below or both (depending on the direction of the test) define the critical region.

Types of errors Committed in hypothesis testing

Dear learner! Try to understand the type of error committed in testing hypothesis from the following decision type and type of hypothesis matrix.

Decision	Null Hypothesis	
	True	False
Reject	Type I error(α)	Correct decision
Accept	Correct decision	Type II error(β)

Type I error: An error committed in rejecting a true hypothesis

Type II error: An error committed in accepting a false hypothesis

Hypothesis Testing Procedure

Dear learner! Let us break the statistical hypothesis testing procedure into the following steps. Assuming μ = population mean, understand the steps with simple examples as below.

1. State Null and alternative hypotheses: The first step of statistical testing is to convert the research question into null and alternative forms. We use the notation H_0 to represent the null hypothesis and H_1 (or H_a) to denote the alternative hypothesis. H_0 is a statement of “no difference.” This is the hypothesis that the researcher hopes to reject. H_1 opposes H_0 . *We retain the premise of the null hypothesis until proved otherwise.*

e.g. $H_0: \mu = 23$

$H_1: \mu \neq 23$ or $H_1: \mu > 23$ Or $H_1: \mu < 23$

2. Selecting a significance level or error threshold

If we wish to reach a “yes-or-no decision,” fixed level testing must be pursued. (This is not always necessary, and is sometimes unwise.) To pursue fixed-level testing, we set an error threshold for the decision. The error threshold, called alpha (α), is the probability the researcher is willing to take of incorrectly rejecting a true H_0 . For example, the researcher may be willing to take a 1% chance of incorrectly rejecting a true H_0 . In such instances, $\alpha = .01$.

3. Selecting the appropriate Test Statistic

A test statistic is calculated. There are different test statistics depending on the data being tested and question being asked. Dear learner! These statistics are introduced after the final step of hypothesis testing.

4. Making statistical decision (Rejecting or accepting the hypothesis)

This step is made realized after computing the probability that sample result would diverge as widely as it has from expectations, if H_0 were true. If this probability is equal to or less than α value (or if the calculated test value is greater than the table value) in case of one-tailed test and $\alpha/2$ in case of two-tailed test, reject the H_0 thereby run the risk of committing type I error. But if greater, we accept the H_0 thereby running the risk of some risk of committing Type II error.

5. Making inferences (drawing valid conclusions)

Based on the finding of step 4, we give the right interpretation here. In conjunction with this, discussion on test of significance is made as follows.

Test of Significance or Statistical tests

The test of significance used for hypothesis testing is of two types: the parametric and the non-parametric tests. The parametric tests are more powerful, but they depend on the parameters or characteristics of the population. The important parametric tests are: the z-test, t-test, χ^2 -test and the F-test. (All the parametric tests are based up on the assumption of the normality (normally distributed data).

The non-parametric tests are population-free tests, as they are not based on the characteristics of population. They do not specify normality of population and equal variances. The tests include Chi-square (χ^2 -test), median test, the sign test, and etc. (**NB:** The χ^2 -test can also be used as a test of goodness of fit and as test of independence in which case it is a non-parametric test.) They are the only usable tests with nominal data; they are also the most appropriate tests to use with ordinal data.

A. The parametric tests. Now let us start with the z-test.

1. The z Test: The Z-test is used in the case of normally distributed population with known variance or when the sample size is very large. It is widely used to test the significance of several statistics such as mean, median, mode, coefficient of correlation, and others.

Assumptions of the z-test

The main **sampling assumption** of the z test is that observations in the sample are **independent**. This means that data were derived by a simple random sample: each member of the population has an equal (and “independent”) probability of entering the sample. The main distributional assumption of the z test is that the sampling distribution of the *mean* (not the initial data) is **normal**. But modest departures from normality in the population are fully permissible.

- When the sample is small (say, $n < 10$), the population distribution should be close to normal.
- When the sample is moderate (say, $10 < n < 30$), the population should be fairly symmetric and mound-shaped with no outliers.

- When the sample is large (say, $n > 30$), the z test is permissible except in the case of extreme outliers and strong skewness.

Types of Z-test

a. One-sample z Test

The one-sample z test compares a single mean to an “expectation” while using a known or assumed population standard deviation (σ). The test can be done in a one-sided or two-sided way. Dear learner! Let us start by considering one-sided alternative.

Illustrative example (“IQ Data”): Suppose you hypothesize that IQs of children at a particular school are above average. It is known that Wechsler IQ scores are normally distributed with a mean of 100 and standard deviation of 15. A random sample of 9 children ($n = 9$) from the school shows a mean ($\bar{x}$) of 112.8. Now let us proceed with the above steps to test the hypothesis. We want to test whether the children in the sample have an average IQ that is significantly *higher*.

Dear learner! Look at the following solution to test the hypothesis.

1. $H_0: \mu = 100$ since an IQ value of 100 represents *no different* than typical.

H_a : A one-sided form to the right (a mean is greater than expected ($H_1: \mu > 100$))

2. Let $\alpha = 0.05$

3. The test statistic for this problem is Z -test.

4 & 5 steps. By applying the formula of Z -test, you find a Z -value of 2.56. The value of the Z -calculated is converted to a p simply by using a Z table. Using the Z table, you find that 0.9948 of the area is to the left of 2.56. We need the area to the right of 2.56. To the right, we find p is equal to 0.0052. Now the decision is to reject H_0 (or accept the H_a) since the p value is less than the pre-specified α . We now say there is *significant evidence* to indicate the children have IQs that are above average.

b. Two-Sided Alternative

The previous illustration used a one-sided alternative to test the data. **Dear learner!** We now use a two-sided test. Two-sided tests allow for unanticipated findings that are either

“up” or “down” from what are expected. Let us once again use the “IQ” illustrative data given above. Now let us follow the testing procedures as follows.

1. $H_0: \mu = 100$

$H_1: \mu \neq 100$.

2. Let $\alpha = .01$.

3. The test statistic for this problem is Z-test.

4 & 5 steps. By applying the formula of Z-test, you find a Z-value of 2.56. In light of the two-sided alternative, we could conceivably reject the null hypothesis in favor of the alternative if $\bar{x}$ is either significantly greater than 100 or significantly less than 100. Therefore, rejection regions lie in both tails of the “null” standard normal probability distribution. Although $\bar{x}$ cannot fall into both tails of the probability distribution, we start by assuming it falls on the upper tail. If the alternative hypothesis was one-sided, the p value would be the area under the curve in the upper extent of the distribution, beyond the Z-calculated. Because this is a two-tailed test, we must account for both tails by doubling the probability in the tail.

If $\bar{x} > 100$, the p value is *double* the area under the curve in the right-tail of Z distribution.

If $\bar{x} < 100$, the p value is *double* the area under the curve in the left-tail of Z distribution.

Thus, for this particular example, $p = 2 \times .0052 = .0104$.

So **dear learner!** How would you interpret this result? Well! The right decision is to *retain* H_0 since the p value is greater than the pre-specified α . But there is a possibility of a type II error (false retention of H_0), which is especially relevant given the small sample size ($n = 9$).

2. One-Sample t Test

Dear learner! Recall that the z-test used to conduct the test depended on knowing or assuming population standard deviation. In most instances, however, it is not reasonable to assume the population standard will be known. Thus, to test H_0 when the population standard deviation is not known, a t tests statistic is used instead of a z statistic. Also in case a sample size is small ($n < 30$), the t -test applies.

Illustrative Example: An educator suspects that the rural female students Mathematics performance is better than urban female students. He took a sample of 18 urban and 14

rural female students and found the following. Do you think the educator's suspicion is true?

	Urban	Rural
Mean	65	58
Variance	170	170

Dear learner! Look at the following solution to test the hypothesis.

1. $H_0: \mu_U = \mu_R$

$H_a: \mu_U \neq \mu_R$

2. $\alpha = .05$.

3. Test Statistic. Since the population standard deviation is unknown and/or $n_1 < 30$ & $n_2 < 30$, we conduct a t -test.

4 & 5 steps. By applying the formula of t -test, you find a p value of ($t_{stat} = 1.506$ with $df = (18-1) + (14-1) = 30$), Where df stands for degrees of freedom. It helps determine the t -value in the t -table.

Decision: Do not reject the null hypothesis (since $t_{stat} < t_{0.05}$ or $p > \alpha$)

Interpretation of the p value. Urban and Rural female students have similar performance in Mathematics at 5% level of significance.

3. ANOVA test (F-test)

In case of performing multiple comparisons by one way ANOVA (Analysis Of Variance), the F -statistic is normally used. It is defined as the ratio of the mean square due to the variability between groups to the mean square due to the variability within groups. The critical value of F is read off from tables on the F -distribution knowing the Type-I error α and the degrees of freedom between and within the groups.

Three primary assumptions of ANOVA are that:

1. Each group is obtained randomly, with each observation independent of all other observations and the groups independent of each other.
2. The samples represent populations in which the data are normally distributed.

3. $\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \dots = \sigma_k^2$. The assumption of homogeneity of variances is similar to the discussion above under the t -test. The group variances are assumed to be an estimate of a common variance, σ^2 .

Plot of Land	Per Hectare production Data		
	Varieties of Wheat		
	A	B	C
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

Example: Set up an analysis of variance table for the following production data for three varieties of wheat, each grown on 4 plots and state if the variety differences are significant.

Dear learner! Look at the following solution to test the hypothesis.

1. H_0 : The difference in wheat output is insignificant.

H_a : The difference in wheat output is significant

2. Assume the significance level to be 5%.

3. Using the ANOVA formula, $F_{\text{calculated}}=1.5$; F -table (i.e. $F_{0.05}(2, 9)=4.26$

$\Rightarrow$ Since $F_{\text{cal}} < F_{\text{table}}$, accept the H_0 .

$\Rightarrow$ Interpretation: The difference in wheat output is insignificant and is just a matter of chance.

Dear learner! In what case(s) do you employ

i. ANOVA test? _____

ii. t-test? and iii. z-test? _____

B. The Non-parametric test: For our purpose, let us have a discussion on Chi-square test as it is the most common test of such kind. It is used to make comparisons between two or more nominal variables. Unlike the others, the chi-square is used to make comparisons between frequencies rather than between mean scores. This test evaluates whether the difference between the observed frequencies (f_o) and the expected frequencies (f_e) under the null hypothesis can be attributed to chance or actual population differences. Its formula is: $X^2 = [(f_o - f_e)/f_e]$.

Example: Suppose you want to examine the relationship between education and income, and the frequency data are summarized as follows:

Income	School		Total
	Low	High	
Low	30	12	42
High	10	28	38
Total	40	40	80

Dear learner! Look at the following solution to test the hypothesis.

1. H_0 : There is no difference between the two education categories in their pattern of income level
 H_a : There is relationship between the two
2. Assume the level of significance to be 1%.
3. By using χ^2 formula, $\chi^2 = 16.24$ (obtained value); but $\chi^2_{0.01}(1) = 6.64$ (table value)
 $\Rightarrow$ Obtained $\chi^2 >$ table χ^2 value, i.e. H_0 is rejected
 $\Rightarrow$ Interpretation: There is significant relationship between educational and income levels.

Dear learner! Describe and show when to use chi-square method of hypothesis testing?

NB: Dear learner! For a thorough computation of the statistical tests, recall the formulae and procedures followed in your “Statistics for Management” course.

Chapter Summary

In this chapter, you learned about data processing and analysis. Data processing involves the edition, coding, classification and tabulation of collected data. The data collected are edited for their consistency, completeness, accuracy, uniformity, and computations. In data coding, a researcher assigns number, digits or letters or both to various responses for easy tabulation. When the data are arranged into groups or classes according to their resemblances and similarities, it is termed as classification. Finally, tabulation involves the orderly and systematic presentation of numerical data in a tabular form.

Once data are processed, data presentation begins in which data analysis is made. Data analysis involves the computation of certain measures along with searching for patterns of relationships that exists among data group. To help the process of data analysis, scales of measurement are used. This means the process of assigning numbers to objects or observations are made possible. The scales of measurement are classified as nominal, ordinal, interval, and ratio scales of measurement.

Moreover, statistics has a significant place in a social science research because it helps simplify the huge amount of data collected by a researcher. The increasing use of quantitative methods in social researches is also the other reason for the importance of statistics in a research. However, when these statistical methods are misused, it can ultimately make the research work futile.

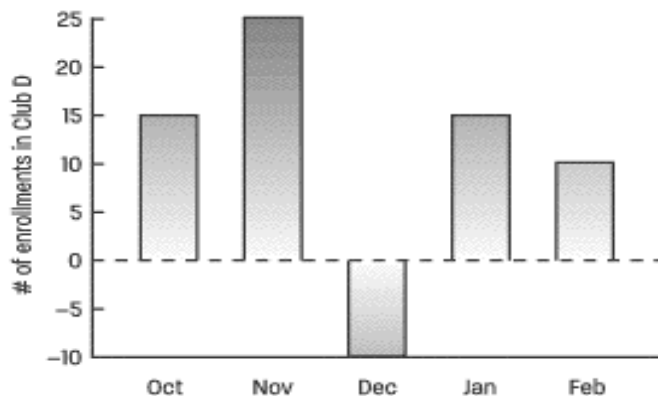
The statistical methods used in a research are observed during the collection, classification, tabulation, and diagrammatic and graphic representation of data; when index numbers are used, and when measures of variability, asymmetry and measures of association are employed. There are two types of statistical analysis: descriptive analysis, and inferential analysis. In descriptive analysis, the nature of an object or a phenomenon under study is described whereas conclusions are reached about the population from the findings of a study. The inferential analysis is of two categories: statistical estimation, and the testing of hypothesis. Whereas statistical estimation involves the estimation of the population parameters from the results of sample data, a hypothesis test attempts to refute a specific claim about a *population parameter* based on the sample data.

Exercises

Part I: Short answer questions

1. What does data processing include?
 - a. editing
 - b. coding
 - c. classification
 - d. tabulation of collected data.
 - e. All
2. If a researcher edits to examine whether all questions in the schedule or questionnaire are answered or not, this edition is made for
 - a. Completeness
 - b. consistency
 - c. Editing for accuracy
 - d. uniformity:
3. _____ refers to assigning of number, digits or letters or both to various responses so as to enable tabulation of information easy.
 - a. Coding
 - b. Tabulation
 - c. Classification
 - d. None
4. The objective of data classification does not include
 - a. Presenting the facts in a simple form
 - b. Bringing out clearly points of similarity and dissimilarity of data
 - c. Facilitating comparison of data
 - d. Bringing out relationship of facts
 - e. None
5. Tabulation is not featured by the following quality
 - a. It clarifies the object of investigation
 - b. It facilitates statistical process
 - c. It facilitates statistical process
 - d. None
6. The process of working-out *statistical distribution*, constructing of *diagrams* and calculating simple measures like *averages*, *measures of dispersion*, *percentages*, *correlation*, etc is called-----
 - a. Data processing
 - b. Data analysis
 - c. Inference
 - d. All

7. Nominal scale of measurement is not featured by
- It is a system of assigning numbers or symbols to events in order to label them.
 - Nominal scale is the least powerful level of measurement
 - The nominal scale places events in order or rank
 - None
8. In which scale of measurement does all statistical techniques can be used and manipulations carried out.
- Ratio scale
 - Interval scale
 - Ordinal scale
 - Nominal scale
9. Analysis of data is not characterized by one of the following
- Analysis of data is one of the most important aspects of research
 - The data to be analyzed and interpreted should be reproducible
 - task of analysis is complete without interpretation
 - None
10. From the following table, answer the question that follows



- What month is with the greatest increase in enrollments?
- October
 - November
 - December
 - January
 -) February

Part II. Short answer questions

1. What is descriptive analysis?
2. Define what inferential statistics is.
4. Clearly show the procedures of hypothesis testing
5. Distinguish between the applications of the following tests of hypothesis
 - a. z-test
 - b. t-test
 - c. χ^2 -test
 - d. F-test

Chapter Eight

Interpretation and Reporting the Research Result

8.1. Introduction

Dear learner! As you remember in the 6th and 7th chapters, you learned about data collection, and data processing and analysis. After data collection and analysis, a researcher has to accomplish the task of drawing inferences followed by report writing. Interpretation has to be done carefully so that misleading conclusion will not be drawn and the whole purpose of doing research will not be vitiated. Through interpretation, the researcher can expose relations and processes that underline his/her findings. If hypotheses are tested and confirmed, the researcher may arrive at generalizations. But in case the researcher had no hypothesis, he/she would try to explain his/her findings on the basis of some theory. All the analytical information and consequential inferences may well be communicated, preferably through research report, to the users of research results who may be individuals or groups or some public or private organizations. Accordingly, in the 8th chapter, issues including meaning of and rationale for interpretation, techniques of interpretation, precautions in interpretation, significance of report writing, steps in writing report, layout of research report, types of report, mechanics of writing report, and precautions for writing research report will be discussed in detail.

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8.2. Learning Objective

Dear learner! After thoroughly going through this chapter, you are expected to;

- ✓ Explain what data interpretation is and its rationale
- ✓ Be familiarized with the techniques employed for data interpretation
- ✓ Understand the pre-requisites of interpretation
- ✓ Develop know-how for research report writing in which you are able to:
 - Define research report writing
 - Identify the various types of reports
 - Understand the steps followed in report writing
 - To be clear with the layout of the research report

- Understand the oral approach of disseminating a research result

8.3. Data Interpretation

8.3.1. Meaning and Need of Interpretation

The task of drawing conclusion or inferences and of explaining their significance after a careful analysis of selected data is known as interpretation. As a matter of fact, interpretation is the main function of a researcher and/or statistician and the function of collection and analysis is just an auxiliary function as a necessary precedent of interpretation. It must, therefore, be clear that if the methods of statistics are not properly and correctly applied as the science demands, then the inferences which will be drawn from a set of data will also be wrong. On the other hand, if the data have been collected and analyzed properly according to accepted principles of science, there is no reason why the conclusions, which emerge from such data, are not found to be fairly accurate. Hence, it is said that “statistics are like clay of which you can make either a god or a devil”.

Interpretation of data refers to that important part of the investigation, which is associated with the drawing of inferences from the collected facts after an analytic study. It is the interpretation that makes it possible for us to utilize collected data in various fields of activity. The usefulness of the collected data lies in its proper interpretation. It provides certain conclusions about the problem under study.

Mistakes in the analysis of data are either deliberately or unconsciously committed. Deliberate mistakes and errors are due to bias and prejudices, which can be completely eliminated if the researcher is very careful in the selection of his/her staff who are given the task of collection and analysis of data.

The significance of interpretation may also be examined in the case of statistical forecasting. Here it becomes a more crucial problem because if the forecast value is wrongly interpreted, it can lead to highly damaging situations at a future date. The image that emerges from the forecast becomes the guiding force in taking the decision regarding a particular activity of whatsoever nature it may be. Business forecasting plays a dominant role in the modern business decisions. Incorrect forecasts may cause heavy

losses in business. To avoid such a situation, interpretation has to be most objective and unbiased, otherwise modern business will run in to difficulties.

? **Dear learner!** Can you confidently define what data interpretation is now? If so, how can you define it? _____

8.3.2. Techniques of Interpretation

The task of interpretation is not an easy task. Rather it requires a good skill on the part of researcher. Interpretation is an art that one learns through practice and experience. The researcher may seek guidance from experts for accomplishing the task of interpretation. There are no existing rules to guide the researcher about how to interpret the data. However, the techniques of interpretation involve the following steps.

1. A researcher must give reasonable explanation of the relation which he/she has found and he/she must interpret the lines of relationship in terms of the underlying processes and must try to find out the thread of uniformity that lies under the surface layer of his/her diversified research findings.
2. Extraneous information or data, if collected during the study, must be considered while interpreting the final result of research study, for it may prove to be a key factor in understanding the problem under consideration.
3. It is advisable, before embarking upon final interpretation, to consult some one having insight into the study and who is frank and honest and will not hesitate to point out omissions and errors in logical argumentations. Such a consultation will result in correct interpretation and, thus, enhance the utility of research result.
4. A researcher must accomplish the task of interpretation only after considering all relevant factors affecting the problem to avoid false generalization. He/she must not be in hurry while interpreting results, for quite often the conclusion, which appear to be alright at the begging, may not at all be accurate.

8.3.3. Pre-requisites of Interpretation

In order to draw correct and valid inferences from the data, several safeguards shall have to be taken. Even when the researcher is a trained statistician, he has to ensure the

fulfillment of a minimum number of prerequisites before proceeding to interpret the data. These pre-requisites are discussed below:-

1. The Data should be homogeneous:- This requirement is considerably important in the process of interpretation. Unless data are homogeneous, comparability of different situations cannot be ensured. Efforts should be made to safeguard this comparability character of the data at the collection stage itself. If, for example, data on employment are to be collected from different places, we must use one unit of employment throughout. Data may be collected in man-hours or man-days and the definition of a man-day should be same in all situations. Homogeneity of data should also take in to account the similarity of features in two different areas. We may collect statistics on man-days worked in a given agricultural operation in two regions that are agro-climatically and socio-politically different and these statistics may not offer scientific comparisons even when the unit of measurement is same.

If the data are not homogeneous, inferences drawn can easily lead to wrong findings. For investigating wage rates at various places of work, we may compare the like categories and not the unlike ones. Homogeneity of data is, therefore, an important pre-requisite for interpretation.

2. The data should be adequate: - Many a times, people rush to conclusions with only inadequate data. Such conclusions are bound to be dangerous and can often be wrong and misleading. Whatsoever perfect an interpreter may be, the interpretation is bound to lead to some false conclusions. For any given situation, insufficient and incomplete data should not be analyzed to draw results. For interpretation to be meaningful, data must be complete and adequate. Many times it has been observed that when an adequate set of observations has been collected on two variables, the correlation will be very strong (or sometimes very weak) but for the same variables, quite opposite results are obtained by taking a smaller number of observations. Inadequate data often makes it difficult, if not impossible to analyze data scientifically and make meaningful interpretations. Data must be adequate before inferences are drawn and interpretations are done.

3. The data should be appropriate: - The suitability of data is of paramount importance in the process of analysis and interpretation. Expertise of high level is required at the time of designing an enquiry, collecting the necessary data and analyzing it based on sound

statistical basis. Interpretation of the results based on inappropriate data will be a hazardous task because however objective and unbiased the interpretation may be, it is bound to be misleading if the data are not suitable.

4. *The data should be scientifically analyzed*:-The most up-to-date and appropriate methods should be used to analyze the data. A lot depends on the analysis of data. The efficiency and tactfulness with which the statistical methods are used are important. All conclusions and inferences shall depend upon the quality of analysis work. Poor quality of statistical analysis may lead to wrong conclusions and hence to incorrect interpretations. Some times even when data are collected most scientifically, wrong analysis can play havoc with the interpretations.

?

Dear learner! What are the pre-requisites for data interpretation? _____

8.3.4. Conclusions and Generalizations

In the search for truth, the investigator starts with some felt need or difficulty and employing such knowledge of the situation as he/she may already possess or secure from a direct preliminary examination of the object or from previously reported experiences, he/she formulates some tentative conclusion, hypotheses, or suppositions as to the probable nature of the phenomenon under investigation. These hypotheses may be employed in the further examination of the data that are already available. Using these data, each hypothesis is carefully tested and accepted, modified or rejected as the available evidence indicates.

As the research worker comes to the conclusion of his/her investigation, he/she should review carefully the evidence for and against each hypothesis. Each generalization must agree with the facts revealed by the investigation. The investigator will desire to check each generalization carefully, not only against the facts of his/her own study but against the facts and experiences of their investigators.

There are some common instances in which the conclusions of investigation do not accord with the known facts. In the first case, the investigator gives conclusions, which are directly contrary to the facts. This situation arises occasionally where there are strongly preconceived ideas of the results. Then there is a case in which the investigator

assumes that the names given to data-gathering devices accurately indicate what they measure. A third type of situation in which generalizations may not be in accordance with the data is that in which the facts and relationships found to exist under certain conditions are held to be true under all conditions.

A second test of the dependability of the conclusions drawn from any particular investigation is their agreement with previously established generalization. If a conclusion is in conflict with a generalization already held to be true, the disagreement must be resolved, either by showing that the conflict is not real or that the conflicting statement is false.

Attention has already been directed to the fact that the number of hypotheses should not be needlessly multiplied. Every generalization should be reduced to the simplest possible terms consistent with clarity of expression. The terminology must be simple and direct but of such a character as to summarize as precisely as possible the actual facts of the case. Each generalization should be studied with extreme care, in terms of accuracy, clarity and simplicity of expression.

Remembering that the generalizations of one study may become the point of departure in a subsequent investigation, they should be formulated in terms which render them subject to further verification. It is futile to employ mystical forces to explain phenomena. If the projected cause of a phenomenon cannot be verified in fact, however true it may appear; it must be discarded in all scientific explanation. That every generalization be verifiable is a fundamental principle of scientific method which the investigator should consider carefully before casting a conclusion in final form.

8.4. Research Report Writing

8.4.1. Meaning of research report writing

Report writing is a basic task of a researcher since no research can be completed without its report. In fact, a research without a report can be considered as a house without a roof. Any amount of observation or data analysis would be in vain from the manager's point of view if its report is not made available in aid of his/her decision-making process.

A research study culminates in the writing of a report, which is a tool for communicating the various aspects of the study, viz, the problem or topic investigated, the method employed to investigate, the objective, scope and limitations. However, prior to writing the report, its proper planning is essential. Report planning should take into account such details as defining the problem, the purpose of the report, tracking of the material collected, identifying the utility of the report, etc. The ability to write effective reports is one of the most useful skills a researcher can acquire. In this connection, we can say that a report is a communication from someone to another who wants to use that information.

The usefulness of a report is the chief aspect of any report writing. For example, a study dealing with the market segmentation of a company should be in a position to report on the exact distinction between customer groups and needs for the purpose of sub-dividing its market into specific subsets.

The report identifies areas for further research, if any, and for fresh hypotheses. Analysis and inferences, suggestions and observations, footnotes and appendices, etc should also find a place in the report. It should include specific recommendations for a course of action. While all the necessary information materials should be presented in the most appropriate manner, it should be purposeful and should take into account the reader and his/her tastes and preferences when it is written. In the normal course, the report writer should consider such questions as: who will read report? In what way is the report going to be useful? What are the questions the report should answer? How much information is needed? What is the best way of presenting the information? and so on. An effective report stimulates the interpersonal relationships in the organization. In this connection, we can observe that the reader is a human being, not a corporation. Report should provide answers, not create more problems.

? **Dear learner!** Define what a research report is and show its purpose_____

8.4.2. Types of reports

For the purpose of convenience, reports may be classified on the basis of approach under the following titles:-

1. Business report and memorandum
2. Project report
3. Dissertation
4. Commission's enquiry report and
5. Thesis

On the basis of presentation, reports may be classified as:-

2. Inductive reports
3. Deductive reports
4. Step-by step reports
5. Time-sequence reports

A *business report or memorandum* may be defined as a simple business communication from one department head to another, one functional area to another, or from top to bottom in the organizational structure in any specific aspect of day-to-day business activities.

A *project report* is the report of a project undertaken by an individual or group of individuals or any functional area or any aspect or segment of any functional area of a business, industry, or society.

A *dissertation* is a detailed discourse or report on the subject of the study that was undertaken. This is usually used as a document to be submitted for the acquisition of higher degree from a University. A thesis, too, is a detailed dissertation which maintains a valuable proposition or hypothesis, written on the basis of a detail. This too, is used for academic purpose, as a dissertation is.

A *commission's enquiry report* is detailed report prepared by a commission appointed for the specific purpose of conducting a detailed study of any matter of dispute or of anything which requires a greater insight so that it may facilitate a future course of action. It is an expert opinion.

Business reports and memoranda are brief observational reports that assist one in arriving at business decisions, while project reports are detailed reports which are written with a specific purpose in mind, though it may have implications for business decisions.

The *inductive patterns of report writing* are one of the oldest patterns of presenting one's thoughts. Aristotle had pointed out long years ago "A whole is that which has a beginning, middle and an end". This type of report goes from the specific facts, figures, materials of research-to general conclusions, summary, recommendations, etc. Such reports use the well-tested method of starting with introductory material and progressing through factual support to conclusions.

The *deductive pattern of report writing* is the most accepted type of report writing in the business world. It is obviously known as the executive report, the management report, or the action-getting report. This is the pattern of presentation that goes from the general to the specific, i.e., from the more important to the less important. It tends to become more detailed or more specialized in its last pages because it is designed for a reader whose fundamental question is "what do I need to know the act or to decide?" He/she wants to answer at the very outset. The writer has a special obligation to think his/her way through to the central issue or to the main point. It is almost like a newspaper report. One who goes through the title or initial introduction would be able to get a general idea and when one goes on reading, the specific areas are unveiled. The writer has to start on common ground with the reader in the deductive pattern of report writing.

A *step-by-step presentation* doesn't follow any hard and fast rule, it simply presents material blow-by-blow. It is a way of presenting the events or steps in an operation frequently following one another in a point of time. But the step need not necessarily come to a conclusion or an end. A step-by-step pattern is well adapted to reports on manufacturing operations, where a step-by-step analysis is essential. Sometimes, this pattern is well suited to progress reports, which cover only a short period of time.

The *time-sequence pattern of report writing* is almost like that of the step-by-step presentation; but it lays greater emphasis on the time element. It is an easy way of writing a report by merely following the pattern the research material it self offers in a sequence of time. Some readers prefer analysis based upon a sequence of time of events.

8.4.3. Steps in Report Writing

A few steps may be suggested which would enable the researcher to complete his/her task of writing a report clearly and cogently. These steps are:-

1. Plan the project well in advance; fix the target and the final date of completing the report
2. The time for completing data collection and data processing should be well planned, and implemented (ensure that adequate data are kept ready).
3. The time for report writing should be planned, and the task of report writing should not be put off till the last minute. At least, one-third of the total project time should be earmarked for this purpose.
4. Select a structure for the report, arrange in group the data, documents, bibliography, etc in conformity with the structure of the research report
5. Prepare an outline based on the structure, which should cover the main points and sub-points in detail.
6. Prepare a rough point-by-point skeleton for each chapter theme, and crystallize the report in a mental exercise. Do not hesitate to discuss the skeleton with somebody who is well-versed in research and writing, and who can be taken into confidence
7. In the event of any doubt, get it cleared by reading, thinking, recollecting and discussing, before going farther. Put everything in a logical sequence.
8. It is always good to the benefit of the guidance of a good guide at all stages
9. Avoid easy-going and short-cut methods and don't hesitate to write an entire first draft
10. Put the complete rough draft away for several days until you can bring a fresh mind to bear up on it. The researcher is then in a position to view it afresh and make substantial changes
11. Revise the draft thoroughly before the report is typed. Do not hesitate to eliminate any irrelevant and inappropriate portion.

8.4.4. Format/layout of the Research Report

A research report usually follows a fairly standardized pattern. The first step towards preparing a report should be the preparation of an outline of its major divisions and then sub-divisions in the form of headings and sub-headings of proposed chapters, sections or even paragraphs. This outline would guide the researcher in arranging material at his/her disposal in an orderly manner as it sets down the design of the report. The following outline presents the usual sequence of various sections:

A. Preliminary section or Front Matter

- Title page
- Approval sheet
- Acknowledgements (if any)
- Preface or Forward
- Table of contents
- List of tables (if any)
- List of Figures (if any)
- Abbreviations or glossary of technical words used in the report
- Abstract or summary of the entire report

B. Main Body of the Report

1) Introduction

- Background of the study
- Statement of the problem
- Objective of the study
- Significance of the study
- Assumptions and delimitations
- Definition of important terms
- Statements of Hypothesis
- Assumptions underlying the hypothesis

2) Review of related literature or Analysis of previous Research

3) Design of the study

- Procedures used
- Methods of gathering data
- Description of data gathering instruments

4) Presentation and Analysis of Data

- Text
- Tables
- Figures

5) Summary and Conclusions

- Brief Restatement of problem and procedures

- Description of procedures used
- Principles findings and conclusions
- Recommendations for farther research

C. The end matter

- Bibliography
- Appendix
- Index (if any)

Note that when you write your research report, five major sections of the main text of a research report are the chapters on which you organize your report.

Dear learner! For more understanding of the above three main sections of a research report, read the following discussions and examples.

A. The preliminary Section: All the components of the preliminary section are to be written on separate pages and with appropriate capital letters and centering the entire text on the page. *The title page* includes the following items: the title of the report, for whom and by whom the report was prepared, the date of submission, and degree for which the report will be submitted. If it is for a journal, there is no need to include the last two items. A satisfactory title should be brief.

Acknowledgement page is a section that recognizes individuals and institutions to which you are indebted for providing credible assistance in due course of your study.

Table of contents page contains list of major classifications of the report with their corresponding page numbers. Observe the example as follows.

S. No	Contents	Page. No
1.	Acknowledgements	i.
2.	List of tables	iv

List of Tables and figures page contains the list of tables and figures you use in the report with their corresponding page number. Look at the following two examples.

Table No.	Table	Page. No
1.1.	Age Distribution	10
2.7.	Spatial Distribution	80

Fig.No.	Figure	Page. No
Fig.1.1.	Description (of the fig in precise words)	20

Abstract or synopsis is a short summary. The size of the abstract varies from as few as fifty words to as large as a full page. It should exist following the list of figures on the separate page. Information to be included in a summary is Brief statement of the problem, objectives of the study, method and procedures of data collection and analysis, and major findings but recommendations are not included.

B. The main body of the research report

The introduction section normally contains several major subsections. In this part what are included in the proposal part are incorporated.

Literature review is section for documentation with insight theoretical and empirical investigation that had been carried out as related to the study at hand.

Material and methodology part includes detailed description of the manner in which decision has been made about the type of data needed for the study, the tools and approaches used for their collection, and the method by which they have been collected and justification of the method, definition of the population, the sampling techniques used with its justification, the size of the sample and the rationale for the size, statistical tools used for data analysis and the rationale will be dealt in detail in this section.

Findings and Discussions part: This is generally the longest section of the report. It is an organized presentation of the results. The objective of the findings is an exposition of what the data says rather than drawing interpretations or conclusions. In the discussion section, the researcher presents what interpretations he/she can draw from the data collected and gives

the reader of the report an unambiguous interpretation.

Summary and conclusions: The summary is a brief re-statement of the essential parts including the problem, the hypothesis, description of the problem and findings. The findings state facts while conclusions represent inferences drawn from the findings.

Recommendations: This involves suggested future actions. In academic research, the recommendations are likely to be for further study to test, deepen or broaden understanding in the subject area. In applied research, for decision making, the recommendations will usually be for managerial actions rather than research actions.

C. The appended section

Appendix: complex tables, statistical tests, supporting documents, copies of forms used (questionnaires), detailed description of methodology, instructions to field works and many other evidences that may be important are incorporated in this part.

Bibliography: There should be a bibliographic section if the study makes heavy use of secondary material. For specific reference, use foot notes and endnotes placed at the conclusions of each major section rather than referring to the bibliography. The appended bibliography should contain all the works which the researcher has consulted. It should be arranged alphabetically. There may be several bibliographic entry formats. But the following points should be emphasized on the styles and formats in writing the bibliography.

- All references noted in the text, including the sources for figures and tables must appear in the reference list. (Note: You should include only those references that are cited in the paper).
- Each item in the list should include all pertinent facts about the publication: author, title of the work in question and any journal or book in which it is included, city of publication, publishing company, date, page numbers (for journal and articles).
- Don't give number for references
- When there is more than one work by a single author, place them in reverse chronological order i.e., put the most recent item first and work backward.
- Don't use lines to indicate that the name is repeated; just repeat the name.
- Format reference lists with a "hanging indent", i.e., the first line of the entry is at margin, with second and subsequent lines indented ones.

- Allow one space between the entries.
- Alphabetize references according to the author's last names. For more understanding on other aspects on how to write references, see the detail discussions made in **chapter three**.

? **Dear learner!** Work out on the following questions:

1. List the types of reports you have already leaned _____
2. What are the main components of the out-line of a research report? _____

8.4.5. Further Rules

I. Size and physical layout: The manuscript should be written on un-ruled paper 8.5" X 11" in size. If it is to be written by hand, then black or blue-black ink should be used. A margin of at least one and one-half inches should be allowed at the left hand and of at least half an inch at the right hand of a paper. There should also be one-inch margins, top and bottom. The paper should be neat and legible. If the manuscript is to be typed, then all typing should be double-spaced on one side of the page only except for the insertion of the long quotations.

II. Tables and Figures

Tables:-All tables including those that appear in the appendix should be numbered consecutively to permit their easy identification by the reader. The word *table* is followed by its number.

Ex: Table 1: Age and Sex Distribution

Capitalize the first letter of all principal words of the table title as given in the above example. The title of the table should be at the top of the table. Regarding the source and notes of the table, they are put immediately below the table as: "Compiled by the researcher" (if the data is collected by you as a researcher) or , for e.g., "Philip Kotler, Principles of Marketing, Vth edition, 2000 (if secondary data is used).

Content and placement of tables: If the table is more than half a page, it should be placed center on a separate page. If it is short and occupies less than half a page, place it on a page with textual matter. Long and detail tables should be put in the appendix of the document.

Figures:- Figure is any type of illustrative material other than table. Ex: Graph, line drawing, chart, photograph or map should be labeled as figures. Figures are labeled with the

word “Fig. No”. Figures should be numbered consecutively throughout the report to permit their easy identification. . They should be numbered with Arabic numerical.

Example: Fig.12

In the report, title of the figure is given below the figure or illustration.

8.4.6. Dissemination of a report through oral presentation

The merit of orally presenting the result of a research study is that it provides an opportunity for give-and-take decisions which generally lead to a better understanding of the findings and their implications. But the main demerit of this sort of presentation is the lack of any permanent record concerning the research details and it may be just possible that the findings may fade away from people’s memory even before an action is taken. In order to overcome this difficulty, a written report may be circulated before the oral presentation and referred to frequently during discussion. Oral presentation is effective when it is supplemented by various visual aids. Use of slides wall charts, and blackboards is quite helpful in contributing to clarity and in reducing the boredom, if any. Distributing a broad outline with a few important tables and charts concerning the research results, makes the listeners attentive who have a ready outline on which to focus their thinking. This very often happens in academic institutions where the researcher discusses his/her findings and policy implications with others either in a seminar or a group discussion.

Chapter Summary

Dear learner! Let us summarize what you have already learned in this chapter. Basically, you learned about data interpretation and research report writing. Data interpretation is the task of drawing conclusion or inferences and of explaining their significance after a careful analysis of selected data is made. Data interpretation demands a researcher to have a good skill. Of all, practice and experience play great role in having an effective interpretation.

The second concept was about research report writing. Writing research report is the most crucial step in the research process as it communicates the findings of the study to supervisors and readers or any other users. But when it is badly written, it spoils the entire work.

A research report format contains three sections: The preliminary section, the main text, and the end section. The preliminary section further contains the title page, approval sheets, acknowledgments, abstract, table of contents, and list of tables and figures. The main text section of the report again contains main parts including an introduction, review of related literature, design of the study, presentation and analysis of data, and summary and conclusions parts. Finally, the end matter a research report consists of bibliography, appendices, and indices (if any).

Exercises**Part I. Multiple Choice Questions**

1. One of the following statements is false regarding the use of interpretation of data.
 - a. It makes it possible for us to utilize collected data in various fields of activity
 - b. The usefulness of the collected data lies in its proper interpretation
 - c. It provides certain conclusions about the problem under study
 - d. None of the above
2. Insufficient and incomplete data should not be used for analysis and to make meaningful interpretations. This statement calls for the data to be
 - a. Homogeneous
 - b. Adequate
 - c. Appropriate
 - d. All
3. One of the following statements doesn't support the purpose of a research report
 - a. It communicates the various aspects of a research study
 - b. It strengthens the interpersonal relationships in the organization
 - c. The report writer should only worry about what to communicate but not who will read report
 - d. None of the above
4. One of the following types of a research report is a document submitted for the acquisition of higher degree from a University or any other higher academic institution
 - a. Thesis
 - b. project report
 - c. dissertation
 - d. a or c.
5. One of the following is incorporated in the preliminary section of a research report
 - a. Table of contents
 - b. Statement of the problem
 - c. Methods of gathering data
 - d. Recommendations for farther research
 - e. Bibliography
6. Each table that appears in the text of a report is characterized by
 - a. The word *table* is followed by its number.
 - b. Regarding the source and notes of the table, they are put at the top of the table
 - c. The title of the table should be at the bottom of the table
 - d. All of the above

Part II. Short answer questions

1. Briefly write on the steps involved in the techniques of interpreting data.
2. Narrate the various steps involved in writing a research report

Answers for Exercise Questions at the End of Each Chapter**Chapter One**Part I

- | | | | |
|------|------|------|------|
| 1. d | 3. e | 5. e | 7. a |
| 2. d | 4. c | 6. a | |

Part II

- | | |
|------------------|-------------------|
| 1. See page 6-7 | 4. See page 15-16 |
| 2. See Page 7-8 | |
| 3. See page 9-11 | |

Chapter Two

- | | | | | |
|------|------|------|------|------|
| 1. a | 4. d | 7. d | 10.b | 13.a |
| 2. d | 5. d | 8. a | 11.a | 14.a |
| 3. d | 6. e | 9. e | 12.a | |

Chapter Three

- | | | | | |
|------|------|------|------|-------|
| 1. c | 3. a | 5. b | 7. a | 9. d |
| 2. e | 4. e | 6. e | 8. a | 10. b |

Chapter Four

- | | | | | |
|------|------|-------|----------|------|
| 1. d | 4. d | 7. -- | 11. ---- | 14.e |
| 2. d | 5. d | 9. b | 12. b | 15.e |
| 3. e | 6. e | 10. e | 13. d | |

Chapter Five

- | | | | | |
|------|-----------|--------|-------|-------|
| 1. a | 7. --- | 13. b | 21.d | 27. b |
| 2. c | 8. b | 14. d | 22. a | 28. c |
| 3. c | 9. e | 16.a | 23. c | |
| 4. d | 10. e | 17.--- | 24. a | |
| 5. d | 11. ----- | 18.--- | 25. b | |
| 6. d | 12. ----- | 20. b | 26. a | |

Chapter Six

1.a	5.b	9.b	13. c	17. a
2.d	6.c	10. ----	14. a	
3.a	7.d	11. a	15. a	
4.a	8.b	12. a	16. d	

Chapter Seven

Part

I

1.e	3.a	5.d	7.c	9.c
2.a	4.e	6.b	8.a	10. b

Part II

1. See on page 233 2. See on page 233 3. See on page 235 4. See on page 236-241

Chapter Eight

Part I

11. d 12. b 13. c 14. d 15. a 16. a

Part II.

1. See page 248
2. See page 254-255

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